On track to discovery with the Higgs

Abraham Tishelman-Charny

Thursday, January 22nd 2026
Northeastern University



Overview

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Why do we study particle physics?

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Brookhaven National Laboratory

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→ To understand nature at its most fundamental level: What the universe is made of, and how particles interact.

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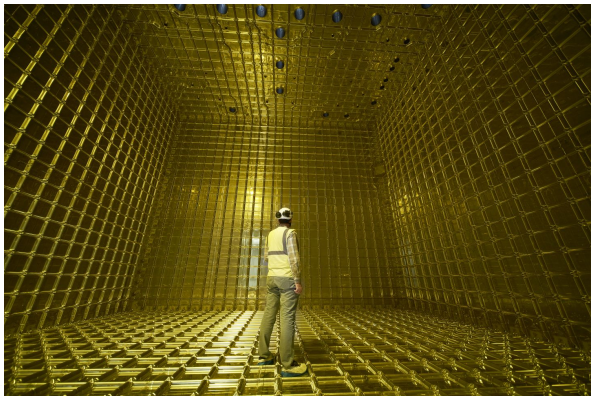
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How do we study particle physics?

Intensity frontier



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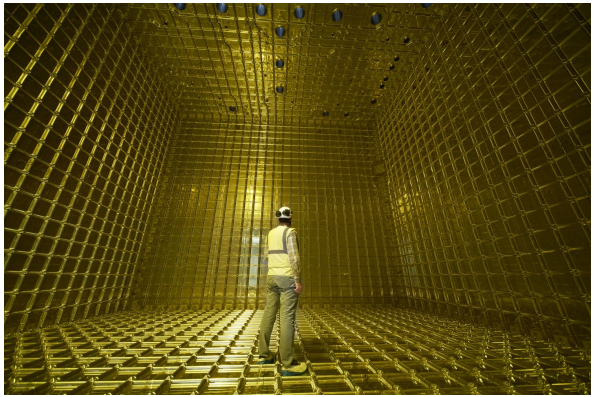
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Cosmic frontier



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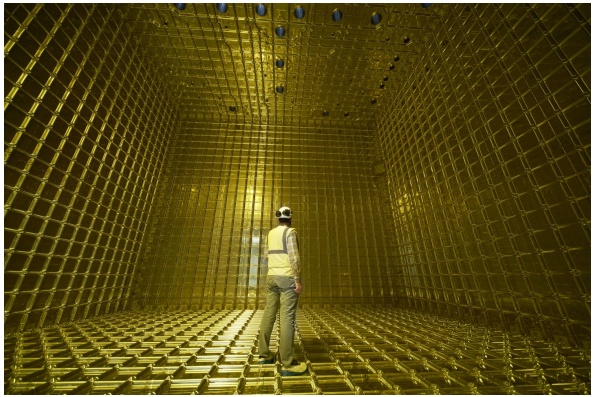
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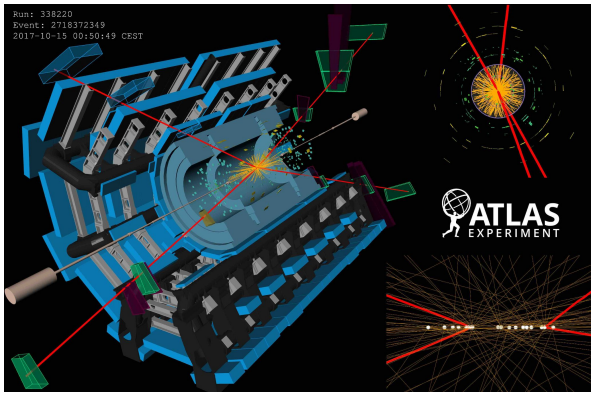
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Cosmic frontier



Energy frontier



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- Why is there more matter than antimatter?

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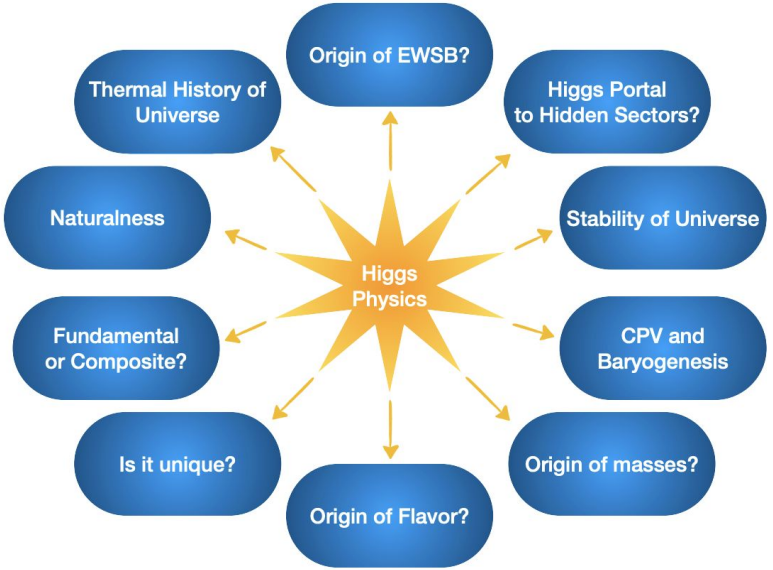
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Many of these questions have potential ties to **the Higgs boson!**



arXiv:2209.07510

Main questions

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Is the observed Higgs boson the standard model one?

Main questions

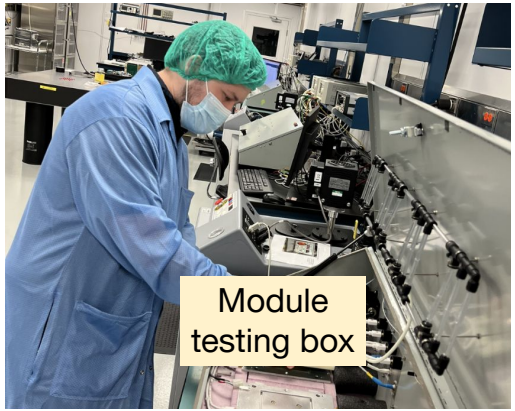
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Is the observed Higgs boson the standard model one?

Do other scalar bosons exist?

Outline







LHC

I

Higgs discovery

2009-2026

HL-LHC

II

di-Higgs discovery

2030-2041

Future

III

Future discoveries

2041-...

Next section

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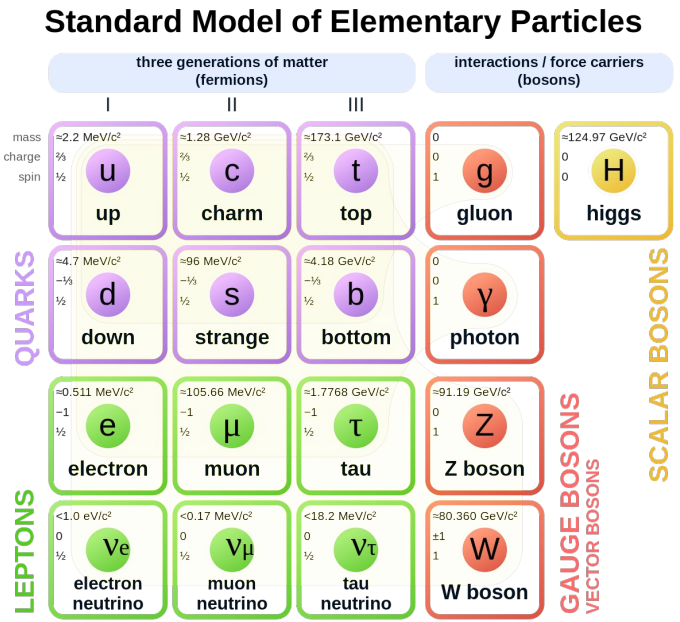
2041-...

The standard model

- What is our **theoretical basis**?

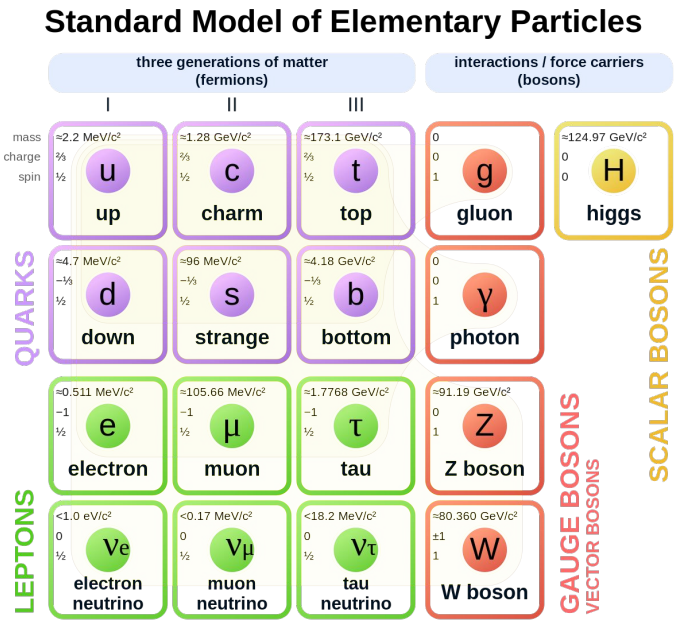
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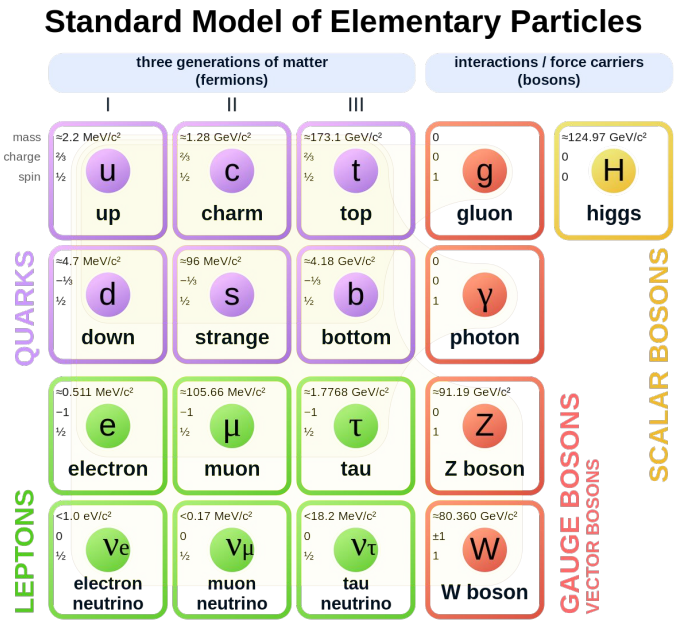
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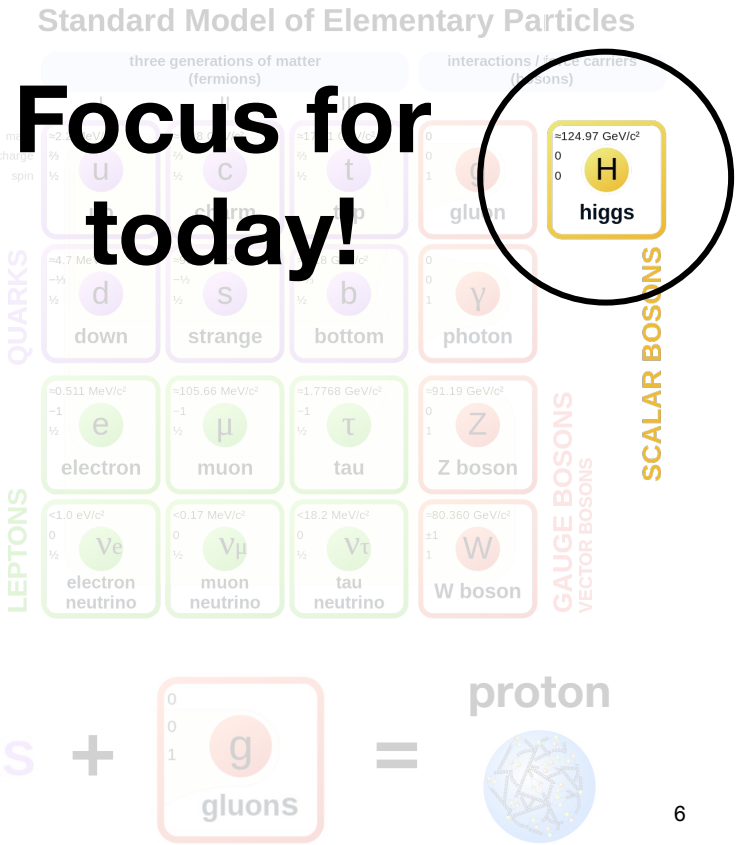
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- Extremely successful! Predicts **vast majority** of observations
- **Does not explain everything**:
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  - Nature of **dark matter**
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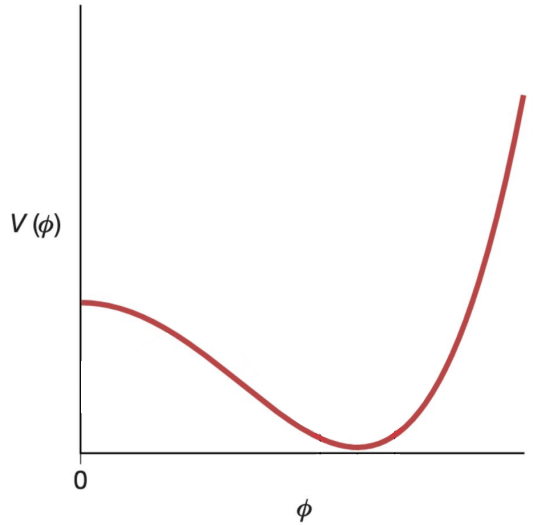
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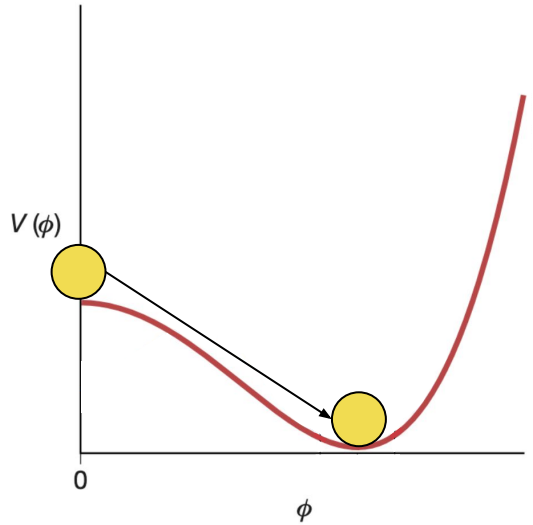
The Higgs potential

- **Higgs potential**: Determines nature of Higgs interactions with **other particles**



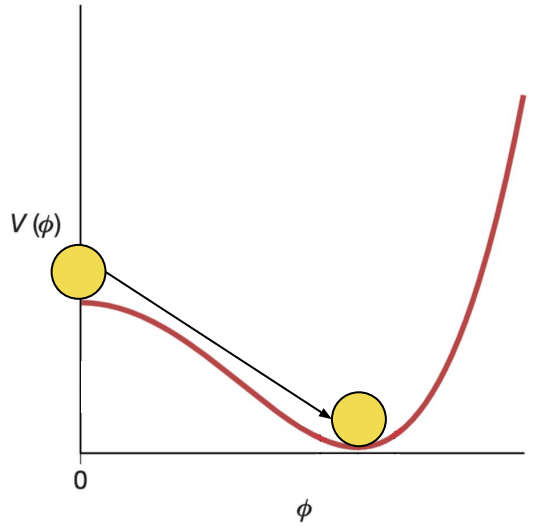
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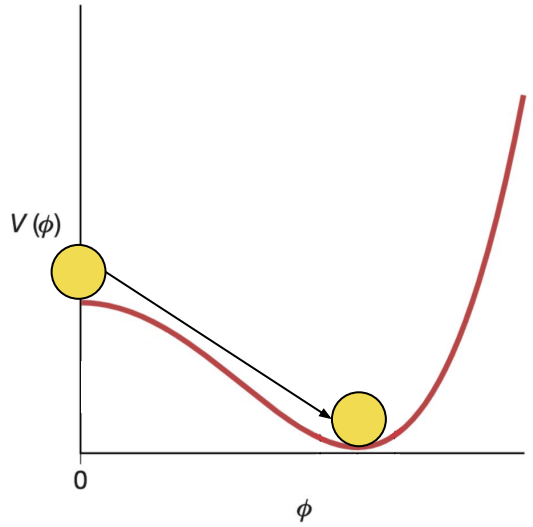


Higgs potential and mechanism

$$V(h) = V_0 + \lambda v^2 h^2 + \lambda v h^3 + \frac{1}{4}\lambda h^4 + \ldots$$

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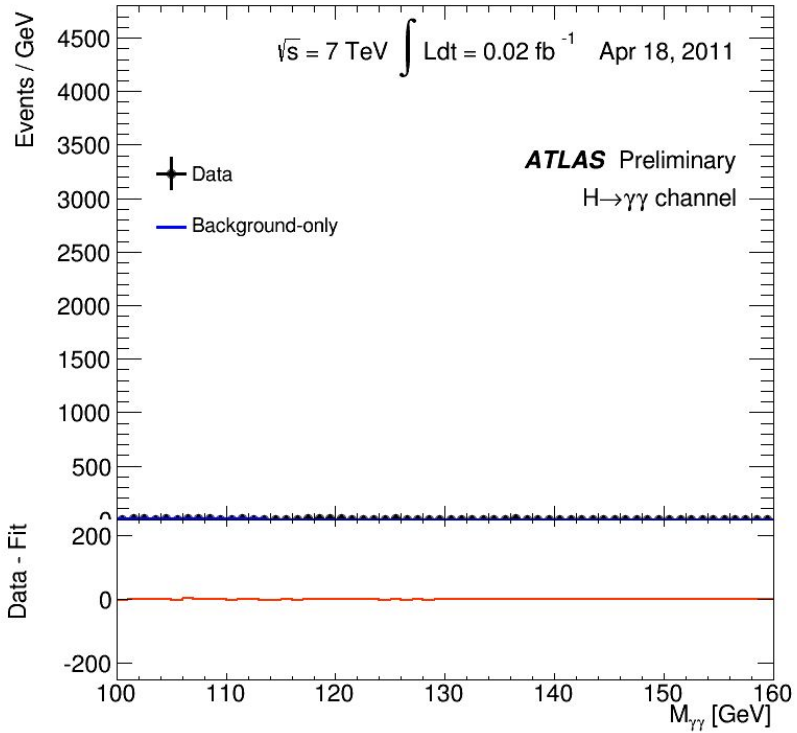
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The Higgs boson

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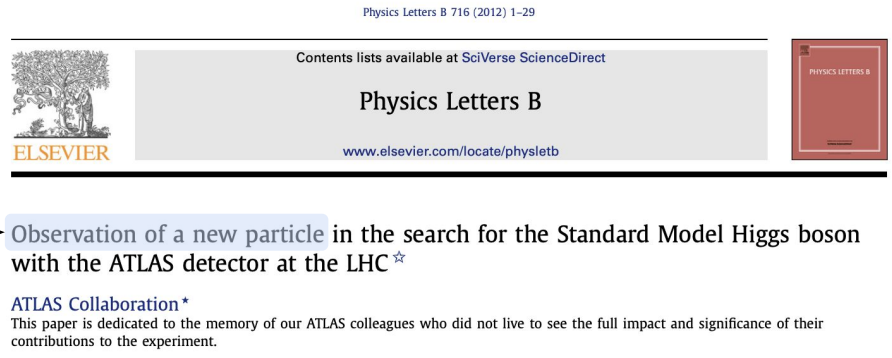
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Physics Letters B 716 (2012) 1–29

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Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC ☆

ATLAS Collaboration *

This paper is dedicated to the memory of our ATLAS colleagues who did not live to see the full impact and significance of their contributions to the experiment.

PLB 716 (2012) 1-29

PLB 716 (2012) 30-61

The Higgs boson

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- Also made **international** news!

The New York Times

Physicists Find Elusive Particle Seen as Key to Universe

Share full article 122



Scientists in Geneva on Wednesday applauded the discovery of a subatomic particle that looks like the Higgs boson. Pool photo by Denis Balibouse

By Dennis Overbye

July 4, 2012

Characterizing the Higgs

- **Is the observed Higgs boson the standard model one?**

Characterizing the Higgs

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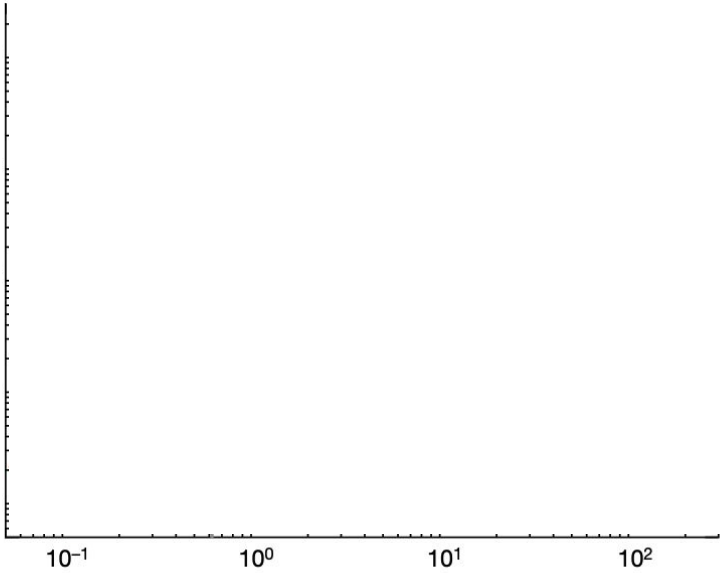
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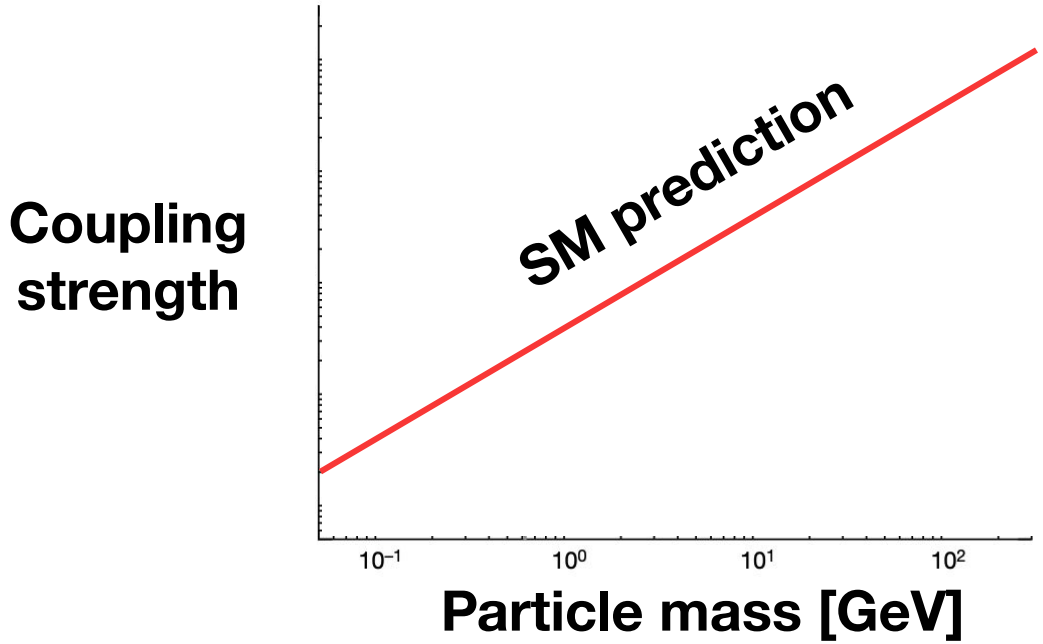
Coupling strength



Particle mass [GeV]

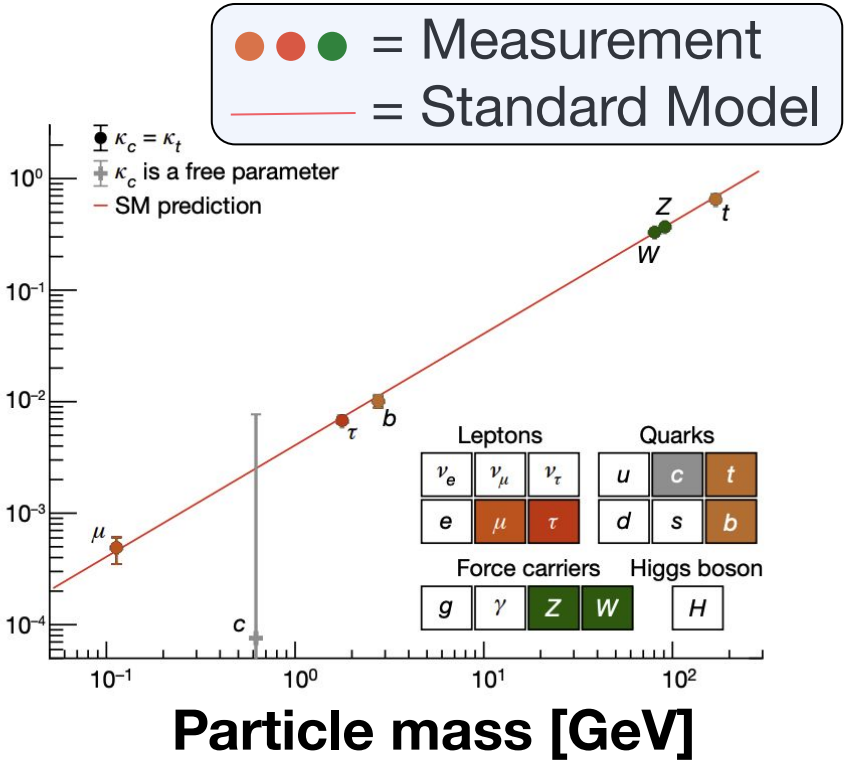
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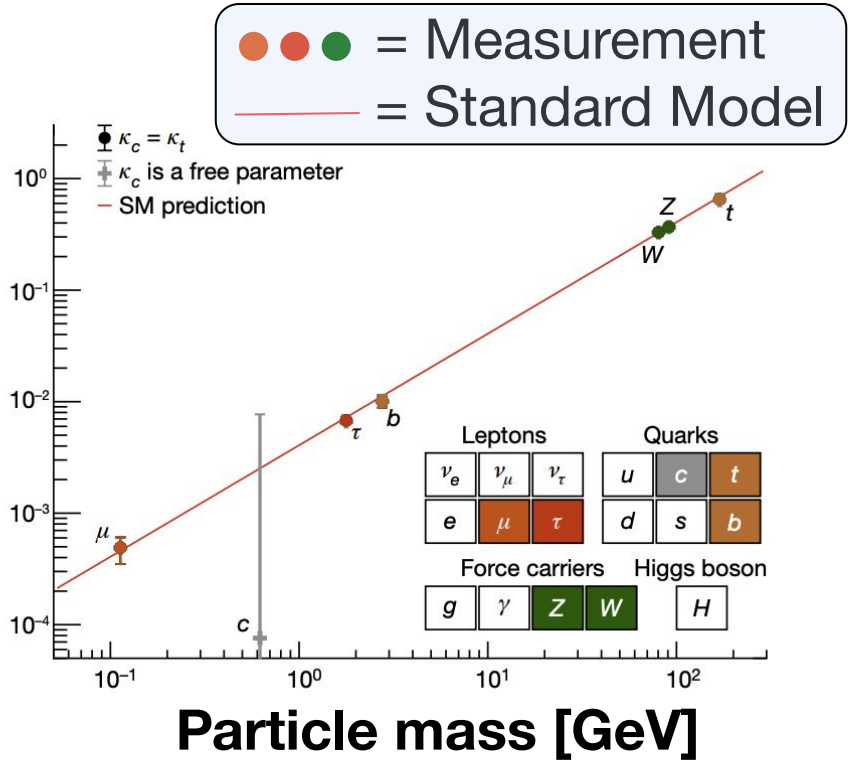


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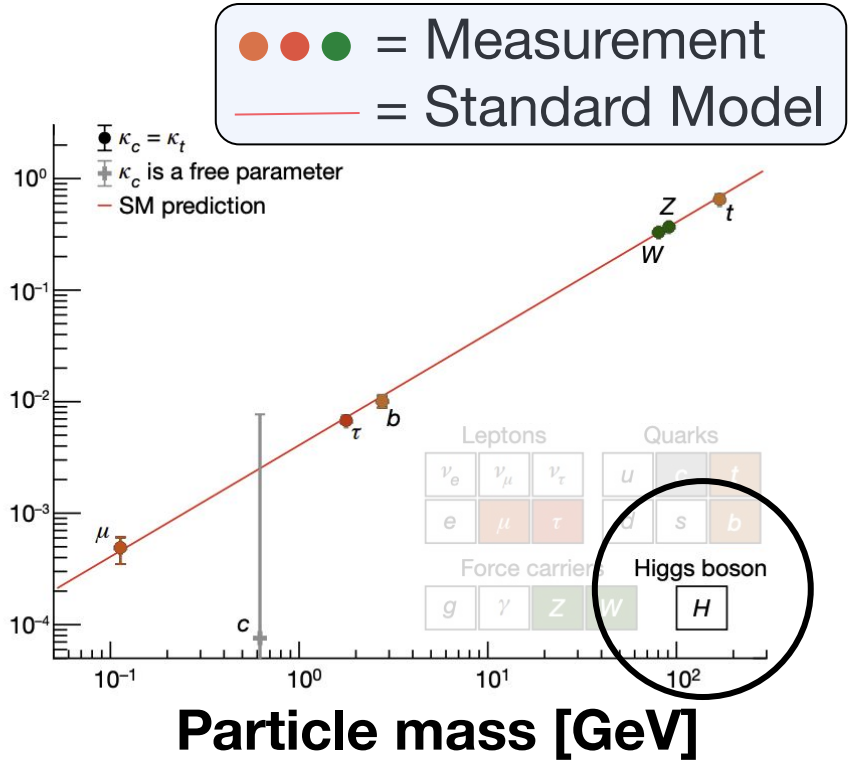
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- Need to characterize, compare to theory.
  - Many measurements since 2012!
- Came a long way, but **<u>there is more to measure</u>**:
  - The Higgs self-coupling



Higgs pair production (HH)

Brookhaven National Laboratory

Can directly access Higgs self-coupling via **Higgs pair production (HH)**

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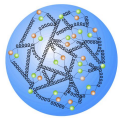
Gluon fusion:

Higgs pair production (HH)

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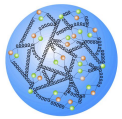
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proton



g

proton



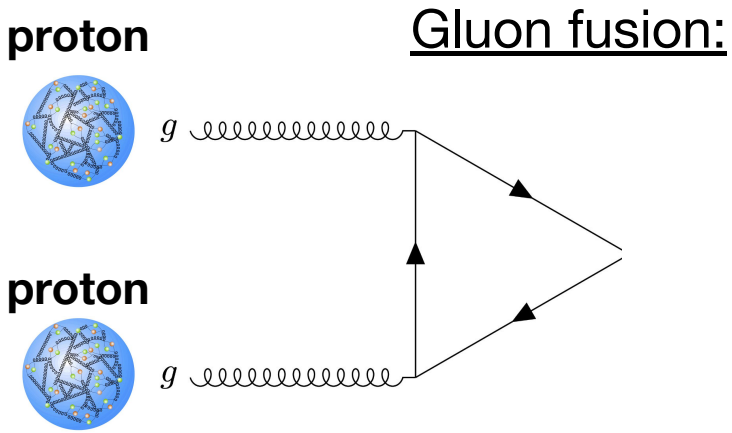
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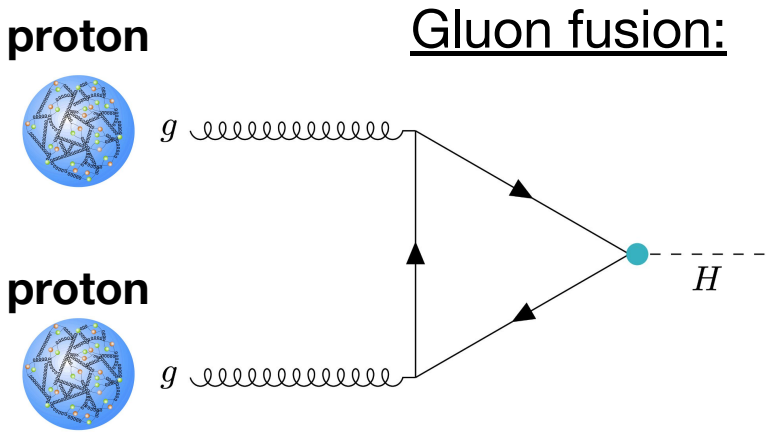
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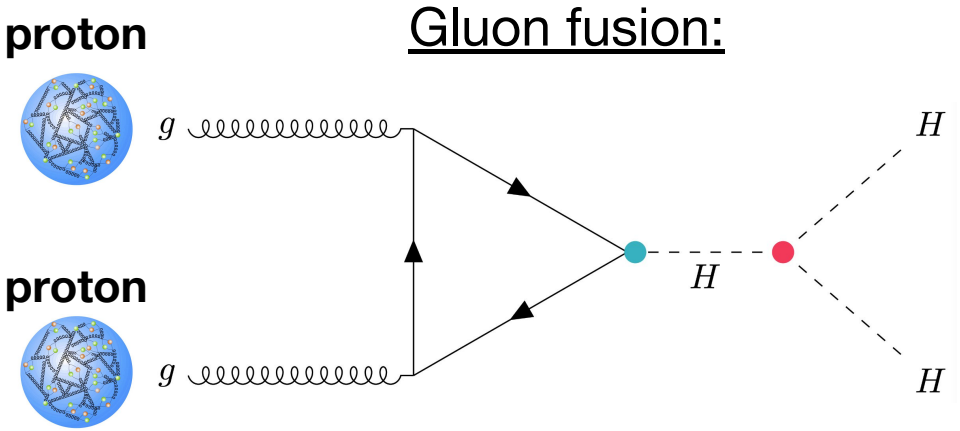
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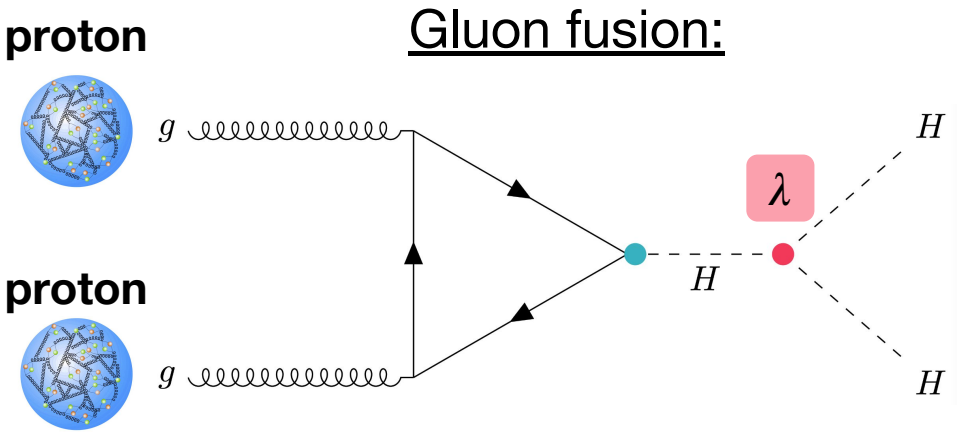
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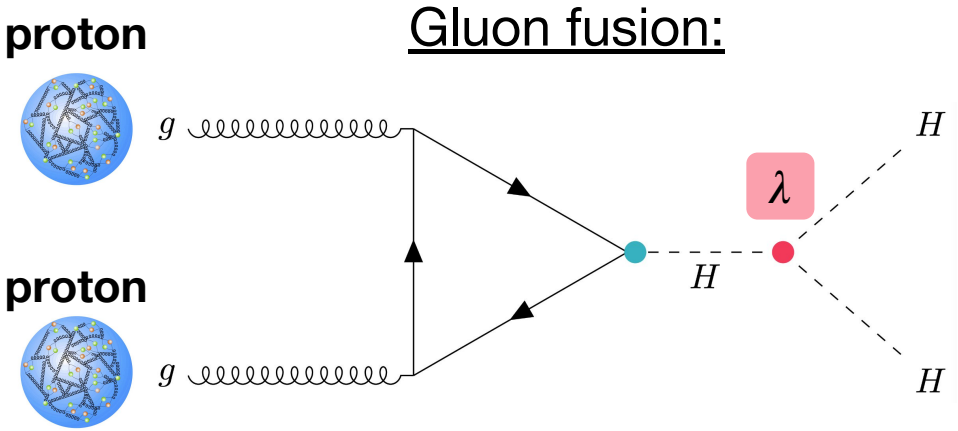
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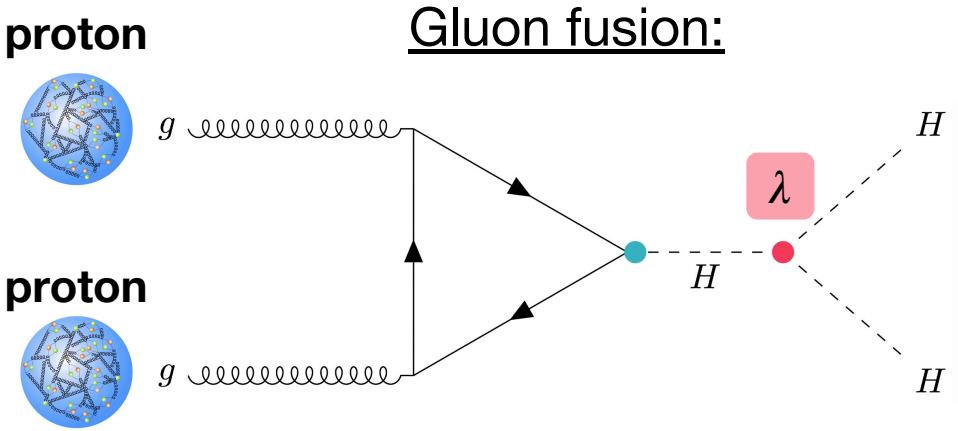


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Higgs pair production (HH)

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Can directly access Higgs self-coupling via **Higgs pair production (HH)**



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- **Rare** process - need to select **final states** with good signal to background ratio

The Higgs as a portal (X→SH)

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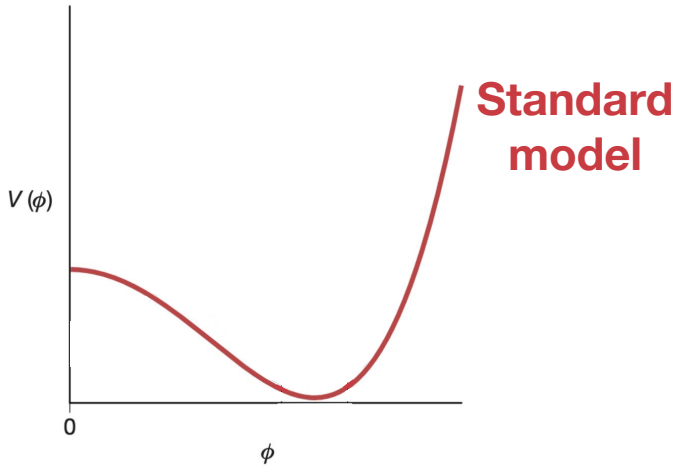
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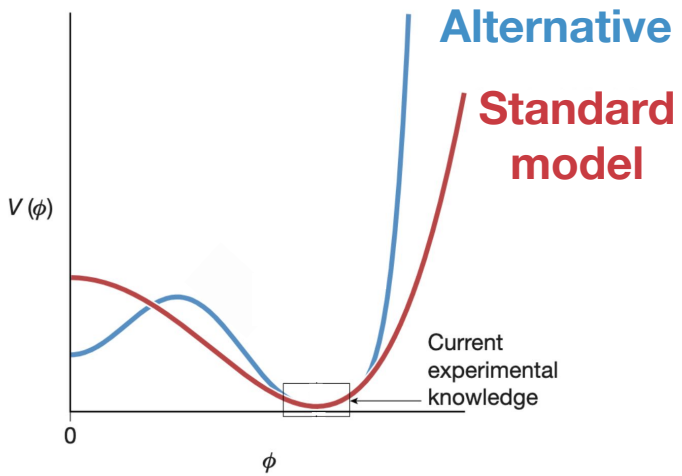


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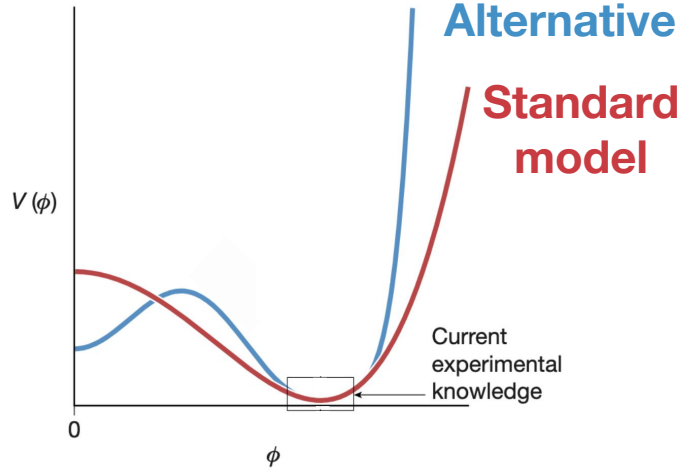


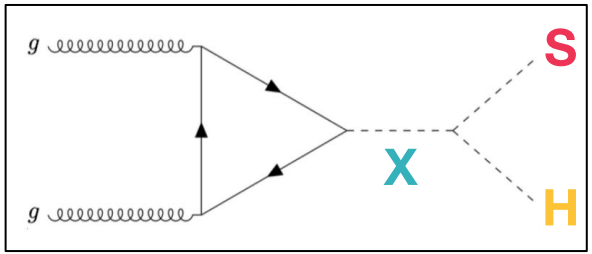
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  - **H:** Higgs boson
  - **X** and **S:** New "Higgs-like" Scalar bosons



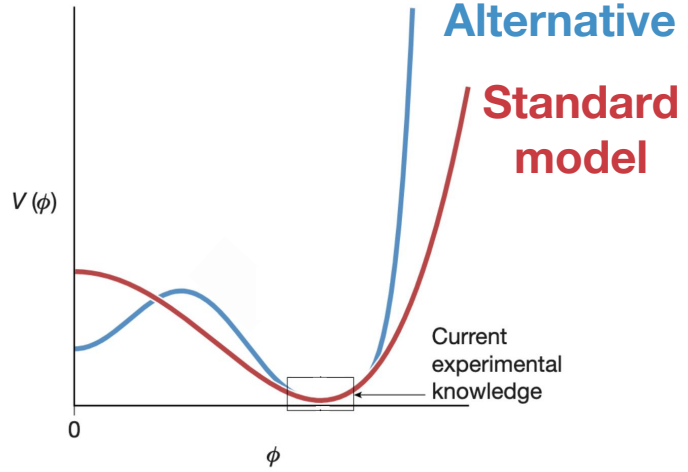


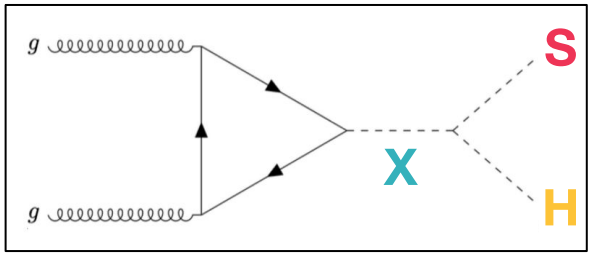
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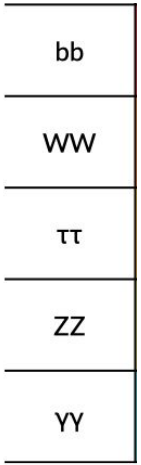
HH/SH final states

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HH/SH final states

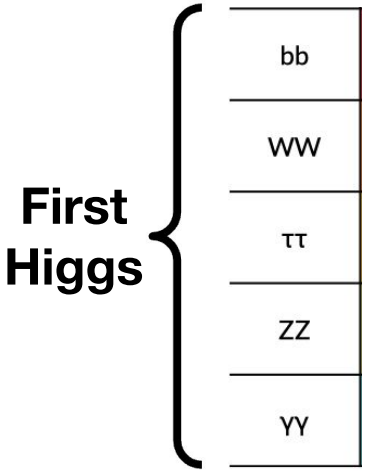
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Some Higgs decay modes



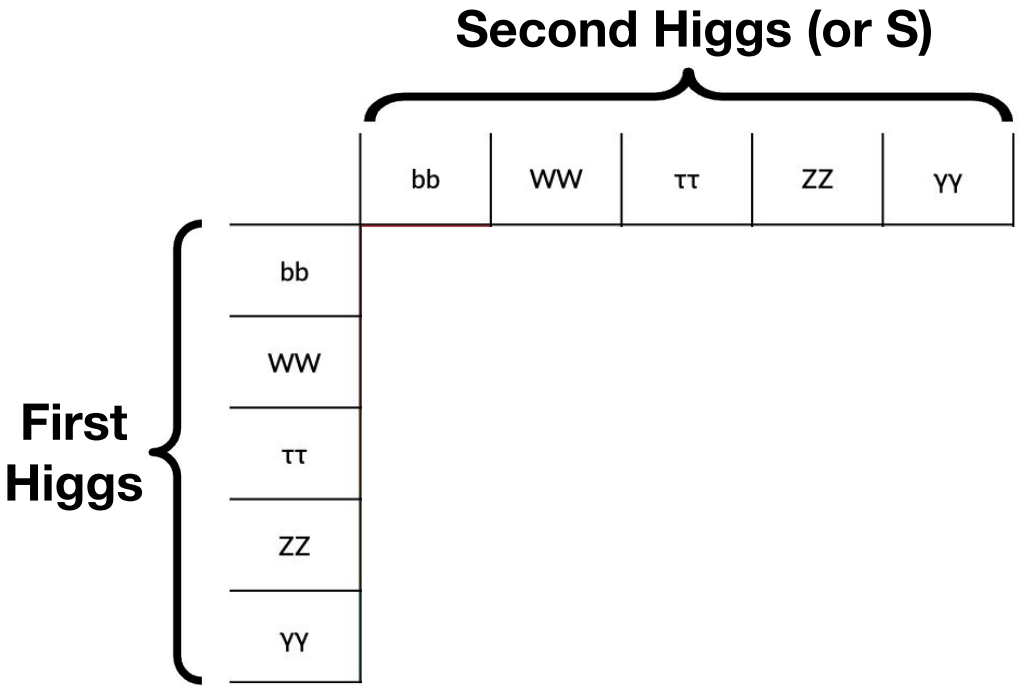
HH/SH final states

- **Higgs boson** has many decay modes
- Therefore, many **HH/SH** combinations



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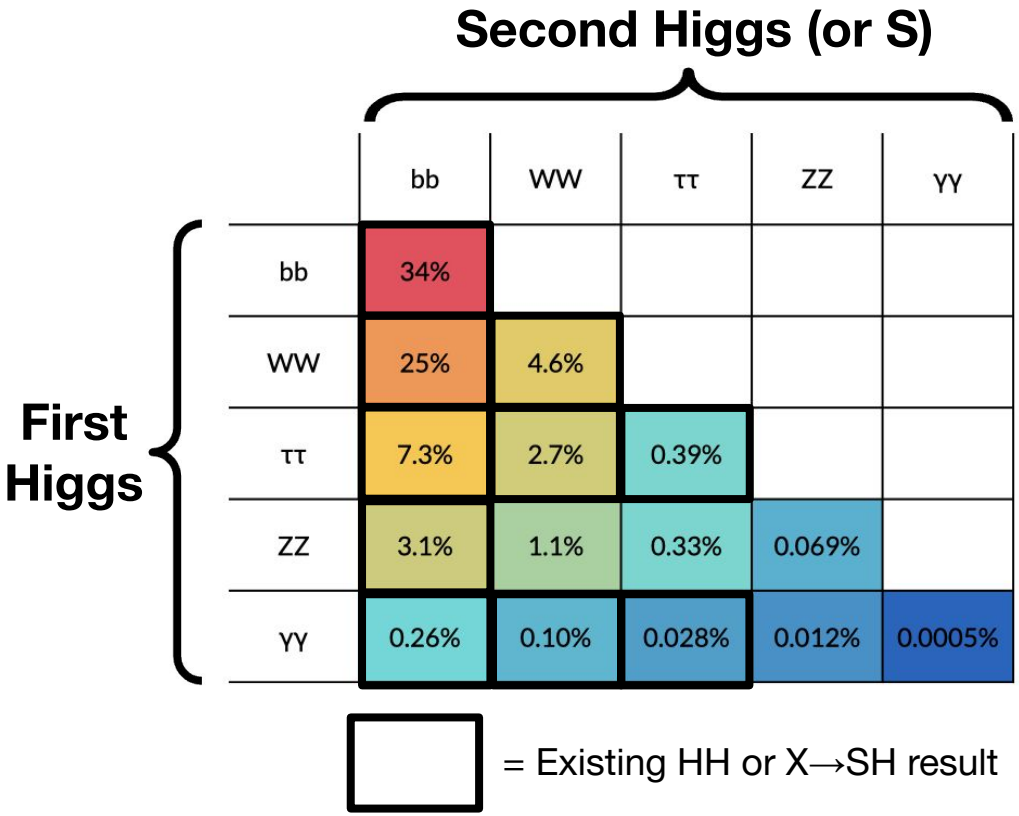
Second Higgs (or S)

First Higgs	bb	WW	ττ	ZZ	γγ
bb	34%				
WW	25%	4.6%			
ττ	7.3%	2.7%	0.39%		
ZZ	3.1%	1.1%	0.33%	0.069%	
γγ	0.26%	0.10%	0.028%	0.012%	0.0005%

Standard model HH likelihoods

HH/SH final states

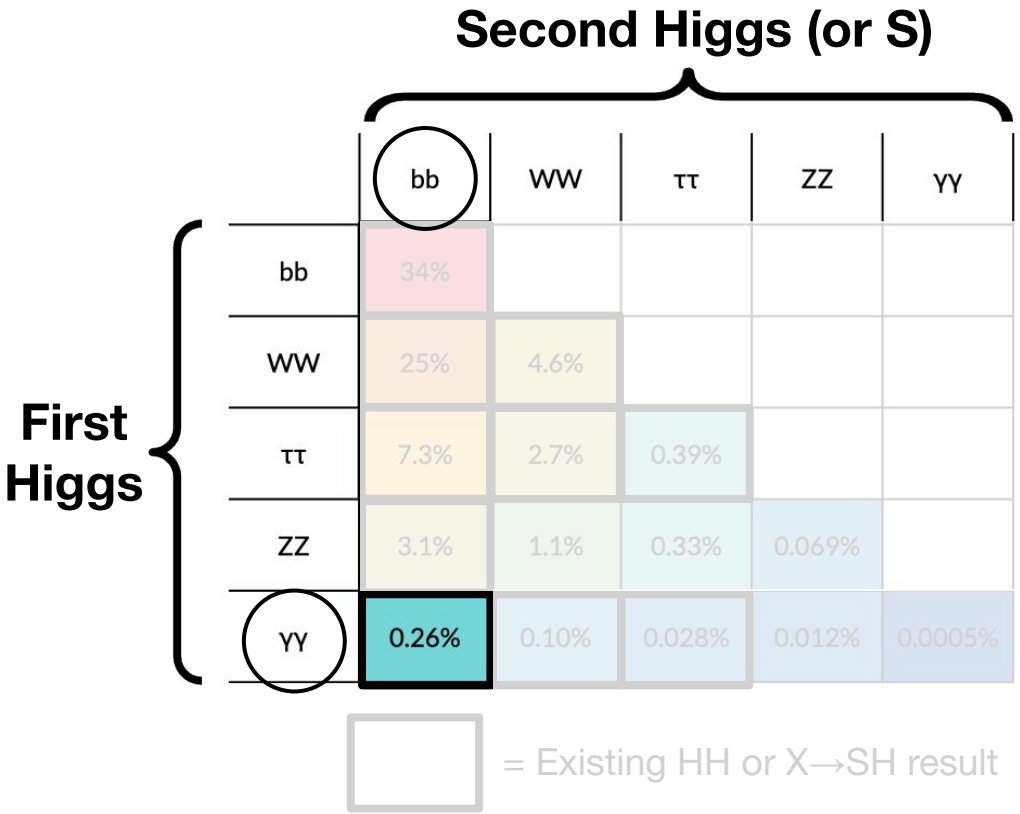
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Standard model HH likelihoods

The (bb)($\gamma\gamma$) final state

Brookhaven National Laboratory

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- **Clean $\gamma\gamma$ signature**. For SM and some BSM scenarios: **High bb branching ratio**

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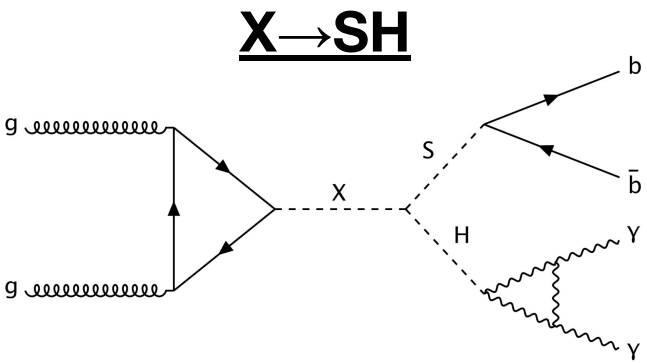
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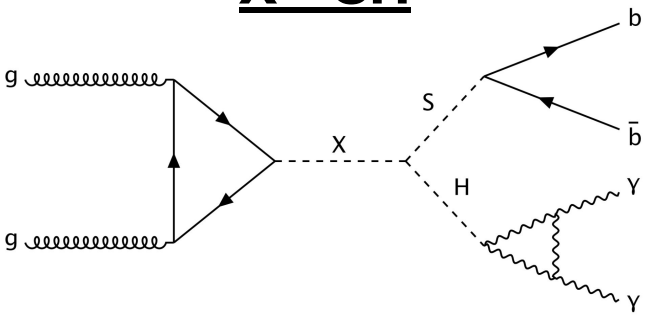
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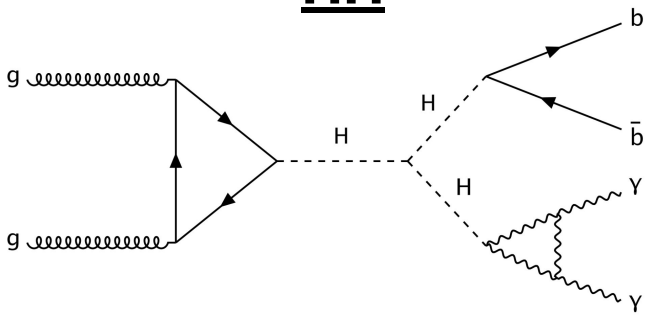
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X→SH

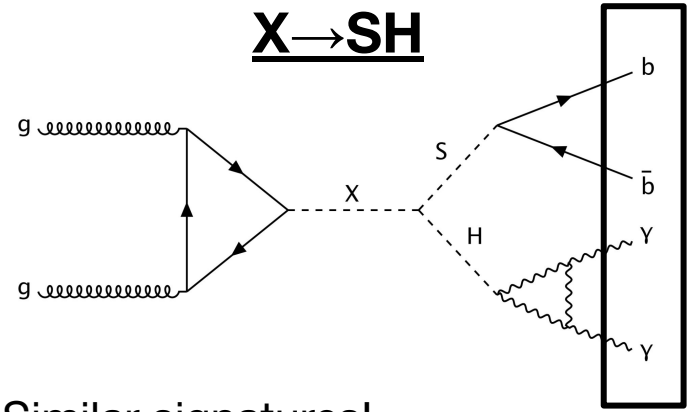


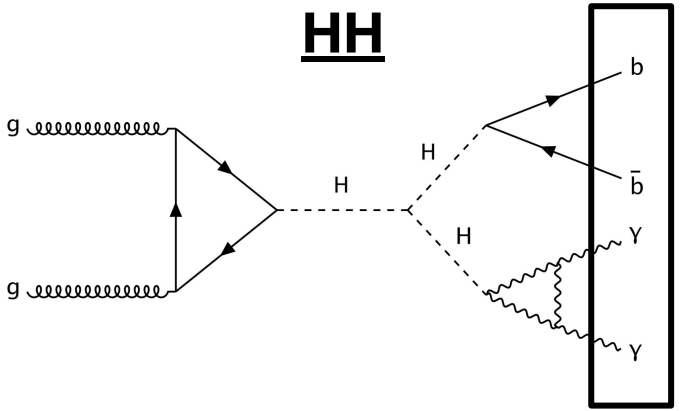
HH



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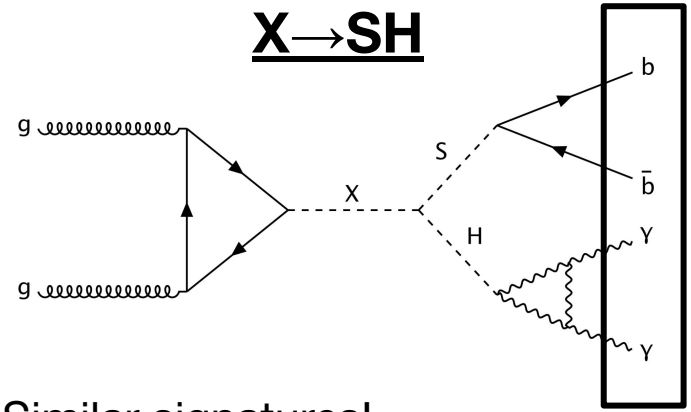


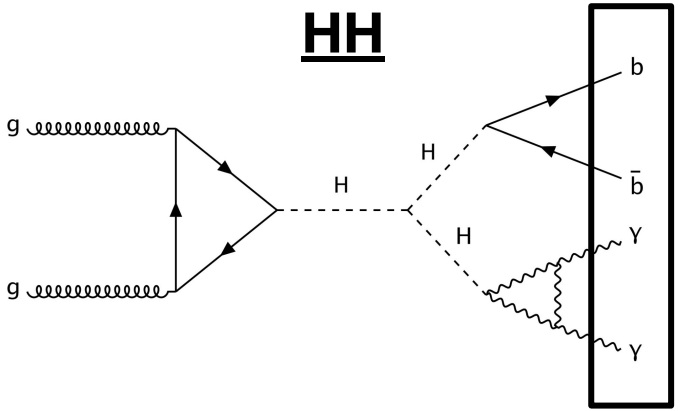


- Similar signatures!

The (bb)($\gamma\gamma$) final state

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- HH: Can achieve good sensitivity on self-coupling measurement.
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- Similar signatures!

- **Kinematics altered** by different **BSM** scenarios:
  - X→SH: Different X and S **masses**
  - HH: Varying of **coupling strengths**

LHC

- Need machine capable of producing **HH pairs**, **X→SH**

LHC

- Need machine capable of producing **HH pairs**, **X→SH**: **Large Hadron Collider (LHC)**



LHC

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- Collides **protons**, heavy ions up to **~ 99.999999%** the speed of light!

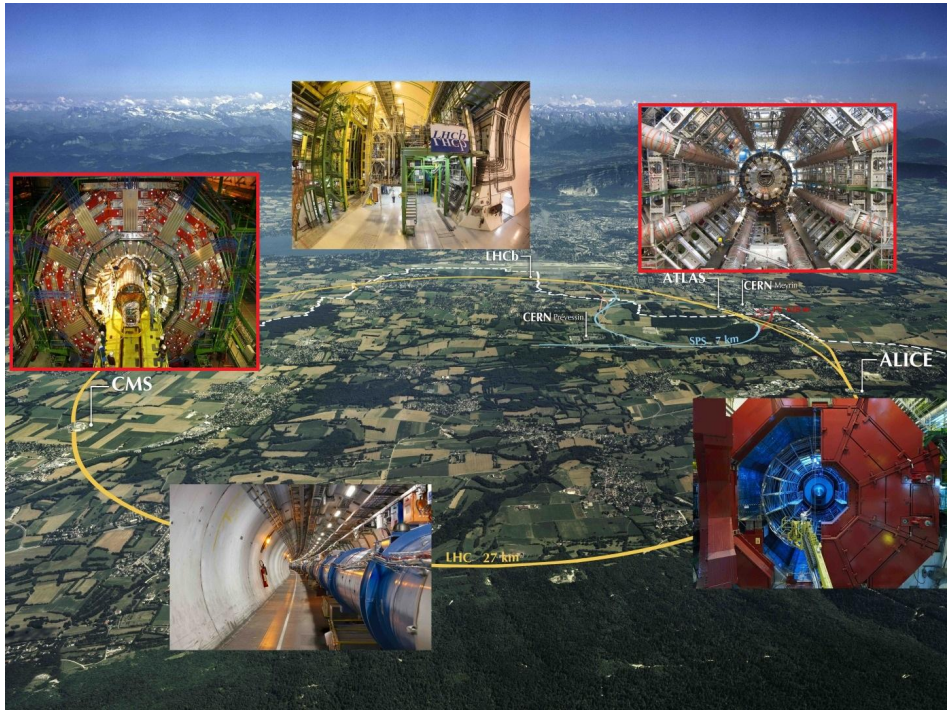


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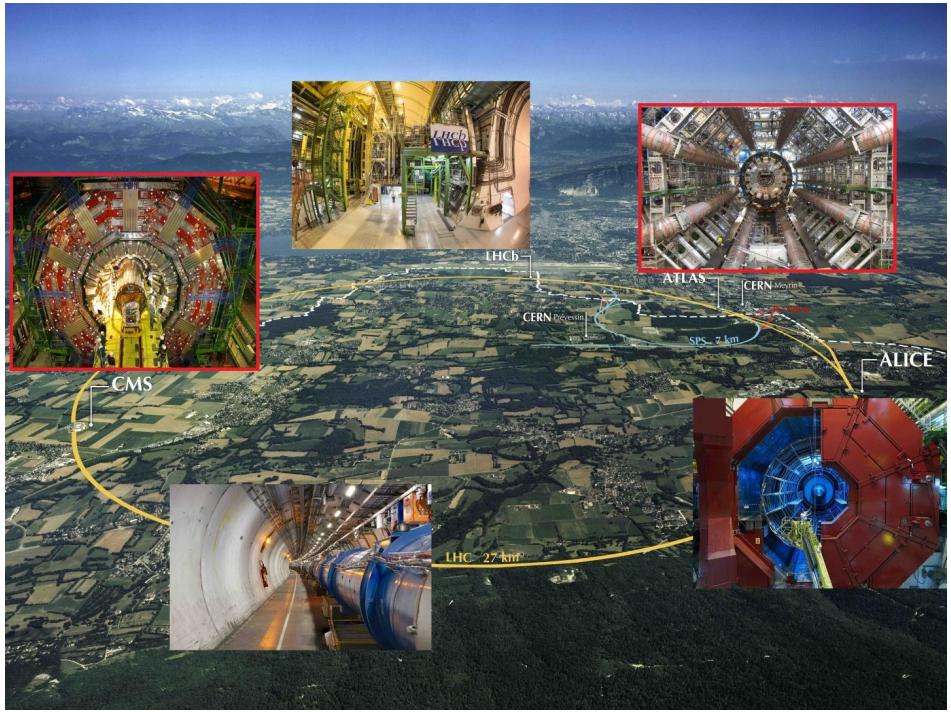
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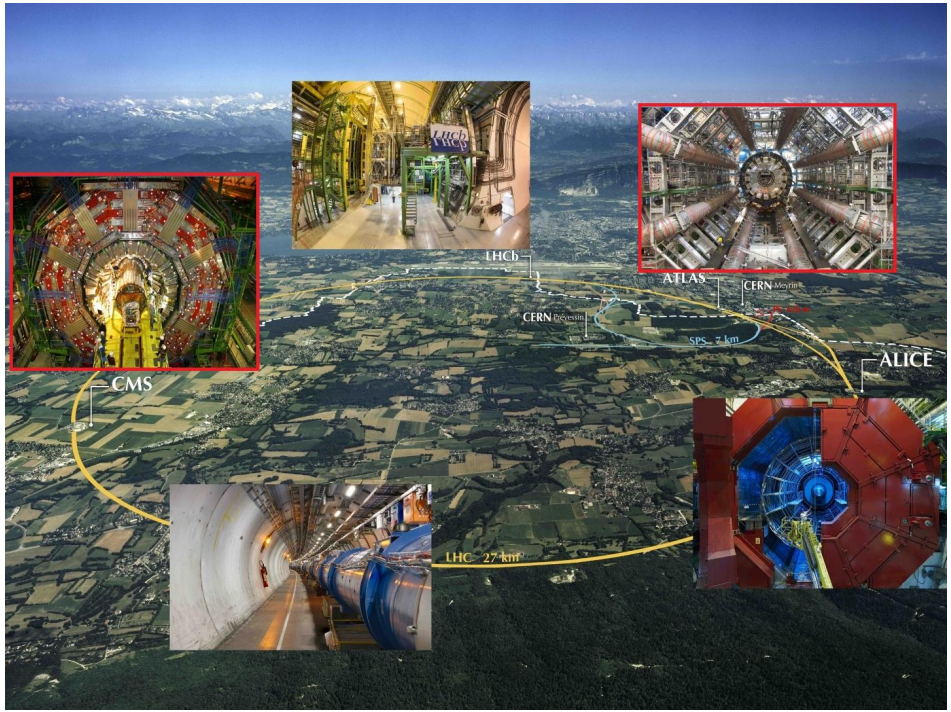
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- **Recent data taking periods:**

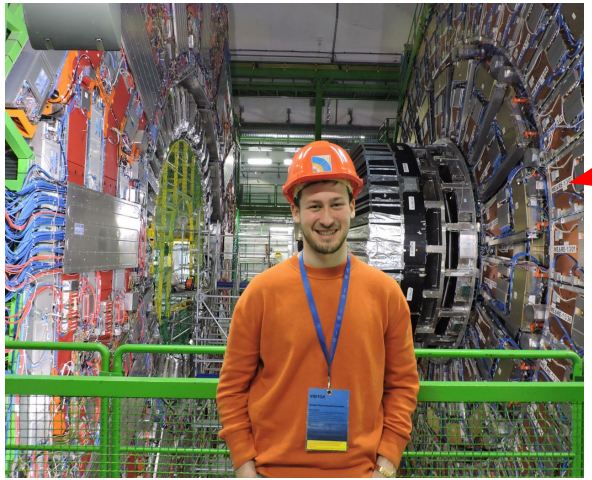
  Run 2: (2015-2018)

  Run 3: (2022-present)



LHC

- Need machine capable of producing **HH pairs**, **X→SH**: **Large Hadron Collider (LHC)**



CMS ~ 2019

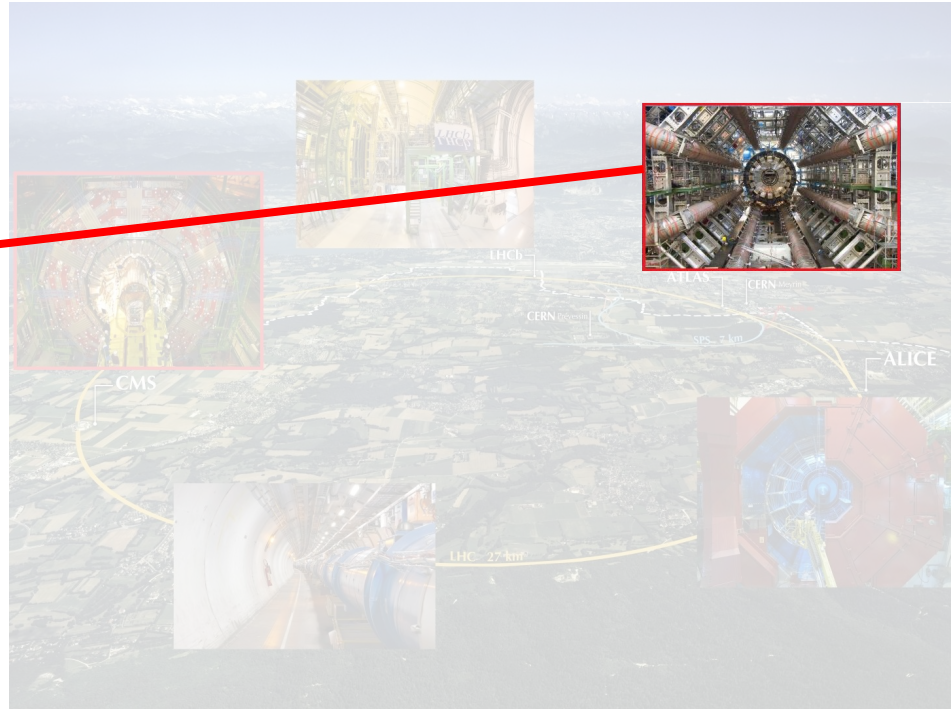


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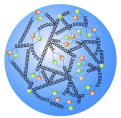
ATLAS ~ 2024



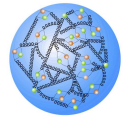
ATLAS

Brookhaven National Laboratory

proton



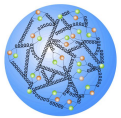
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ATLAS

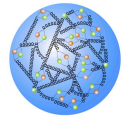
Brookhaven National Laboratory

proton





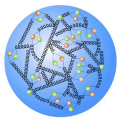
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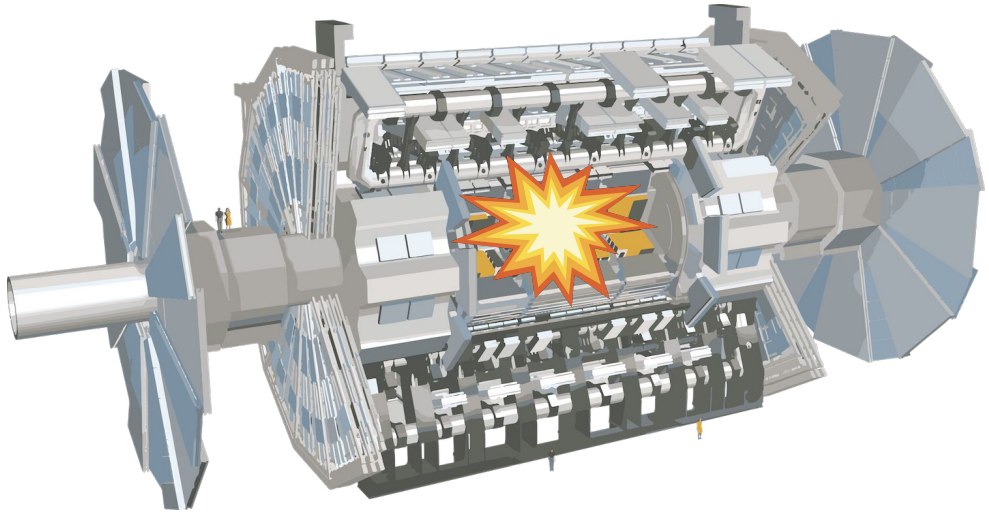


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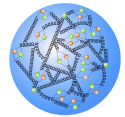
Brookhaven National Laboratory

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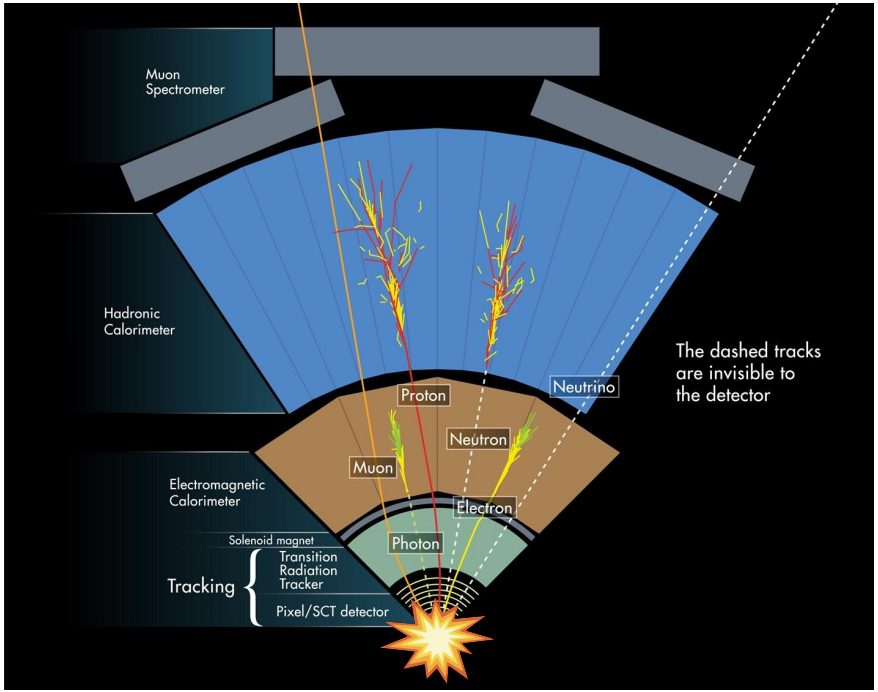




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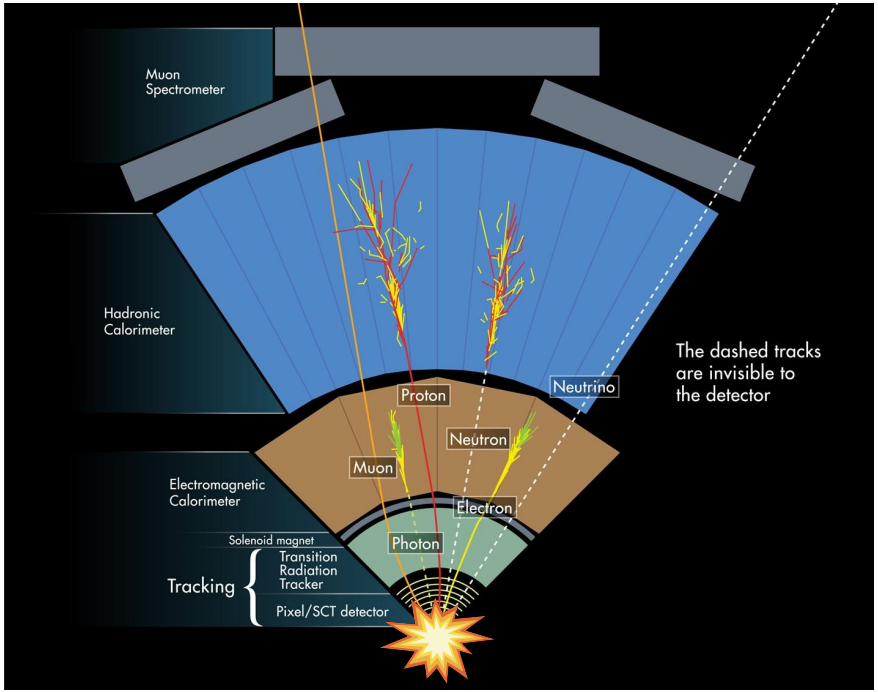


ATLAS


Muon
Spectrometer
Hadronic
Calorimeter
Electromagnetic
Calorimeter
Solenoid magnet
Tracking
Transition
Radiation
Tracker
Pixel/SCT detector
Proton
Neutrino
Neutron
Muon
Electron
Photon
The dashed tracks
are invisible to
the detector

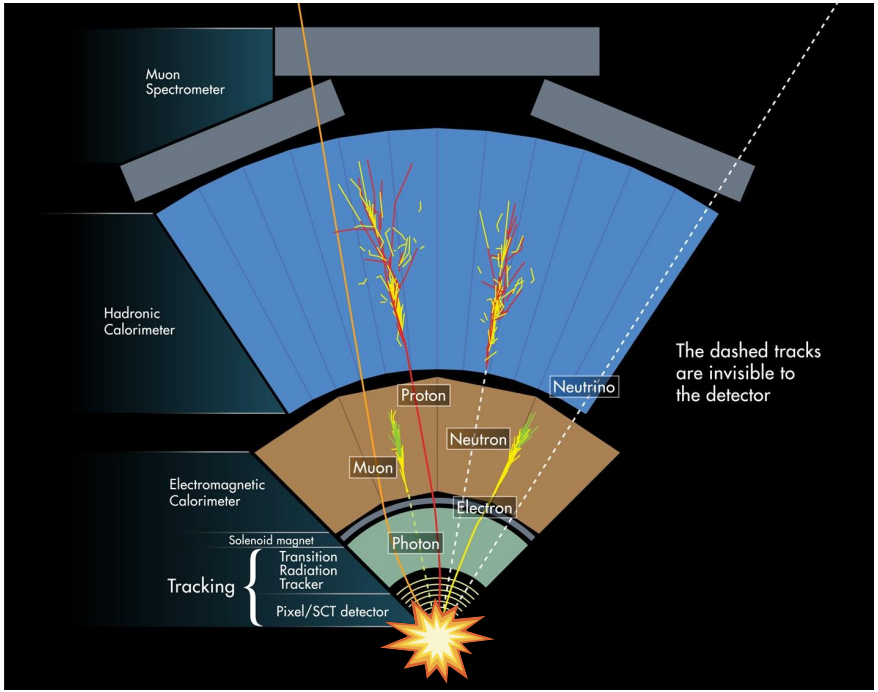
ATLAS

- Different **layers** detect different **particles** (needed for different final states!)



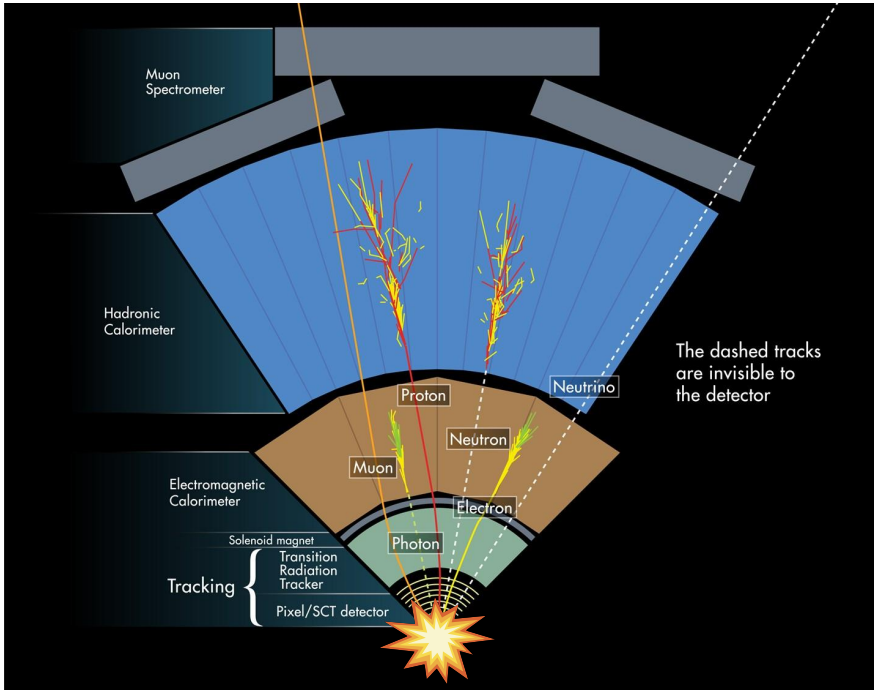
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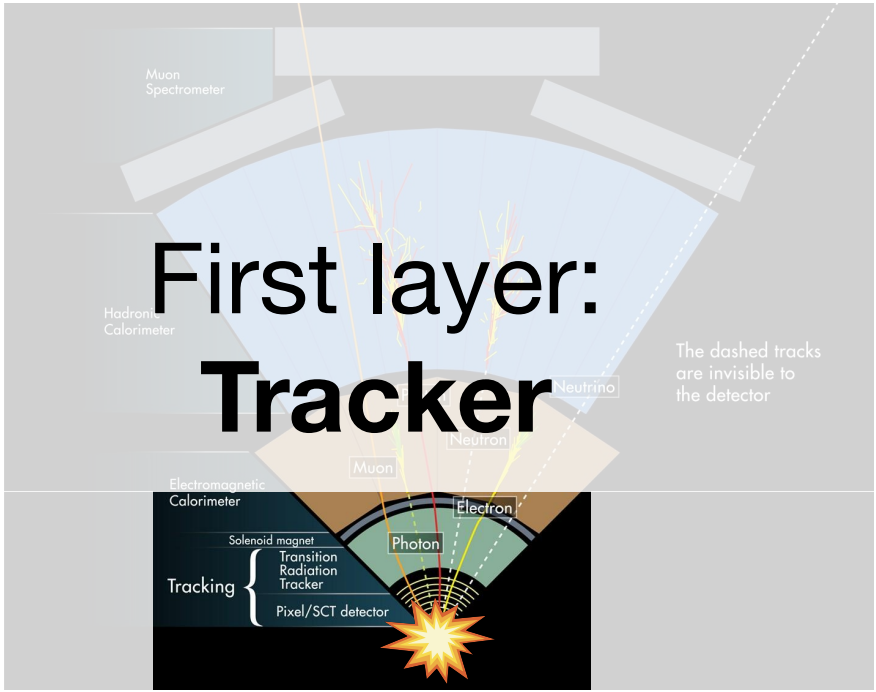
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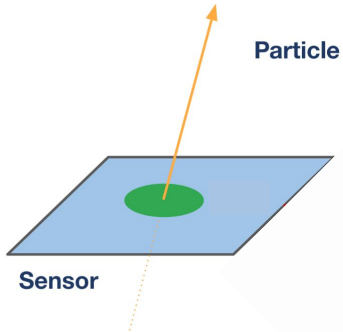
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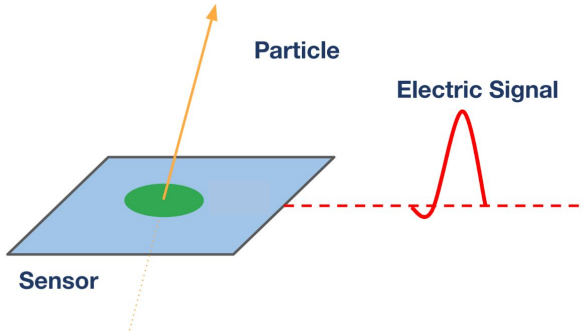
Silicon based tracking

- Particle hits silicon



Silicon based tracking

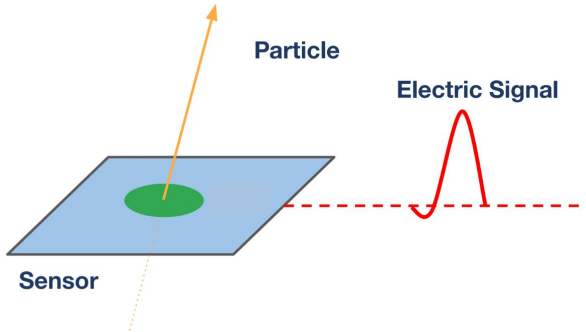
- Particle hits silicon
- Electric signal read out



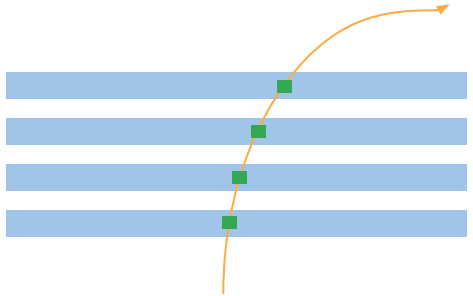
G. D'amen - BNL physics seminar series

Silicon based tracking

- Particle hits silicon
- Electric signal read out
- Series of hits forms a **track**
- Use radius to measure **particle momentum**

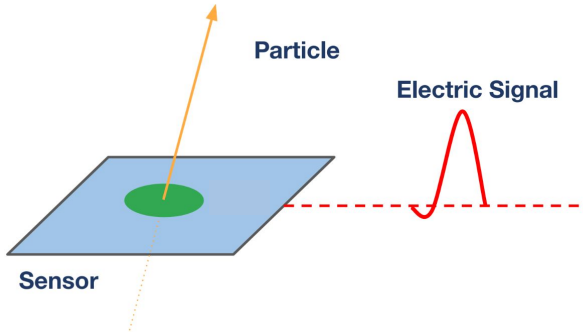


G. D'amen - BNL physics seminar series



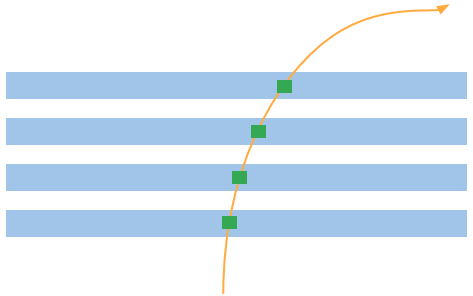
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G. D'amen - BNL physics seminar series

Silicon trackers currently used by ATLAS and CMS!



Identifying bb$\gamma\gamma$

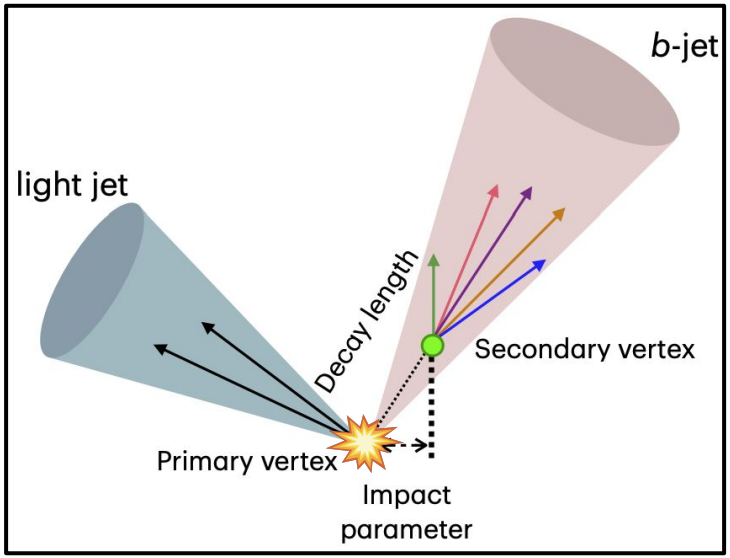
- **Bottom** quark: Relatively **heavy**

Identifying bb$\gamma\gamma$

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Identifying bb$\gamma\gamma$

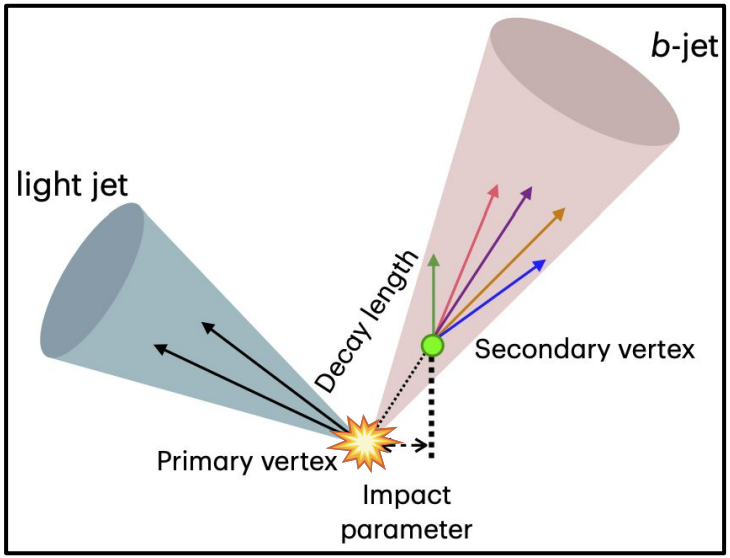
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- **Distinctly different** signature! Can use to differentiate from **“lighter”** jets



arXiv:2505.19689

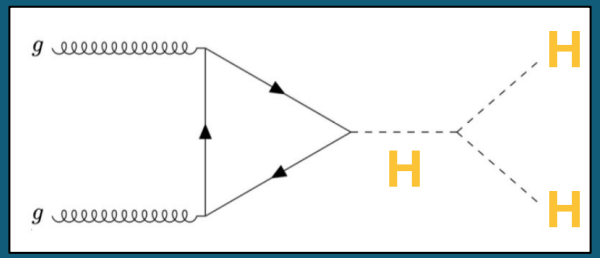
Identifying bb$\gamma\gamma$

- **Bottom** quark: Relatively **heavy**
- Propagates before decay: **b-jet**
- **Distinctly different** signature! Can use to differentiate from **“lighter”** jets
- For both **HH** and **X→SH**:
  - Use machine learning to identify **b-jets**
  - Identify **two photons** consistent with H→$\gamma\gamma$



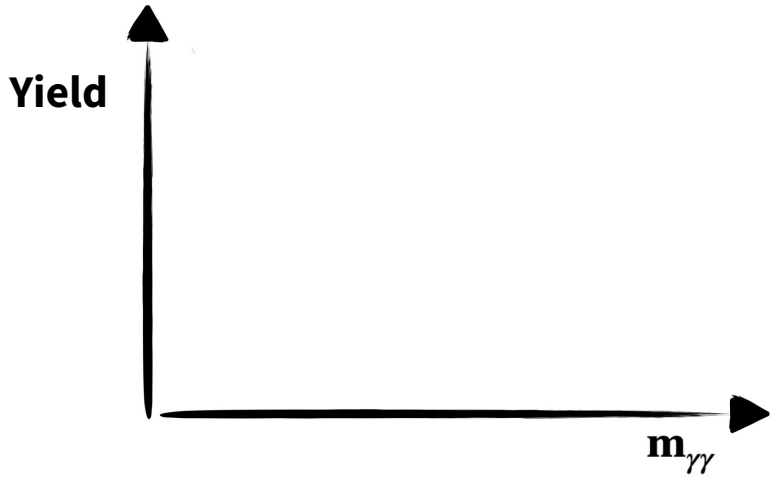
arXiv:2505.19689

Run 2 + Run 3 HH→bb$\gamma\gamma$



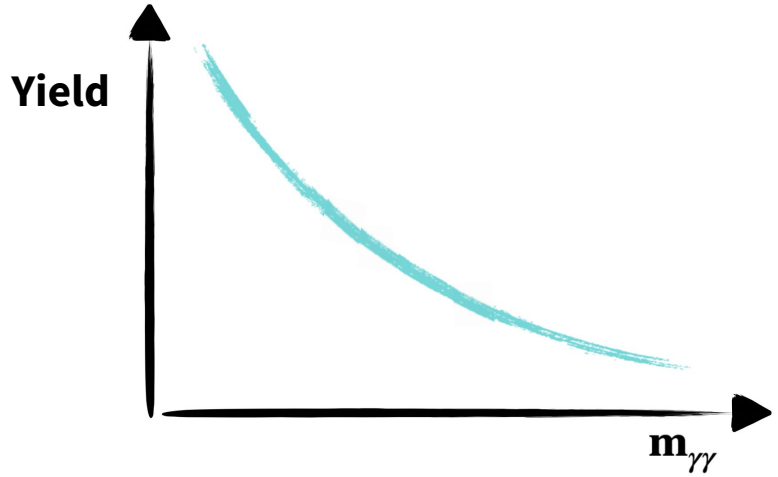
HH→bb$\gamma\gamma$: Strategy

- Take advantage of clean **di-photon mass** signature



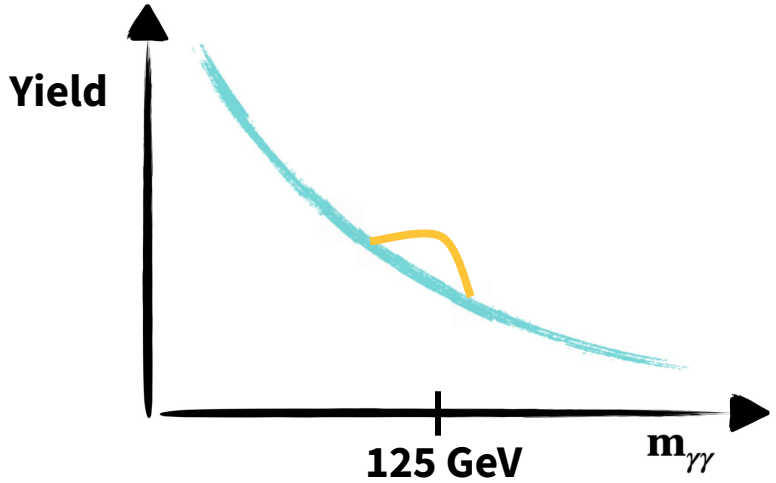
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- Three physics signatures:
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  - HH (Signal)

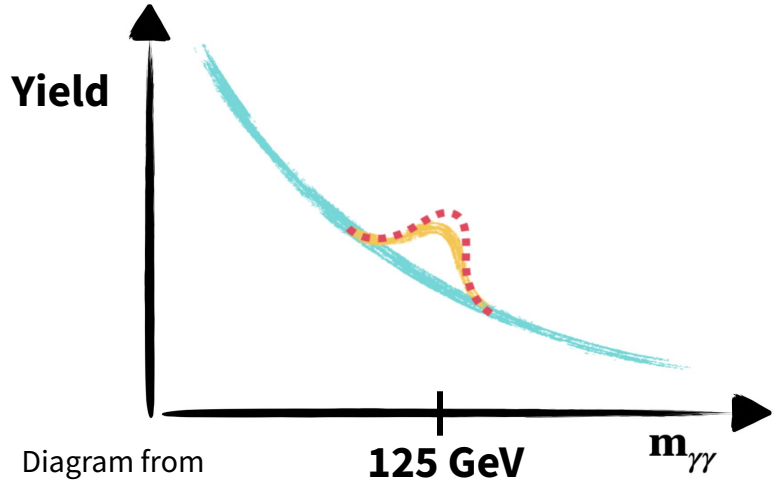


Diagram from Elena Mazzeo

HH→bb$\gamma\gamma$: Strategy

- Take advantage of clean **di-photon mass** signature
- Three physics signatures:
  - Continuum background ($\gamma\gamma$+jets, tt$\gamma\gamma$)
  - H (Resonant background)
  - HH (Signal)
- Need to **separate** single Higgs and continuum backgrounds from HH

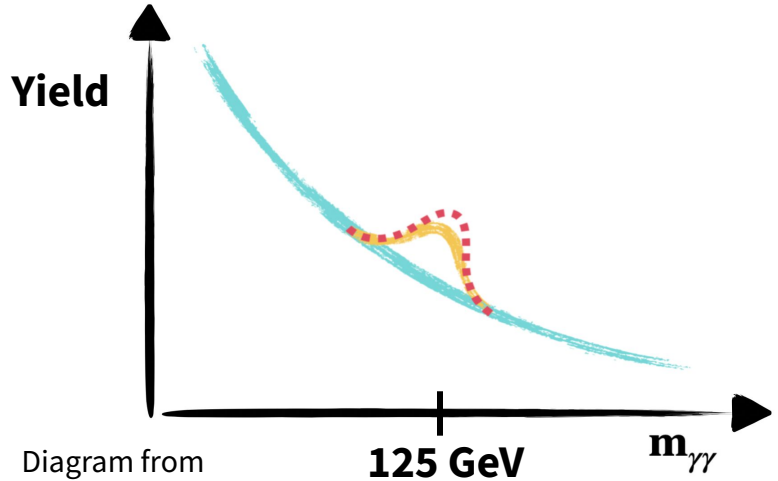


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  - H (Resonant background)
  - HH (Signal)
- Need to **separate** single Higgs and continuum backgrounds from HH
- HH and H modelled with **simulation**. Continuum background modeled with **data**

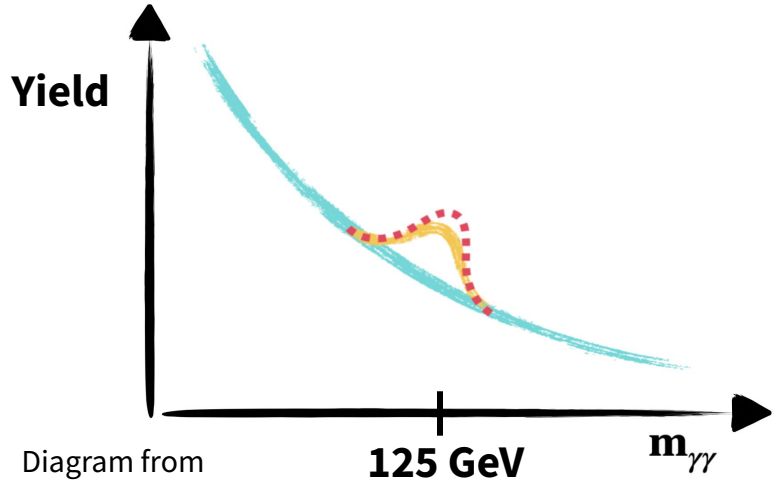


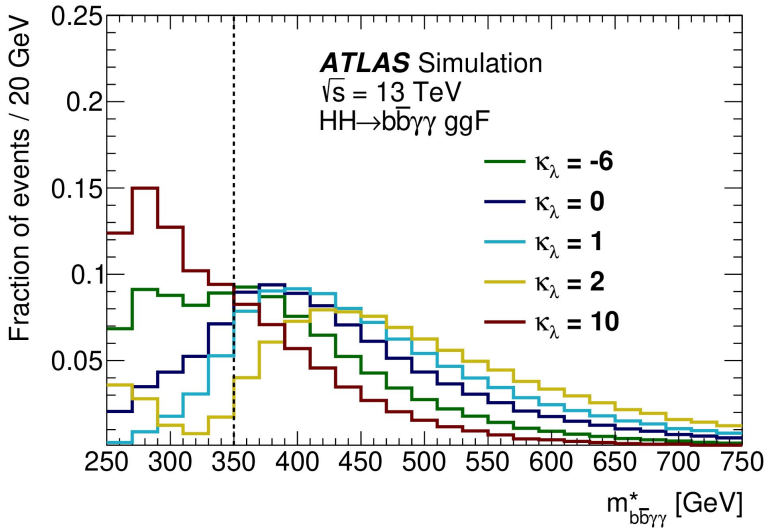
Diagram from Elena Mazzeo

HH→bb$\gamma\gamma$: Reduced mass

- Define **reduced mass**: ⟶ $m^{*}_{bb\gamma\gamma} = m_{bb\gamma\gamma} - (m_{bb} - 125\ \text{GeV}) - (m_{\gamma\gamma} - 125\ \text{GeV})$

HH→bbγγ: Reduced mass

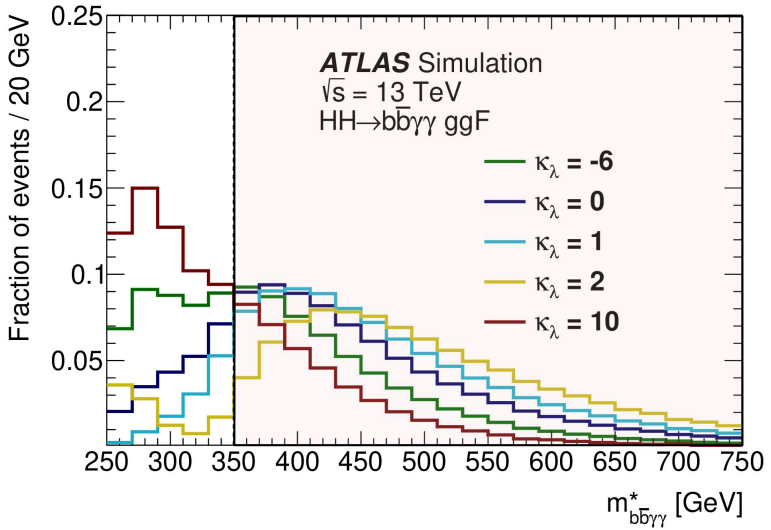
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[Phys. Rev. D 106, 052001]

HH→bbγγ: Reduced mass

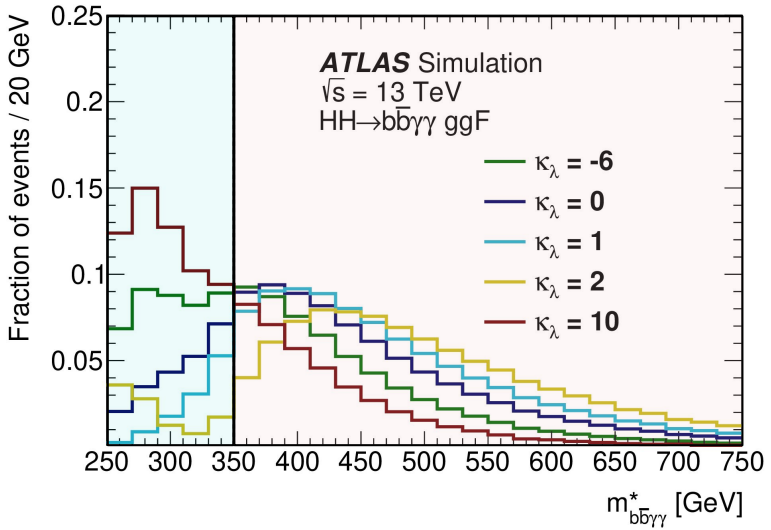
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- Split analysis into 2 regions:
  - **High mass:** > 350 GeV: Targets SM HH



[Phys. Rev. D 106, 052001]

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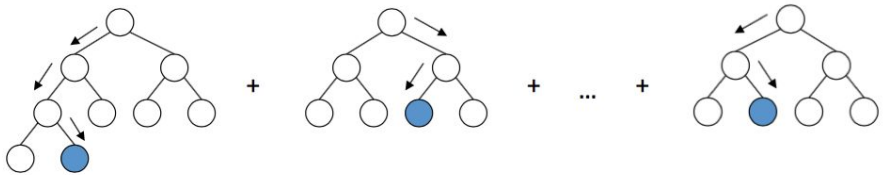
- Define **reduced mass**: → $m^*_{bb\gamma\gamma} = m_{bb\gamma\gamma} - (m_{bb} - 125\ \text{GeV}) - (m_{\gamma\gamma} - 125\ \text{GeV})$
- Split analysis into 2 regions:
  - **High mass:** > 350 GeV: Targets SM HH
  - **Low mass:** < 350 GeV: Targets deviations from self-coupling



[Phys. Rev. D 106, 052001]

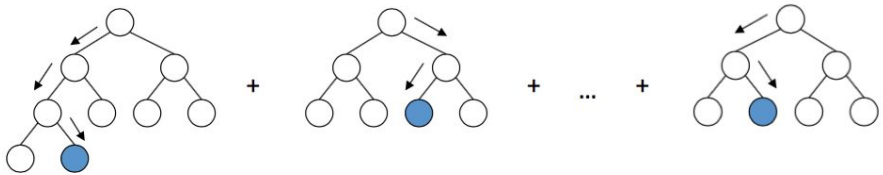
HH→bb$\gamma\gamma$: BDT

- Train **boosted decision tree** to separate **signal** and **background** signatures



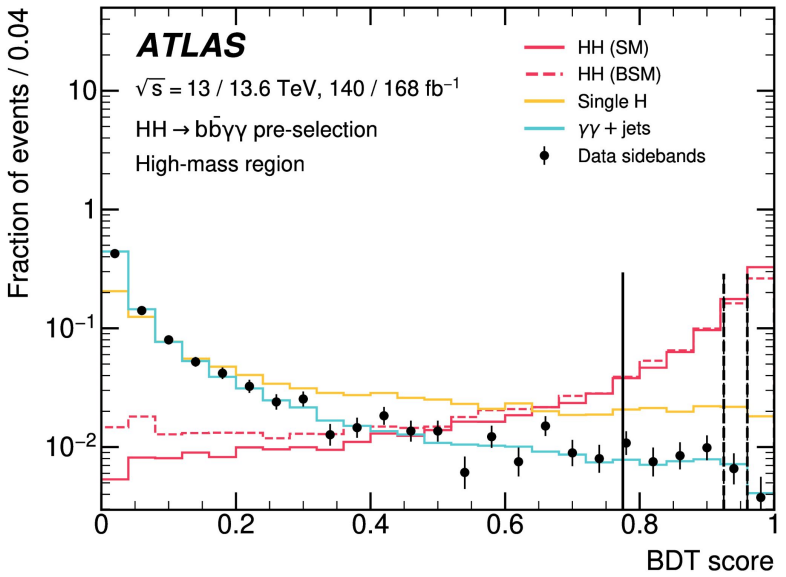
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- Use photon, jet kinematics as main inputs.



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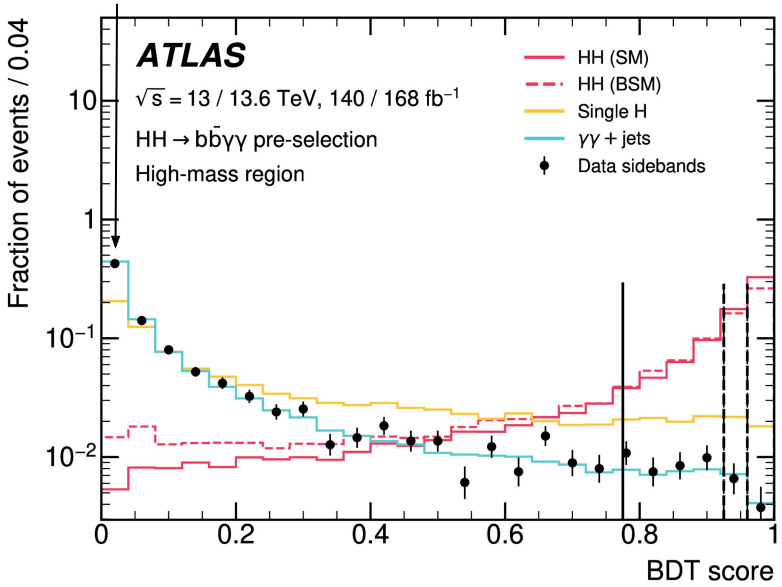


BDT score in high mass region, **data sideband**

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Continuum background, single higgs



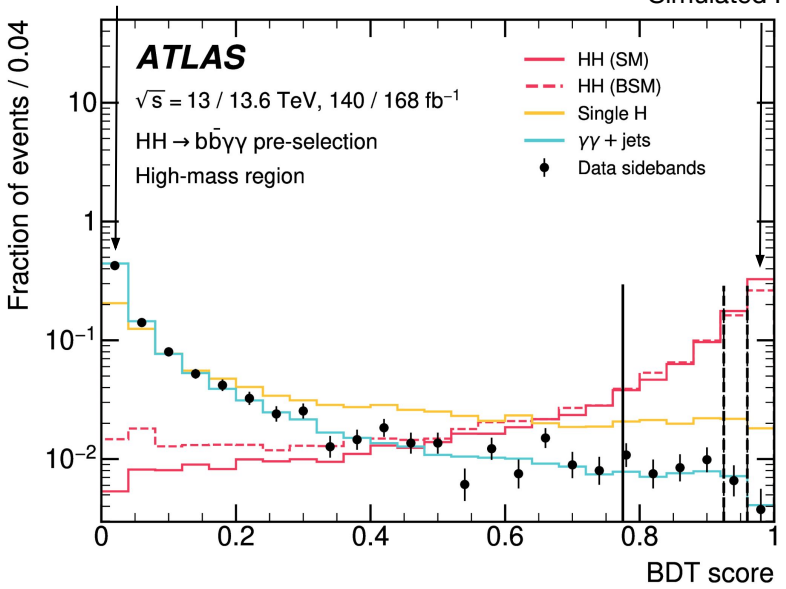
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Simulated HH signal

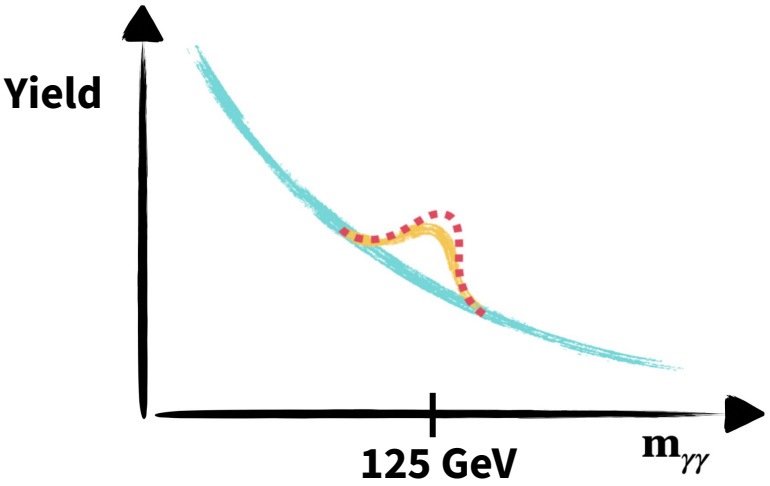


BDT score in high mass region, **data sideband**

HH→bbγγ: Di-Photon mass

Brookhaven National Laboratory

Model HH, H with **double sided crystal ball**
Model continuum background with **data**

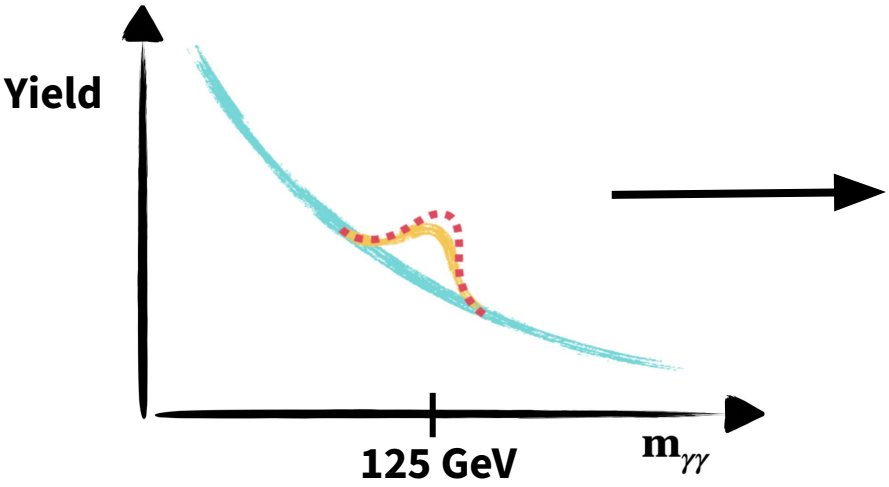


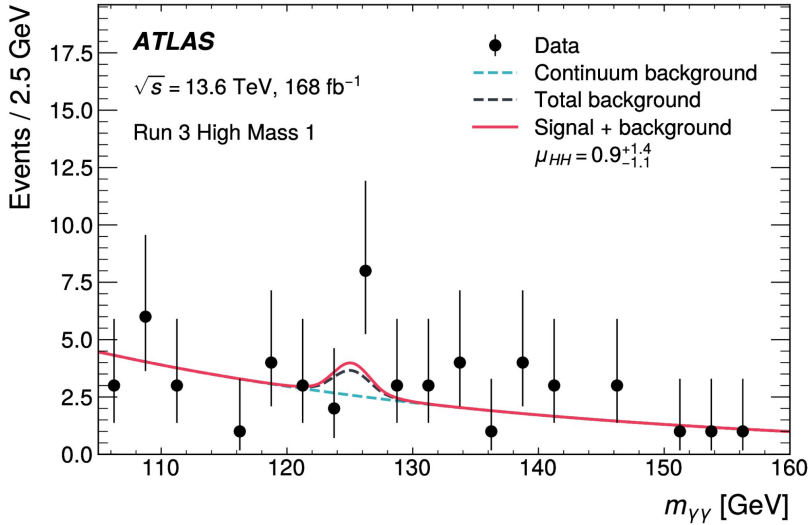
Strategy

HH→bbγγ: Di-Photon mass

Brookhaven National Laboratory

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Strategy

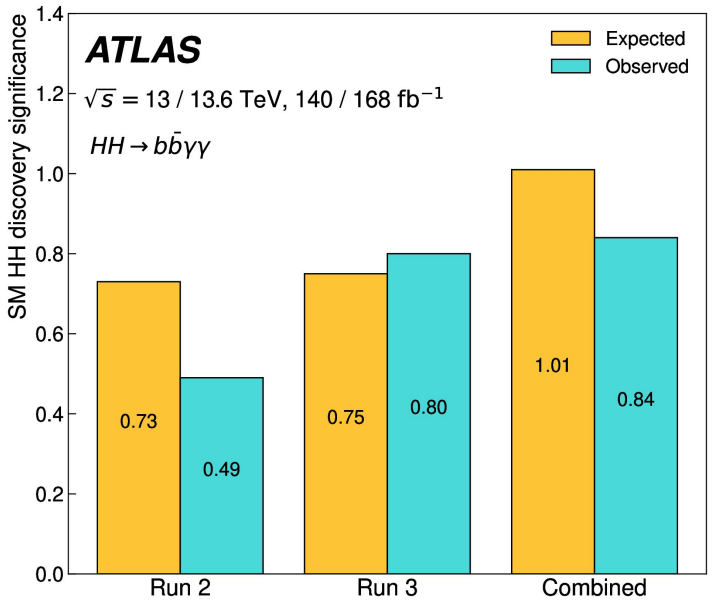
Applied to real data!

HH→bb$\gamma\gamma$: Results

- Observed significance of **0.84σ (1.01σ expected)**

HH→bb$\gamma\gamma$: Results

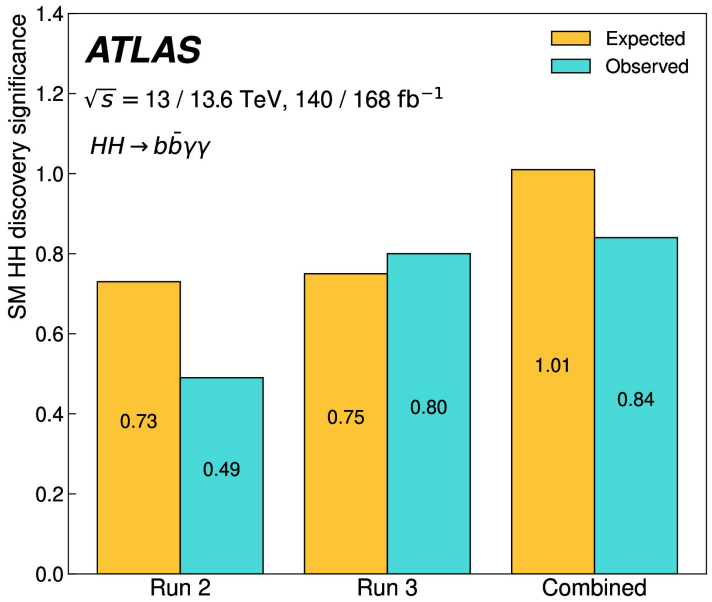
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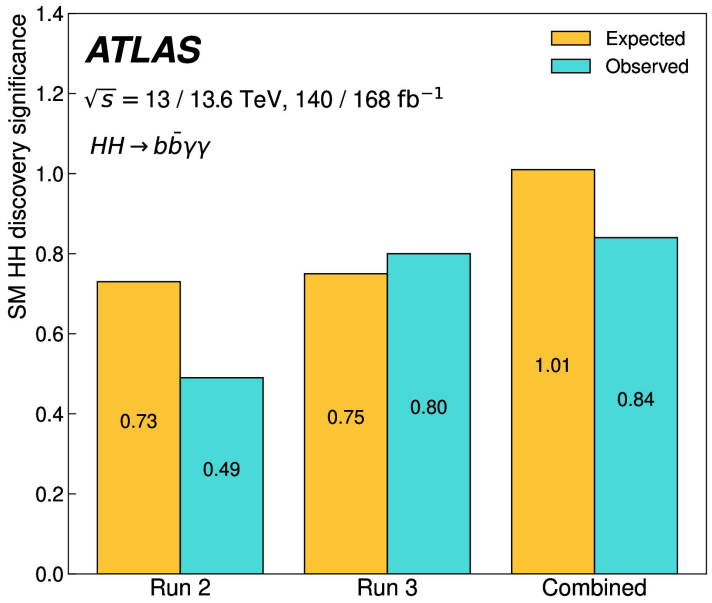


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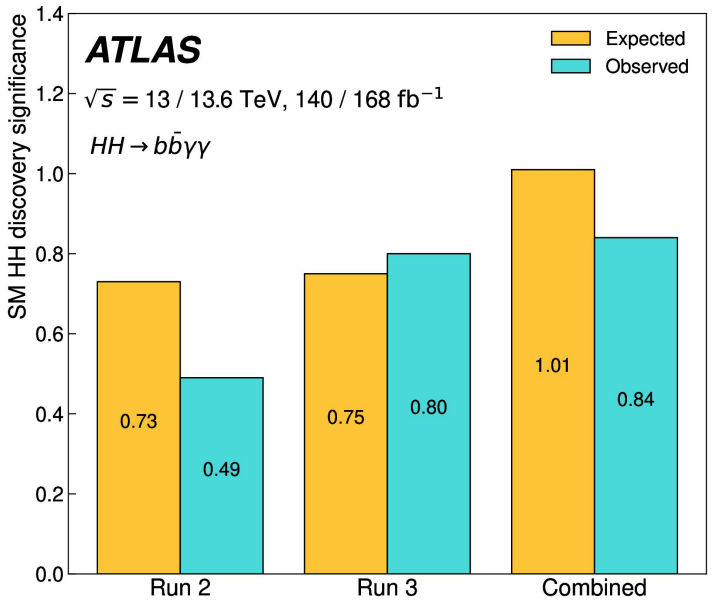
- Constrain self-coupling to ~ **[-2, 7] X SM** at a 95% CL
  - Comparable to **Run 2 ATLAS+CMS combination!** [ATLAS-CONF-2025-012]



Discovery significance results

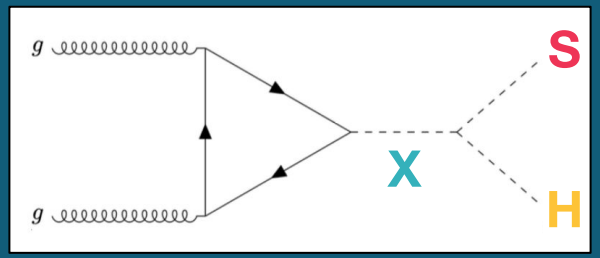
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- Constrain self-coupling to ~ **[-2, 7] X SM** at a 95% CL
  - Comparable to **Run 2 ATLAS+CMS combination!** [ATLAS-CONF-2025-012]
- **Most sensitive single channel HH analysis to date.**



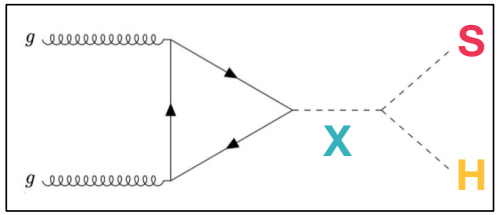
Discovery significance results

Run 2 + Run 3 $X \rightarrow SH \rightarrow bb\gamma\gamma$



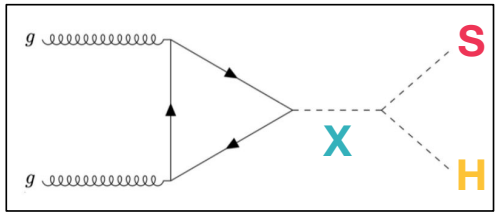
X→SH→bb$\gamma\gamma$: Introduction

- Looking for the **X→SH** process



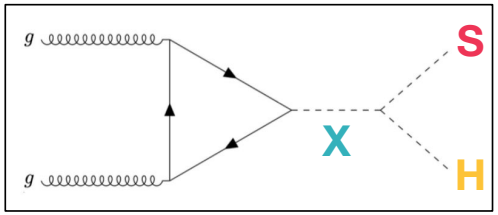
$X \rightarrow SH \rightarrow bb\gamma\gamma$: Introduction

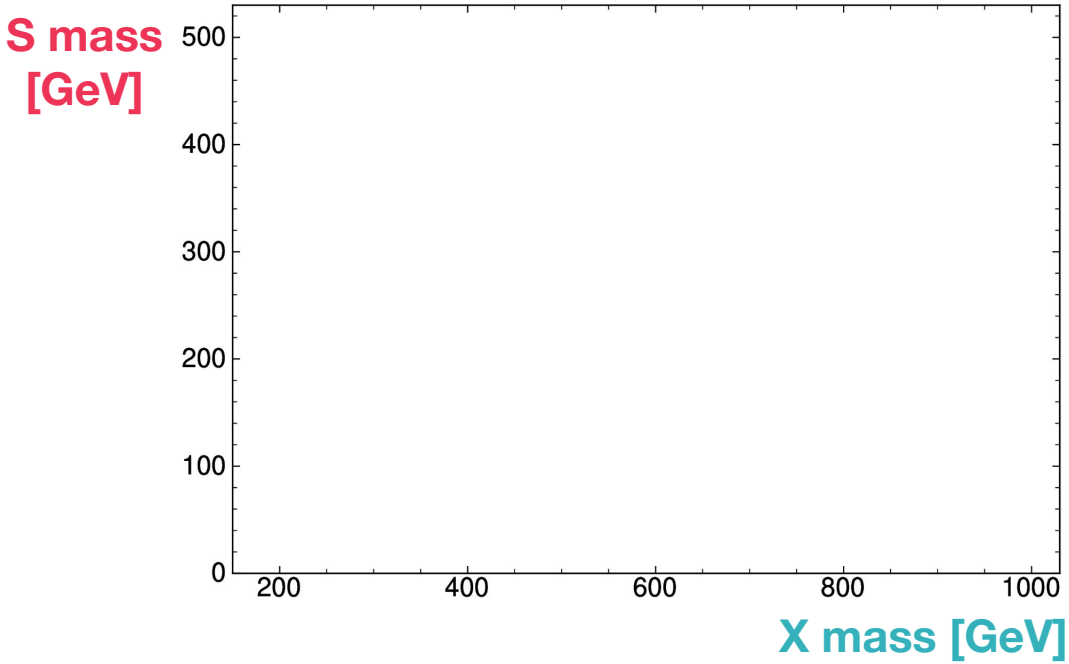
- Looking for the **X→SH** process
  - Don't know exact **mass** of X, S particles, so search through a **range of masses**



X→SH→bb$\gamma\gamma$: Introduction

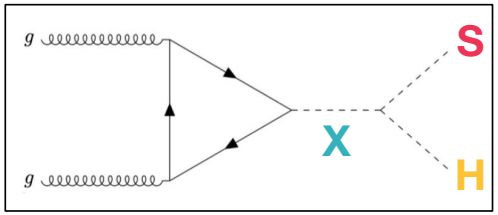
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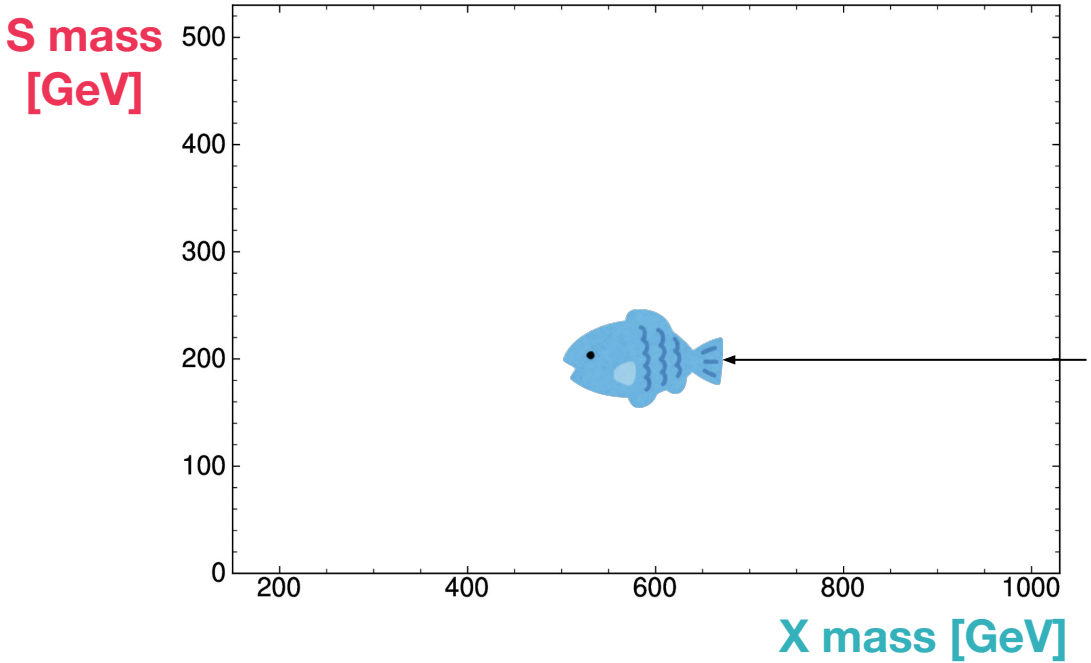




X→SH→bbγγ: Introduction

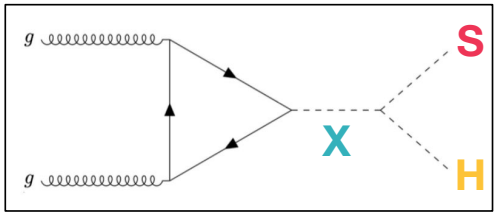
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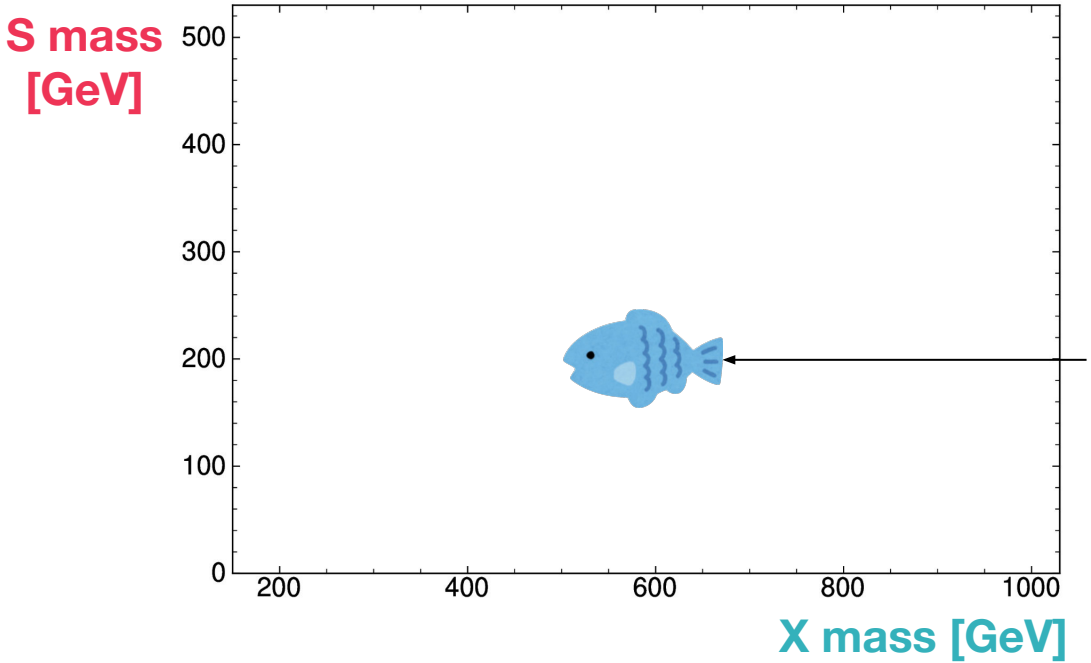




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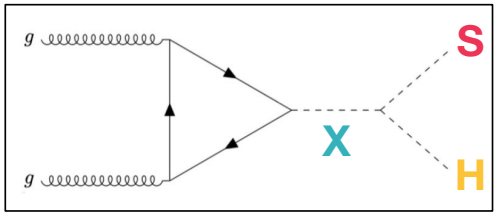


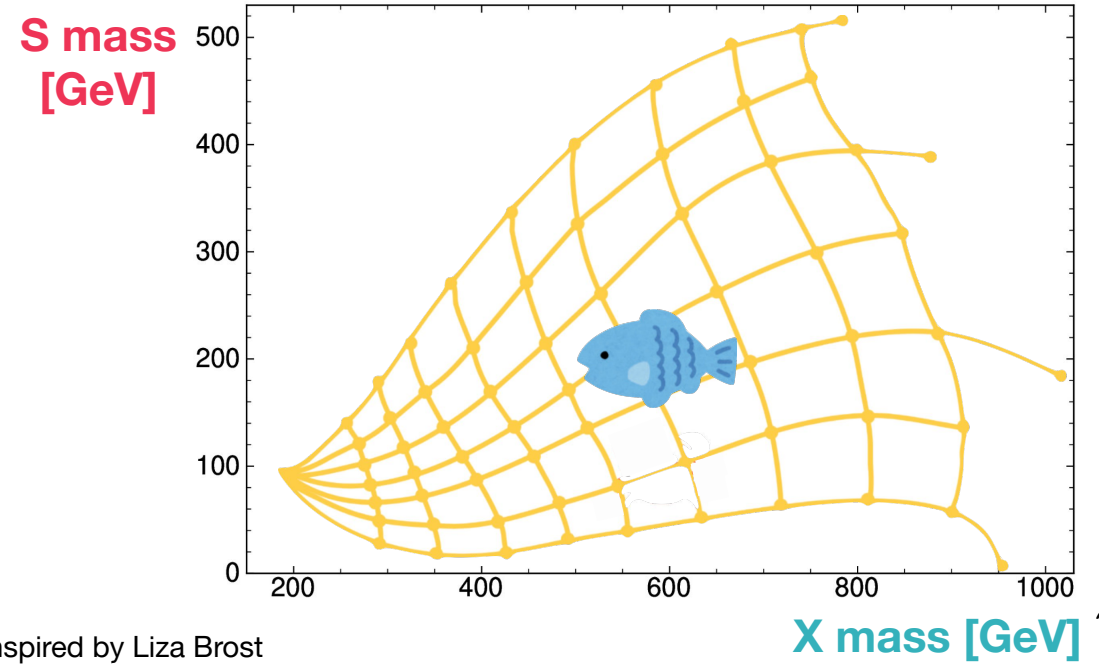
Example:
X mass = 600 GeV
S mass = 200 GeV

Fish = new particles

X→SH→bbγγ: Introduction

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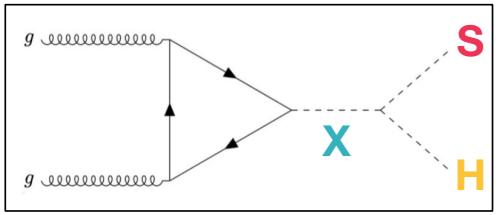


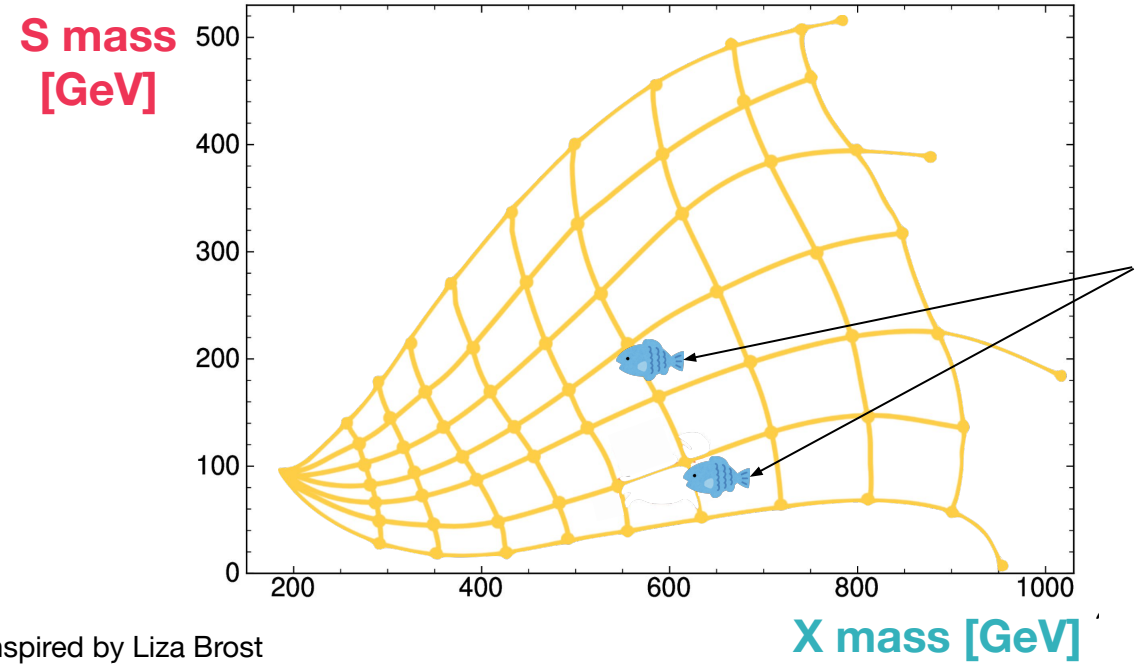
If we "**cast a wide net**", we search for more mass scenarios

Art inspired by Liza Brost

X→SH→bbγγ: Introduction

- Looking for the **X→SH** process
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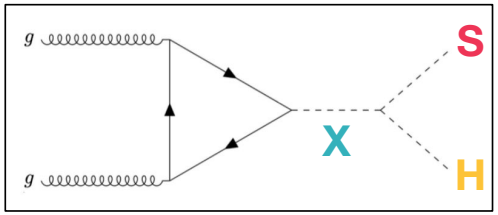


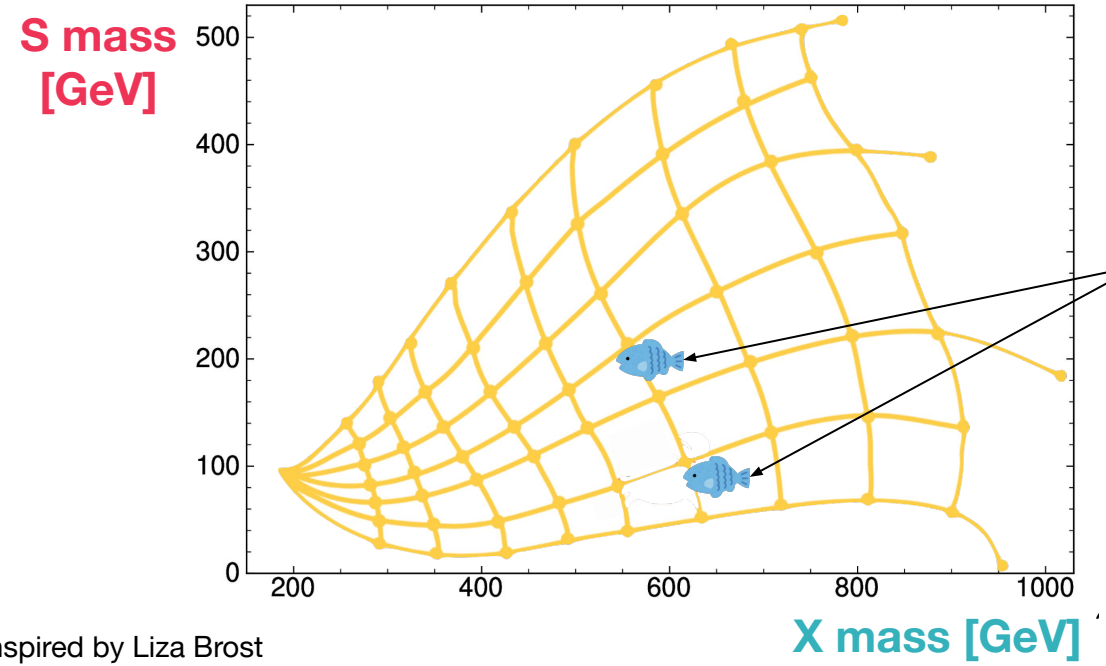
Previous ATLAS and CMS analyses saw **hints of signal** at two mass points

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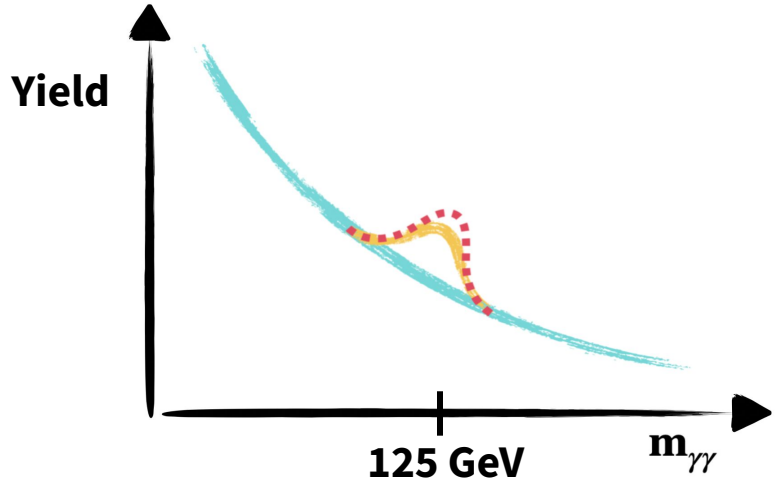


Previous ATLAS and CMS analyses saw **hints of signal** at two mass points
→ Motivation for updated analysis!

Art inspired by Liza Brost

$X \rightarrow SH \rightarrow bb\gamma\gamma$: Signature

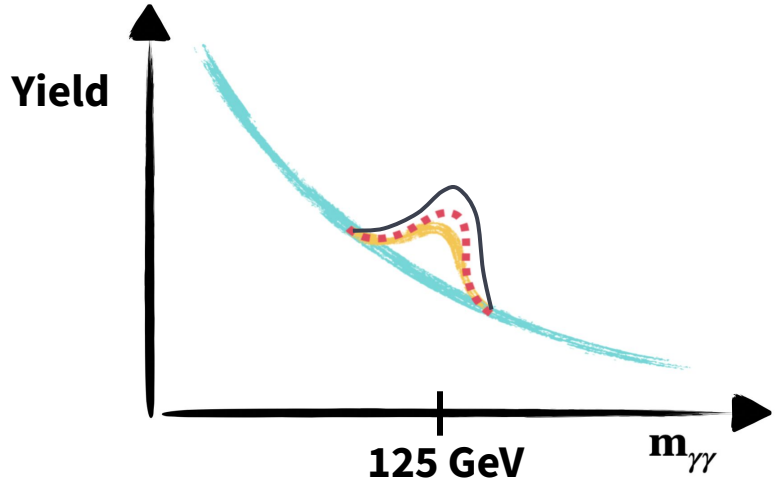
- **Similar signature to HH**: Expect peak at 125 GeV
- Physics signatures:
  - Continuum background ($\gamma\gamma$+jets, tt$\gamma\gamma$)
  - H (Resonant background)
  - HH (Resonant background)



$X \rightarrow SH \rightarrow bb\gamma\gamma$: Signature

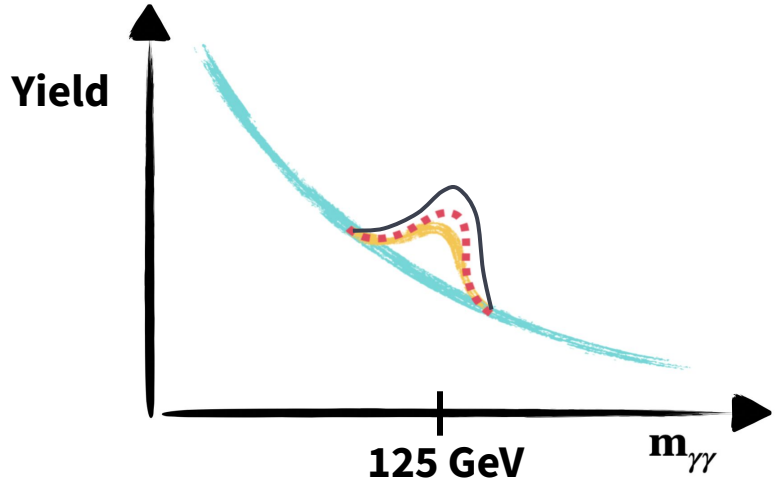
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  - H (Resonant background)
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  - X→SH: (Signal)



X→SH→bb$\gamma\gamma$: Signature

- **Similar signature to HH**: Expect peak at 125 GeV
- Physics signatures:
  - Continuum background ($\gamma\gamma$+jets, tt$\gamma\gamma$)
  - H (Resonant background)
  - HH (Resonant background)
  - X→SH: (Signal)
- Now treat HH as an additional **background**!



$X \rightarrow SH \rightarrow bb\gamma\gamma$: PNN

- Use a **Parameterized Neural Network (PNN)** to separate signal and background.

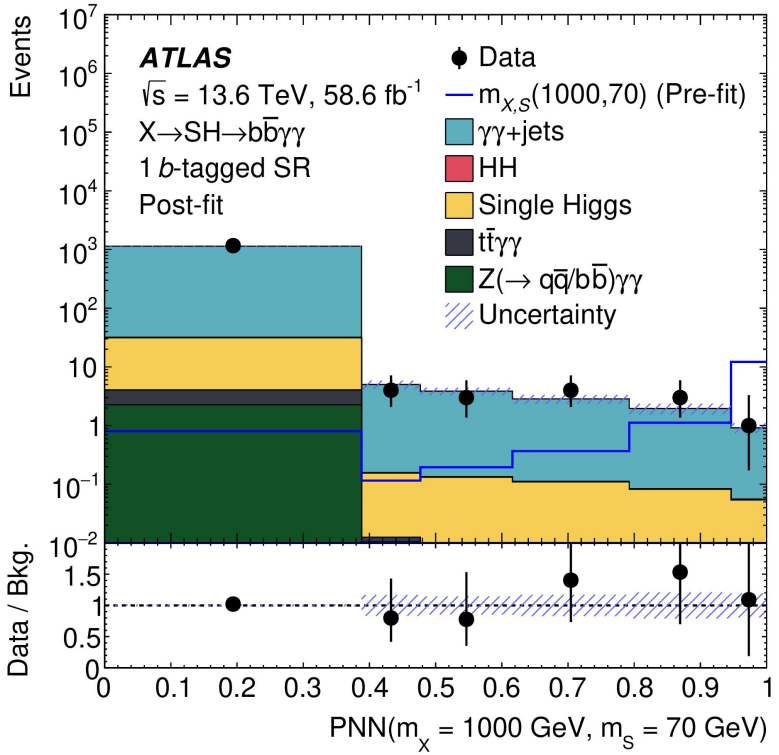
$X \rightarrow SH \rightarrow bb\gamma\gamma$: PNN

- Use a **Parameterized Neural Network (PNN)** to separate signal and background.

- **360 signal mass points**: With single PNN, parameterize on X, S mass.

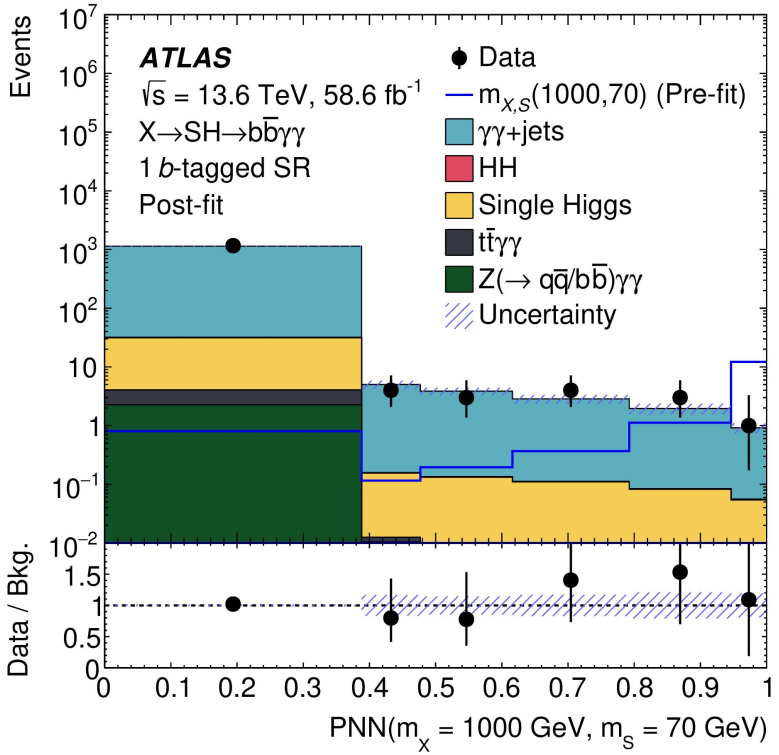
X→SH→bbγγ: PNN

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X→SH→bbγγ: PNN

- Use a **Parameterized Neural Network (PNN)** to separate signal and background.
- **360 signal mass points**: With single PNN, parameterize on X, S mass.
- The PNN does a good job of separating signal and background

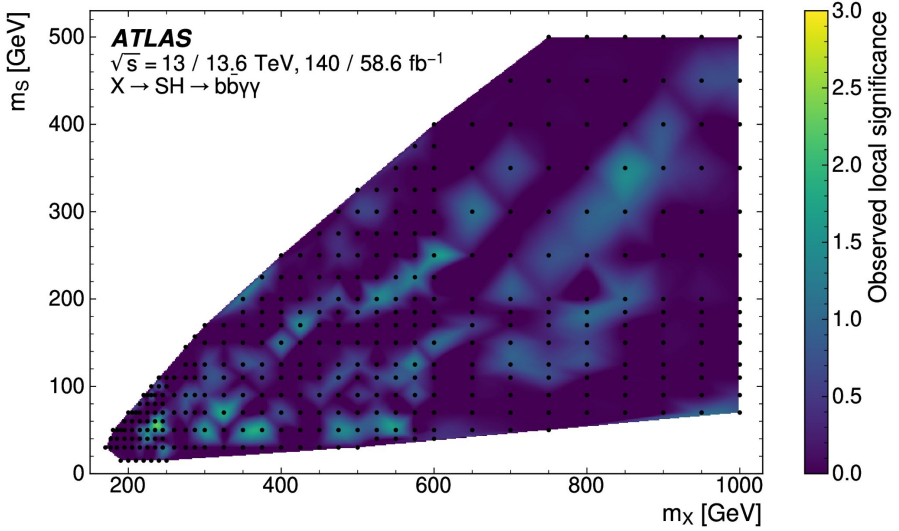


X→SH→bbγγ: Results

Brookhaven National Laboratory

arXiv:2510.02857

Run 2 + Run 3 observed significance

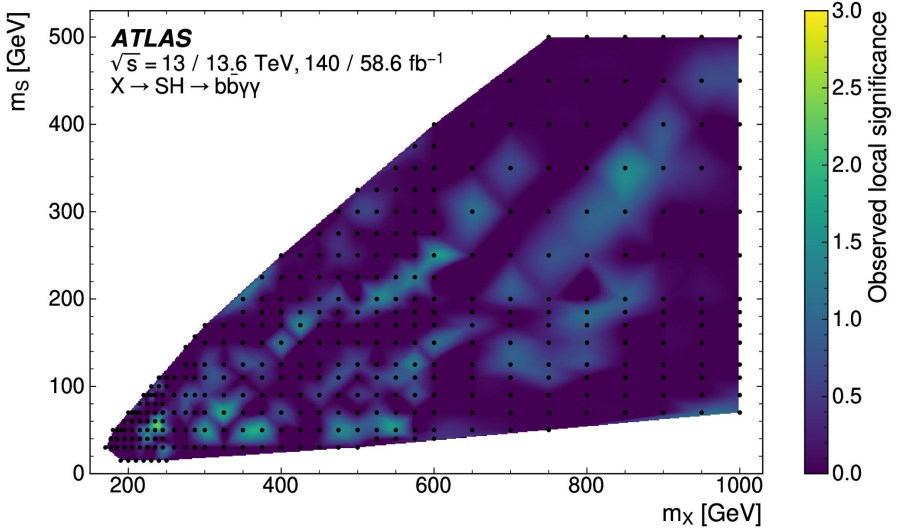


X→SH→bbγγ: Results

Brookhaven National Laboratory

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Run 2 + Run 3 observed significance



= No signal

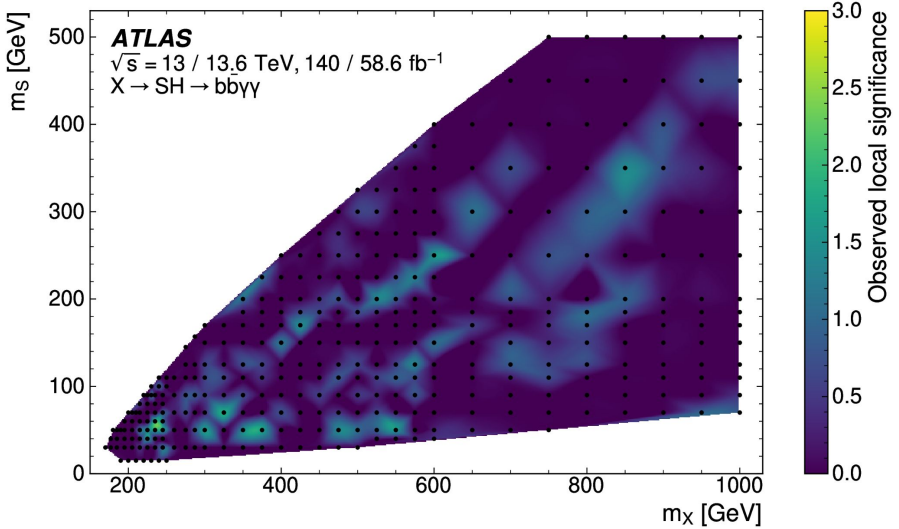
= Very slight signal

X→SH→bb$\gamma\gamma$: Results

- **No significant excess observed:**
  - Maximum **local** significance: **2σ**
  - Global significance ~ **0σ** everywhere

arXiv:2510.02857

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= No signal

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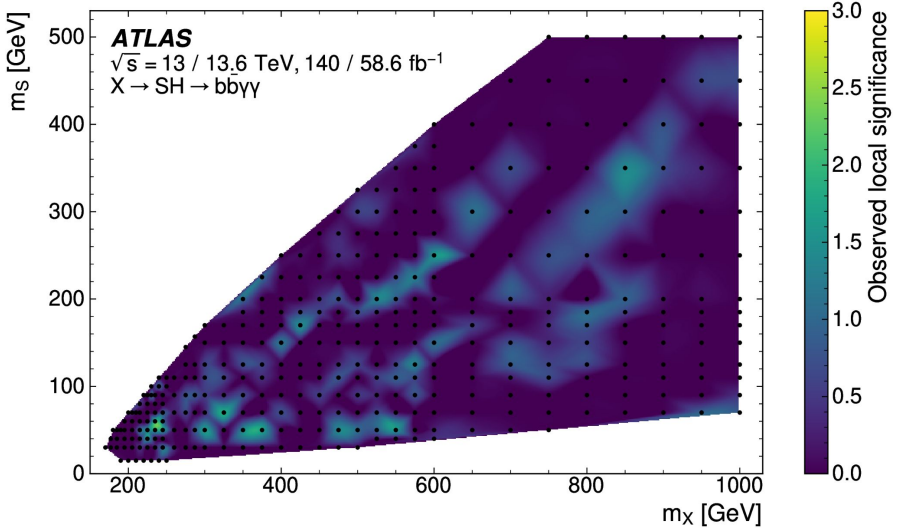
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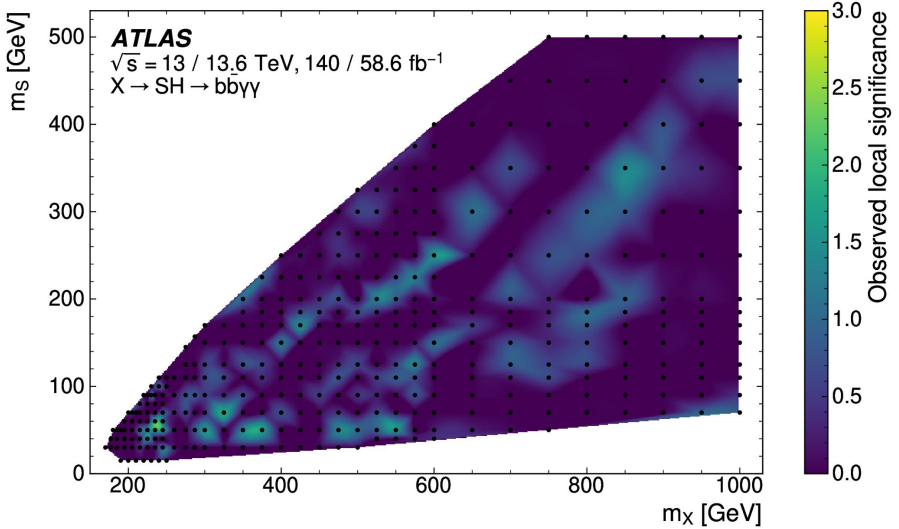
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X→SH→bb$\gamma\gamma$: Results

- **<u>No significant excess observed:</u>**
  - Maximum **local** significance: **2σ**
  - Global significance ~ **0σ** everywhere
- **Do not confirm previous excesses from ATLAS and CMS**
- Improve sensitivity by up to **73%**

arXiv:2510.02857

Run 2 + Run 3 observed significance



= No signal

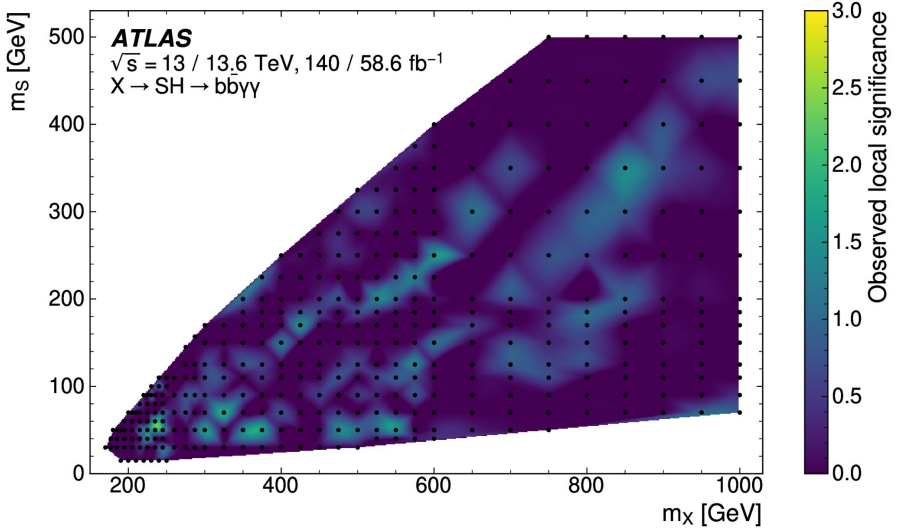
= Very slight signal

X→SH→bbγγ: Results

- **No significant excess observed:**
  - Maximum **local** significance: **2σ**
  - Global significance ~ **0σ** everywhere
- **Do not confirm previous excesses from ATLAS and CMS**
- Improve sensitivity by up to **73%**
- **First Run 3 search** for **BSM particles** coupling to the **Higgs**!

arXiv:2510.02857

Run 2 + Run 3 observed significance



= No signal

= Very slight signal

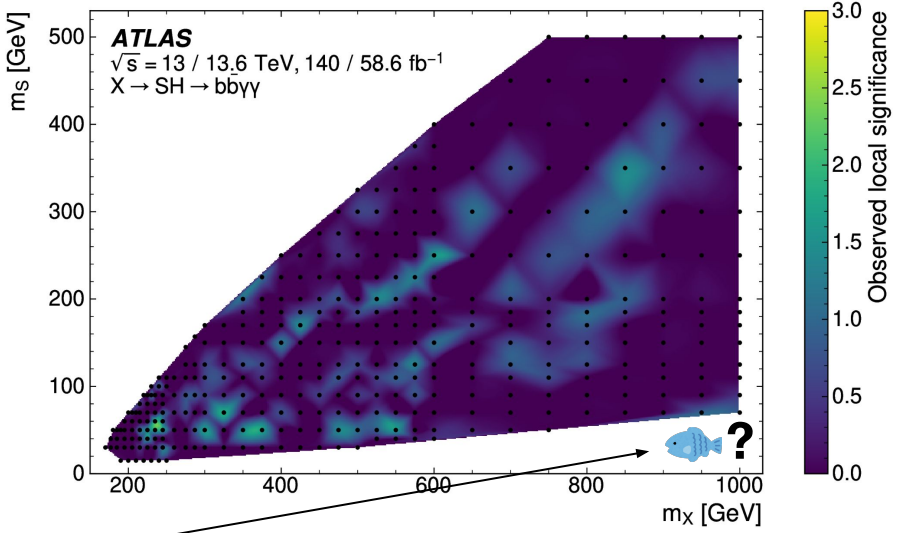
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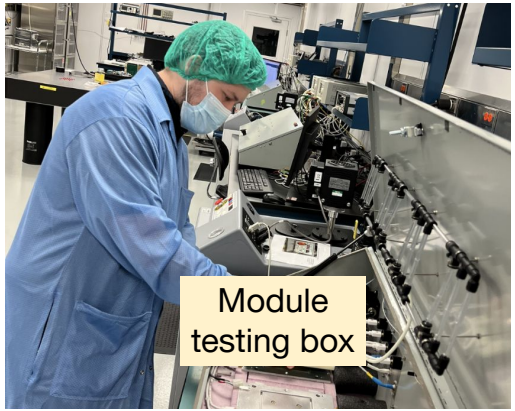
Cast wider net in future analysis?

arXiv:2510.02857

Run 2 + Run 3 observed significance



Next section



LHC

I

Higgs discovery

2009-2026

HL-LHC

II

di-Higgs discovery

2030-2041

Future

III

Future discoveries

2041-...

HL-LHC

- ~2030: Finish upgrade to **High-Luminosity LHC**

HL-LHC

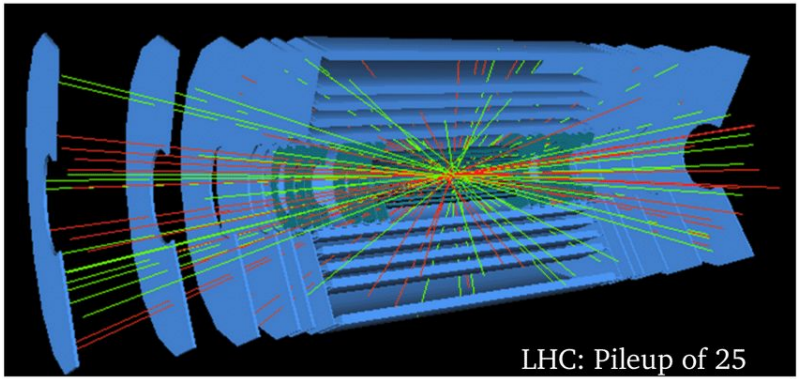
Brookhaven National Laboratory

- ~2030: Finish upgrade to **High-Luminosity LHC**

- **Pro:** Increase datasets by factor of **~10**

HL-LHC

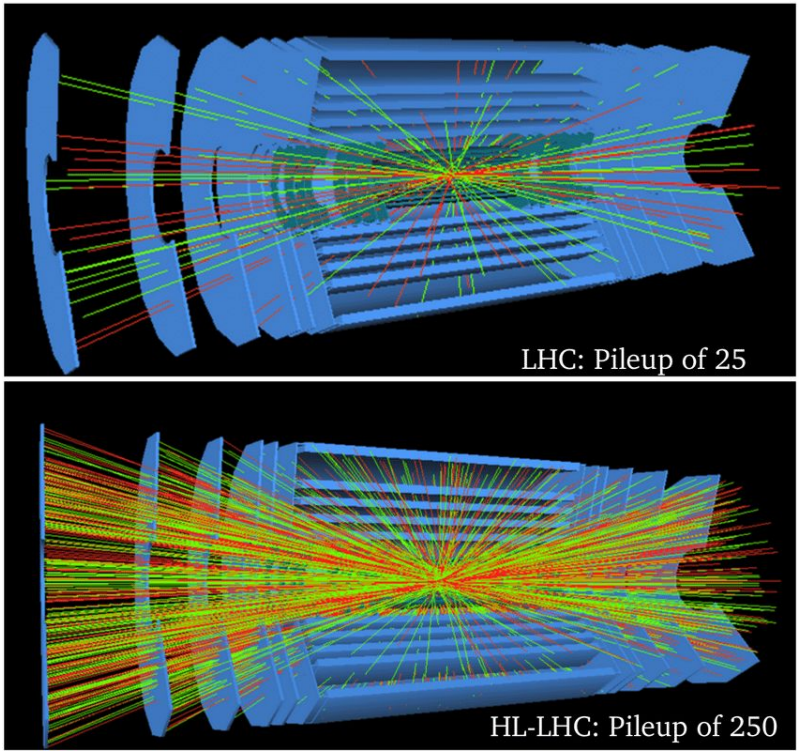
- ~2030: Finish upgrade to **High-Luminosity LHC**
- **Pro:** Increase datasets by factor of **~10**
- **Con:** Extremely challenging data-taking environment



Development and Evaluation of Novel, Large Area, Radiation Hard Silicon Microstrip Sensors for the ATLAS ITk Experiment at the HL-LHC

HL-LHC

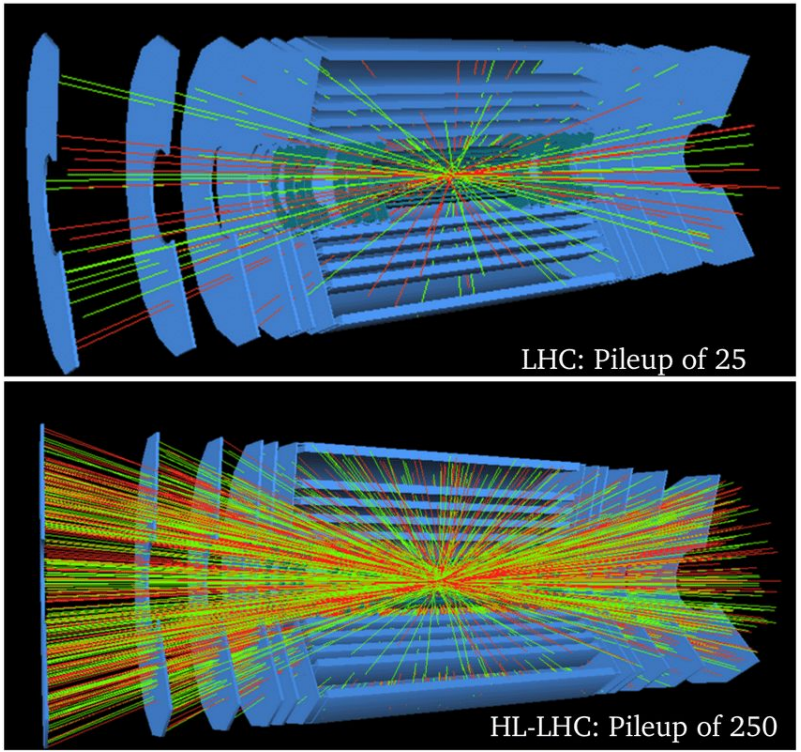
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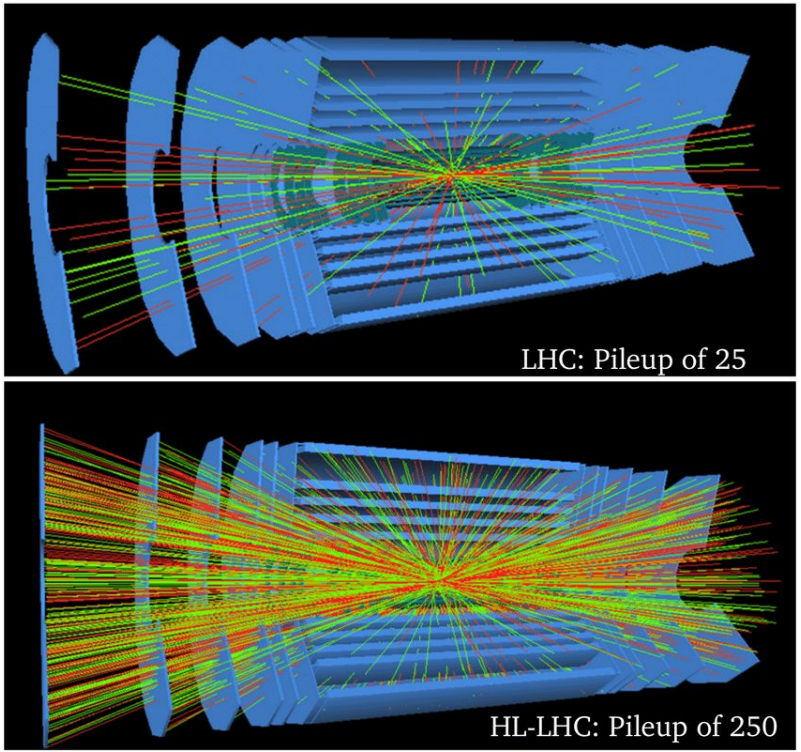
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- **ATLAS cannot cope with this!** How will we enable discovery at HL-LHC?



Development and Evaluation of Novel, Large Area, Radiation Hard Silicon Microstrip Sensors for the ATLAS ITk Experiment at the HL-LHC

HL-LHC

- ~2030: Finish upgrade to **High-Luminosity LHC**
- **Pro:** Increase datasets by factor of **~10**
- **Con:** Extremely challenging data-taking environment
- **ATLAS cannot cope with this!** How will we enable discovery at HL-LHC?
  - **Extensive detector upgrades needed**



Development and Evaluation of Novel, Large Area, Radiation Hard Silicon Microstrip Sensors for the ATLAS ITk Experiment at the HL-LHC

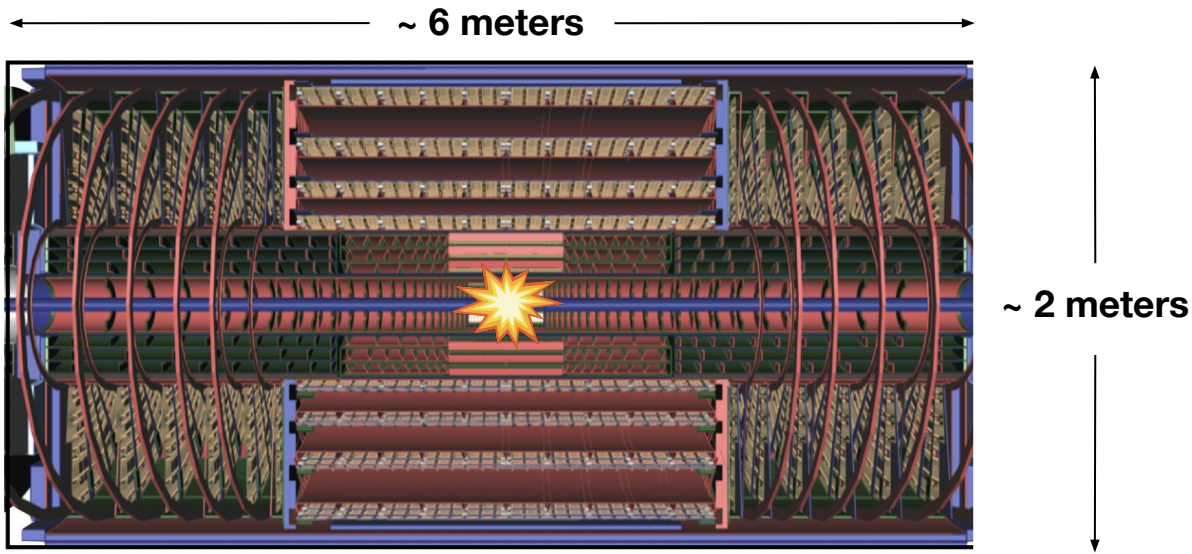
The ATLAS ITk

Brookhaven National Laboratory

- To continue excellent **b-tagging** in this environment, need a performant **tracker**

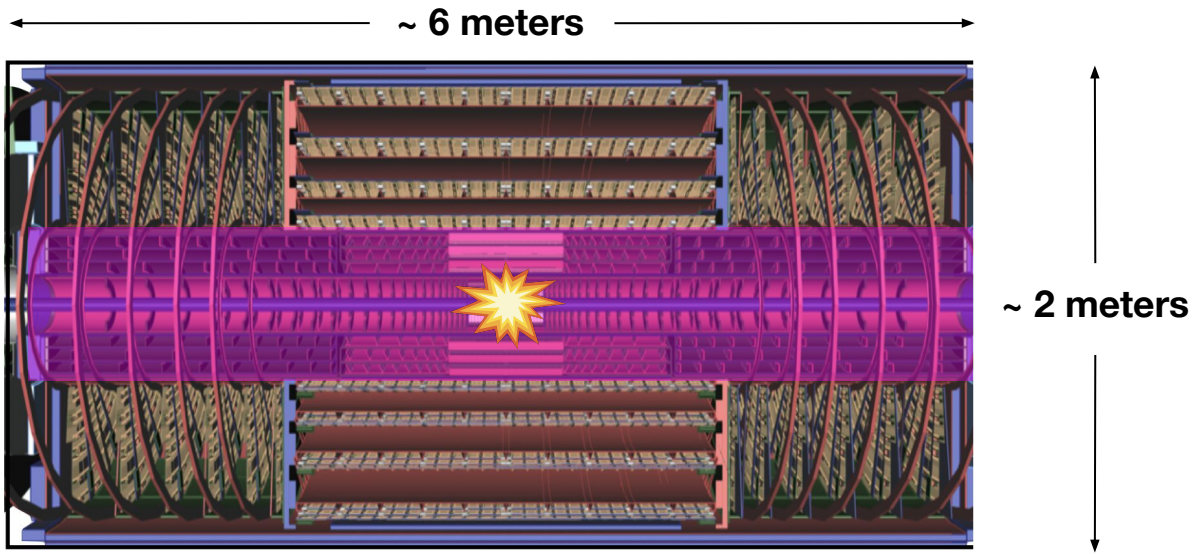
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The ATLAS ITk

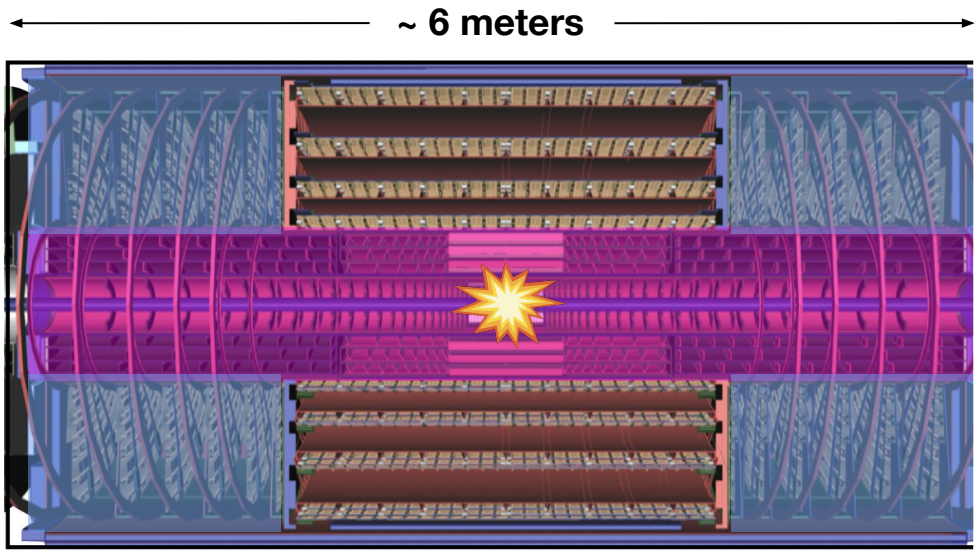
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ITk pixels

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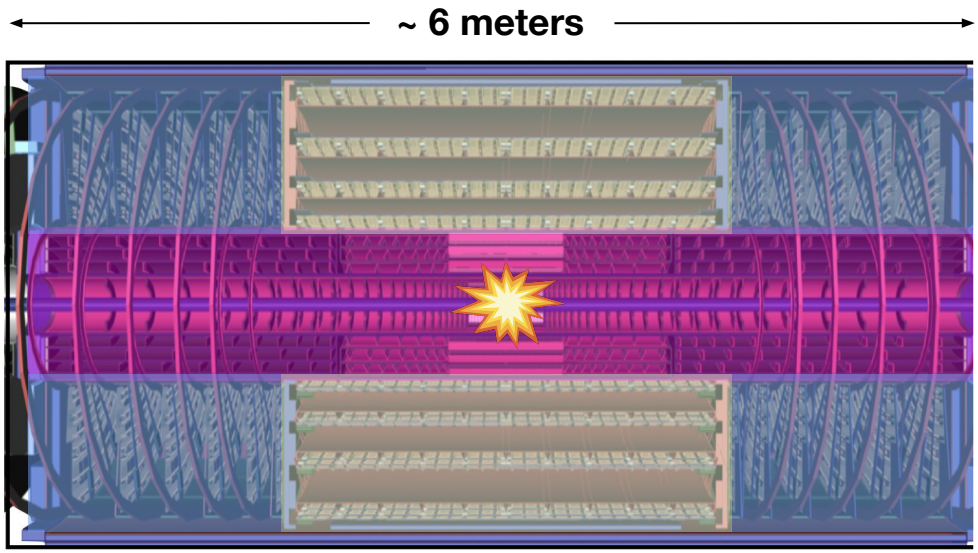


ITk pixels

ITk strips endcaps

The ATLAS ITk

- To continue excellent **b-tagging** in this environment, need a performant **tracker**
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ITk pixels

ITk strips endcaps

ITk strips barrel

Modules

- Individual detection unit of the ITk strips: a “**Module**”

Sensor



Birds-eye view

Modules

- Individual detection unit of the ITk strips: a “**Module**”

Sensor



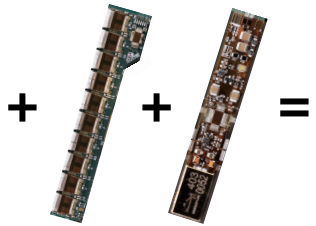
Birds-eye view

Modules

- Individual detection unit of the ITk strips: a “**Module**”

Sensor

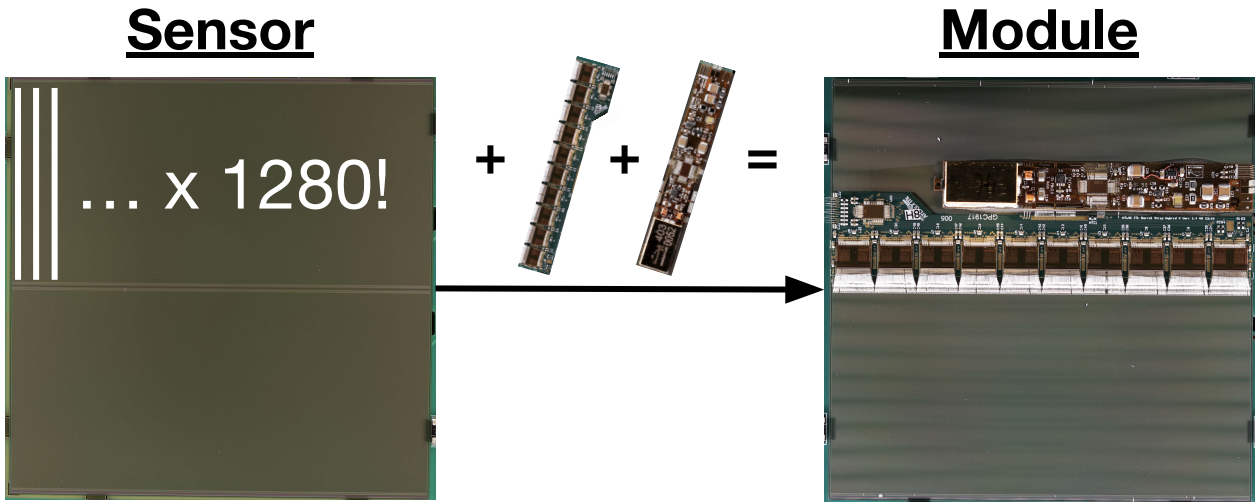




Birds-eye view

Modules

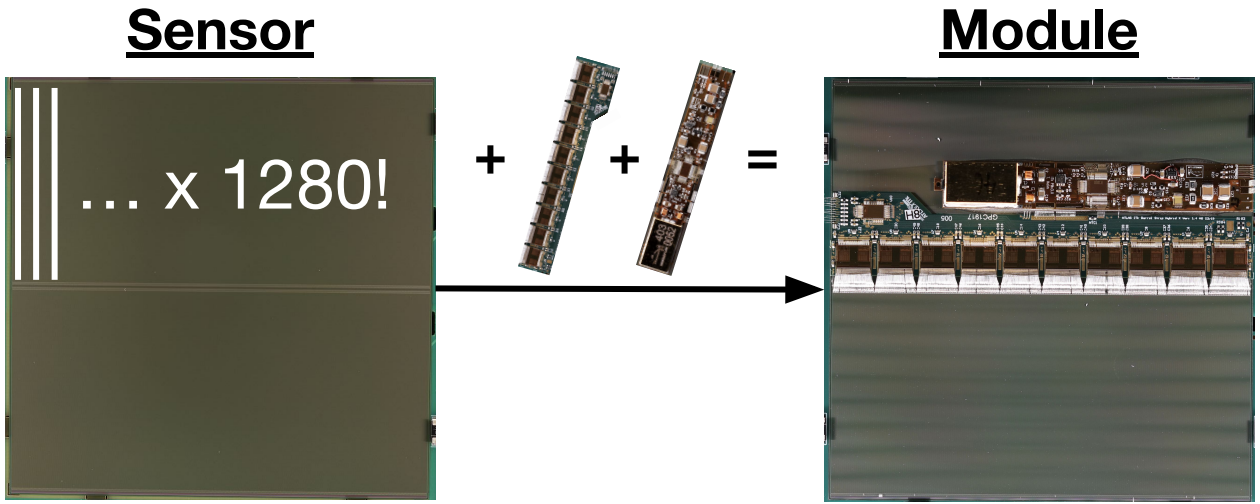
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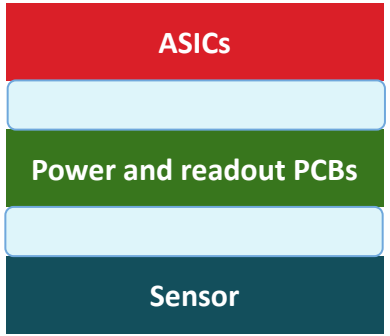
Birds-eye view

Modules

- Individual detection unit of the ITk strips: a “**Module**”



Birds-eye view



Side view

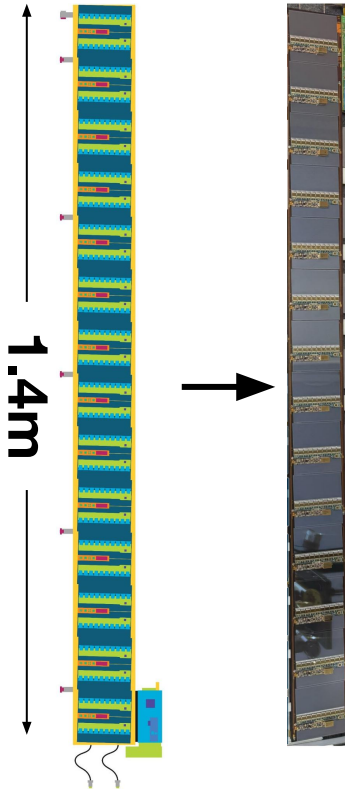
Staves and Quality Control

Brookhaven National Laboratory

- Modules attached to **local support structures** - finished product called a **stave**

Staves and Quality Control

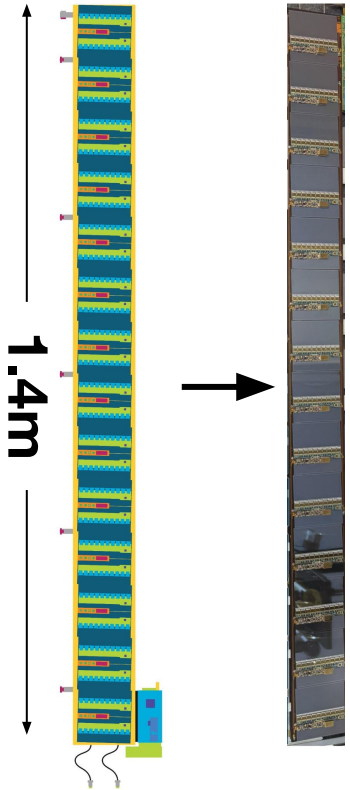
- Modules attached to **local support structures** - finished product called a **stave**



Each stave has **14 modules per side**

Staves and Quality Control

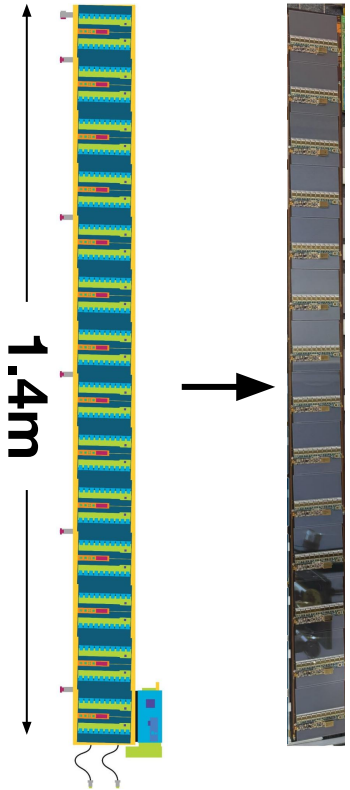
- Modules attached to **local support structures** - finished product called a **stave**
- BNL will assemble and test **196** staves to complete **half** of the Barrel (**~5500** modules)



Each stave has **14 modules per side**

Staves and Quality Control

- Modules attached to **local support structures** - finished product called a **stave**
- BNL will assemble and test **196** staves to complete **half** of the Barrel (**~5500** modules)
- **Every module** and **stave** undergoes **stringent quality control**
  → Ensure a quality detector



Each stave has **14 modules per side**

Path of a module



Path of a module



Produce ~ 1800 modules each!

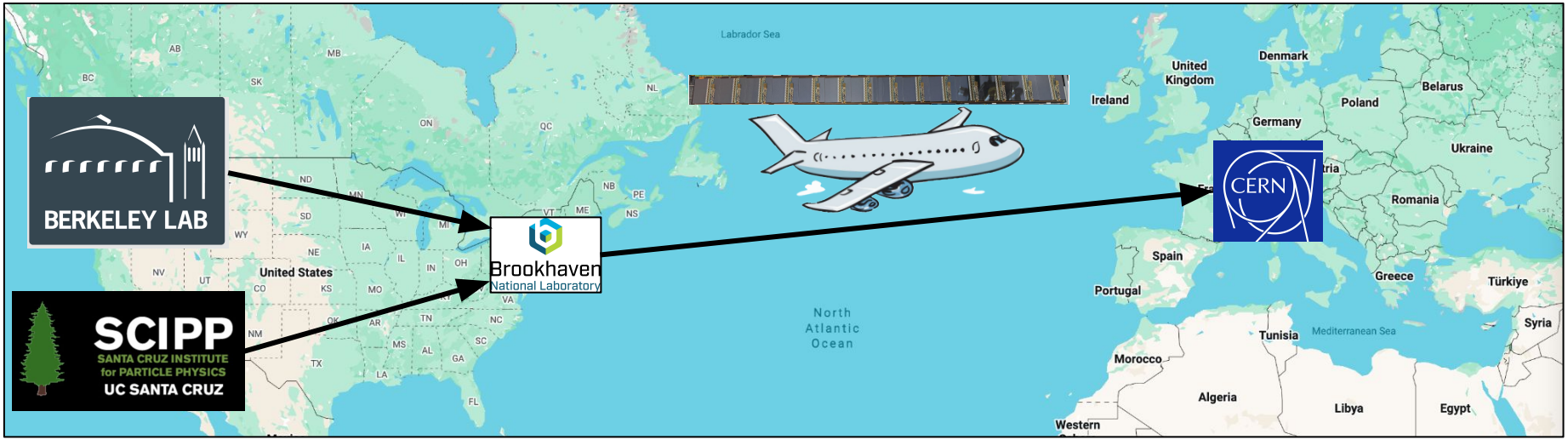
Path of a module



Produce ~ 1800 modules each!

Produce ~ 1800 modules, load 196 staves

Path of a module



Produce ~ 1800 modules each!

Produce ~ 1800 modules, load 196 staves

Send to CERN for full ITk integration

Stave testing

- After passing, **load** module onto **stave**

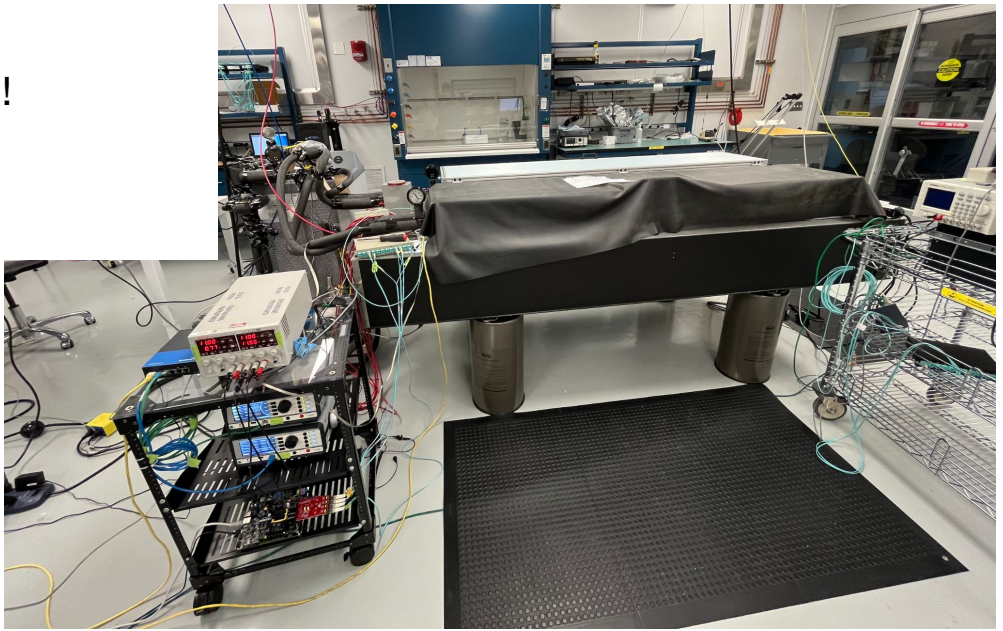
Stave testing

- After passing, **load** module onto **stave**
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BNL stave testing setup

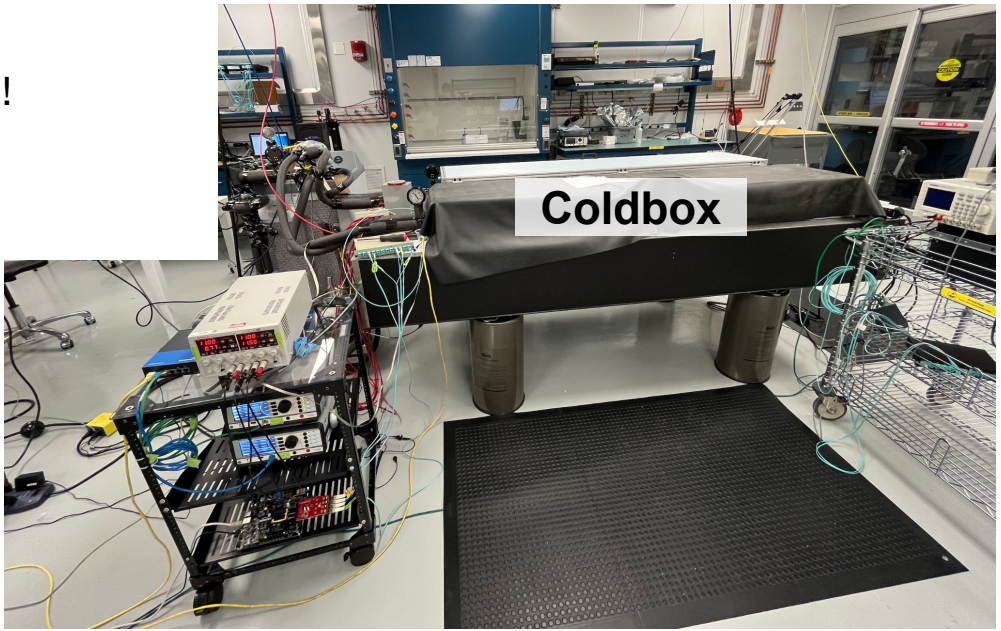


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BNL stave testing setup

- Coldbox (Stave inside)

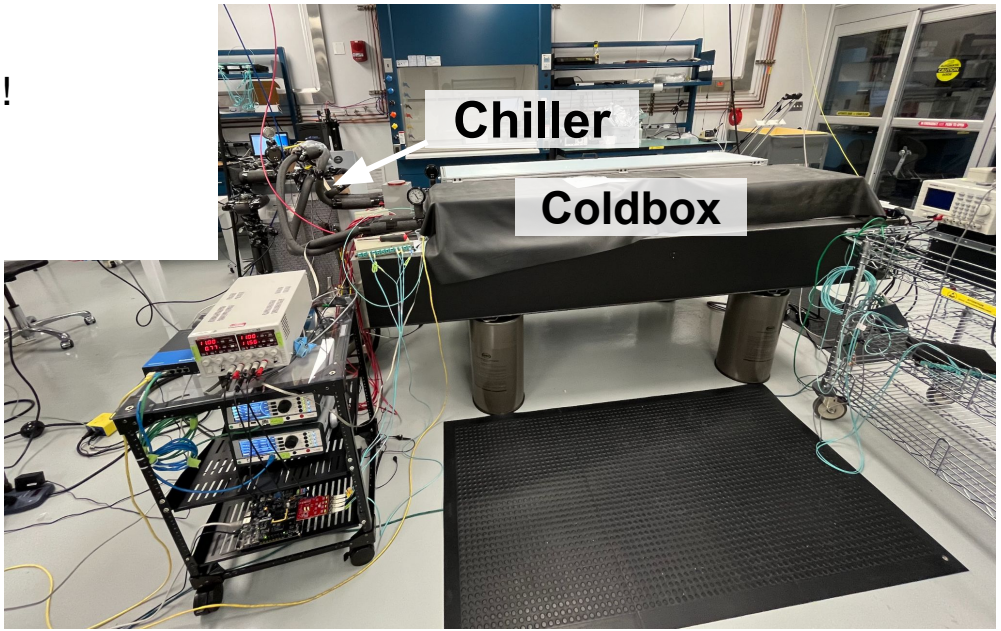


Stave testing

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BNL stave testing setup

- Coldbox (Stave inside)
- Chiller (temp control)

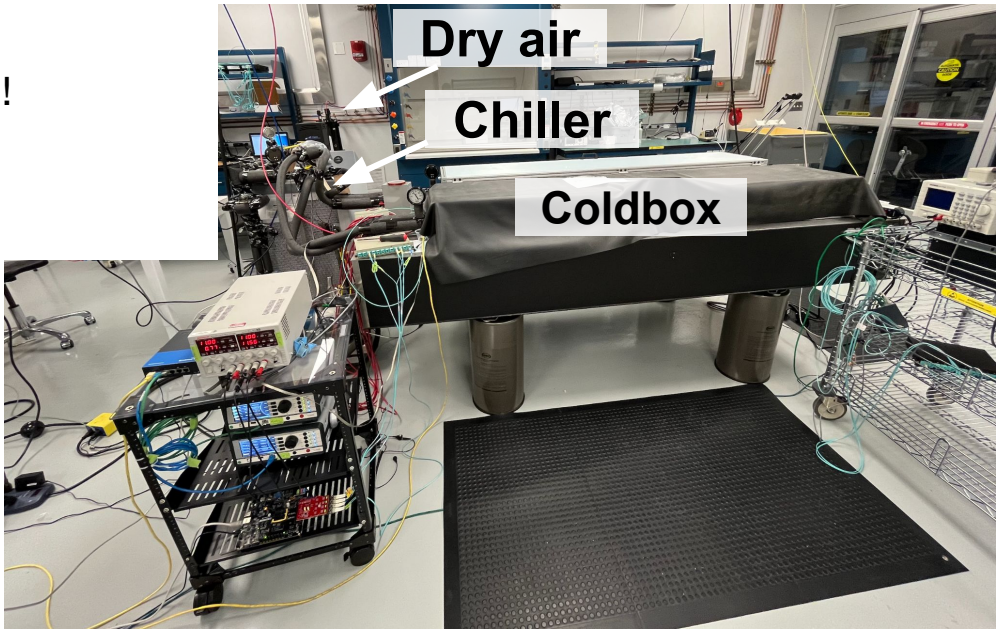


Stave testing

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- Coldbox (Stave inside)
- Chiller (temp control)
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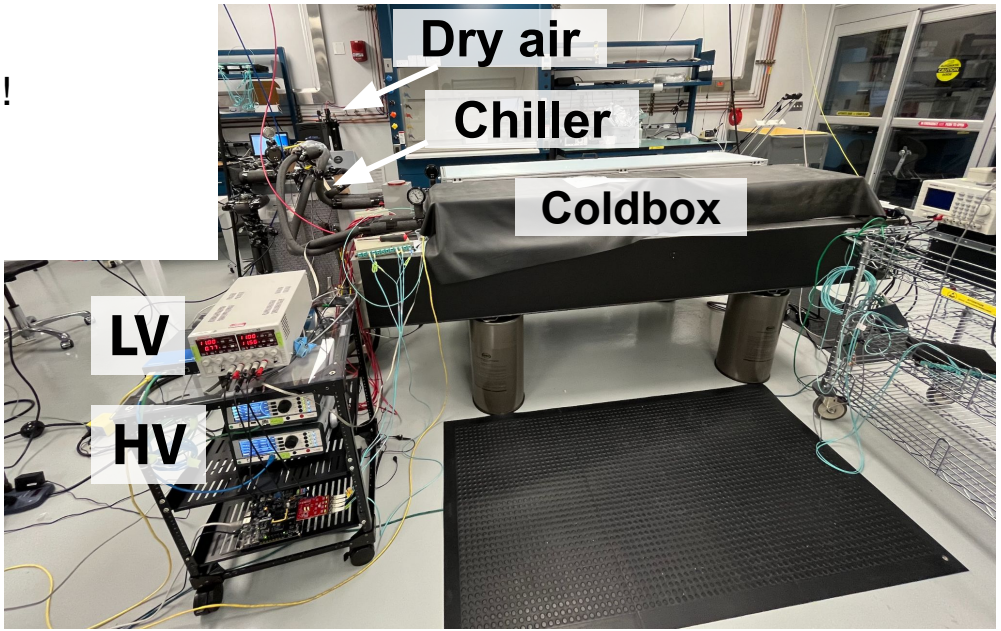


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BNL stave testing setup

- Coldbox (Stave inside)
- Chiller (temp control)
- Dry air (humidity control)
- Power supplies to power electronics, bias sensors

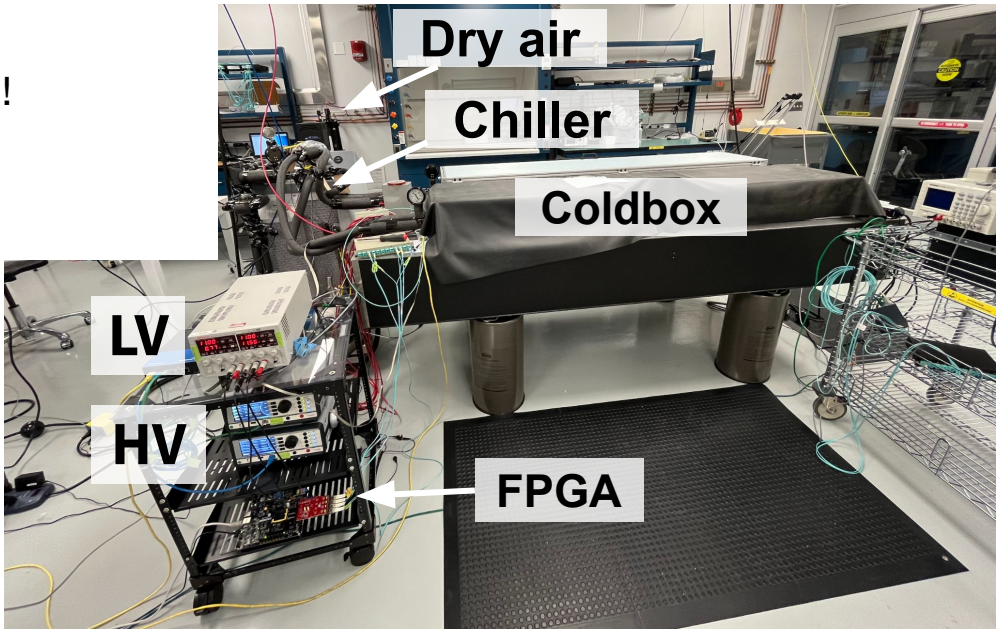


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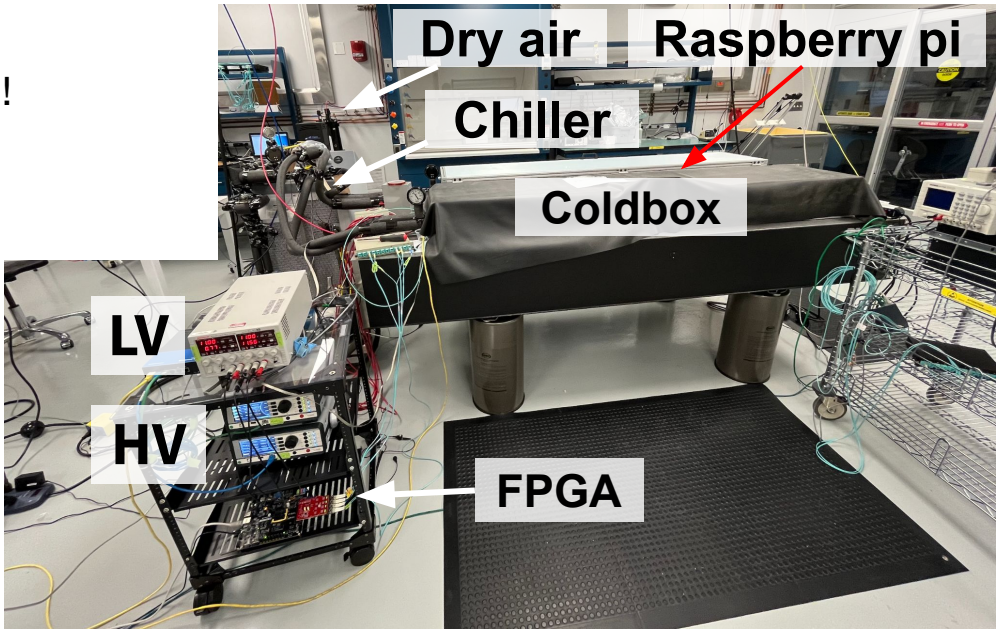


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- Stave tester



Fracturing

- **Pre-production:** Modules behaved well before going onto staves

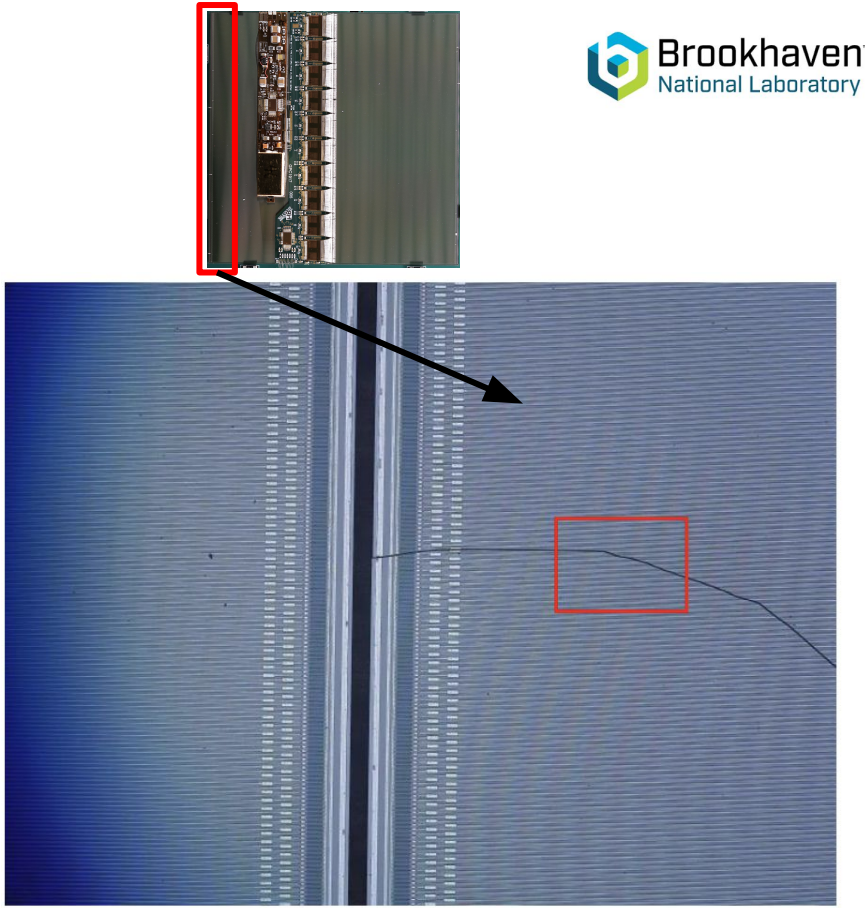
Fracturing

- **Pre-production:** Modules behaved well before going onto staves

- **After putting modules onto staves**, and **thermal cycling**, started observing electrical breakdowns

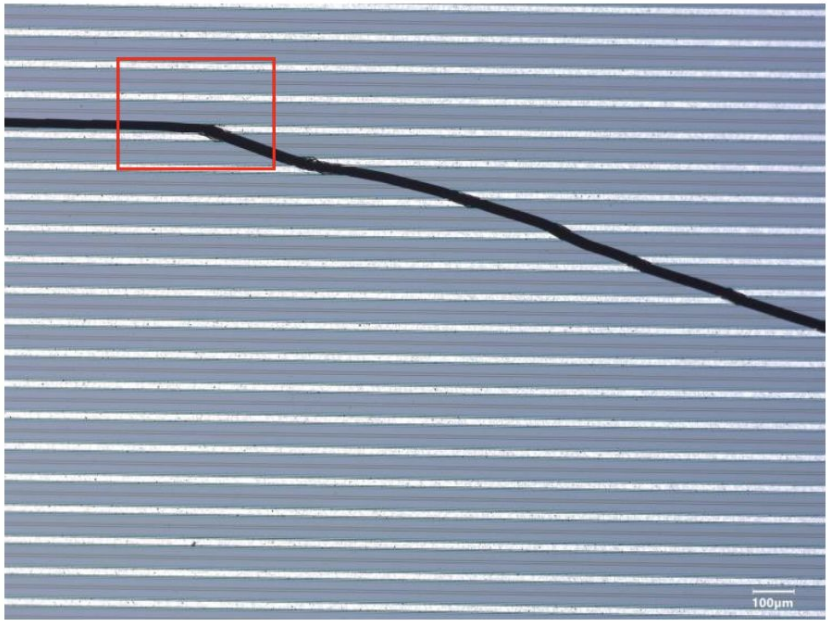
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- Upon investigation, found sensors **physically fracturing**



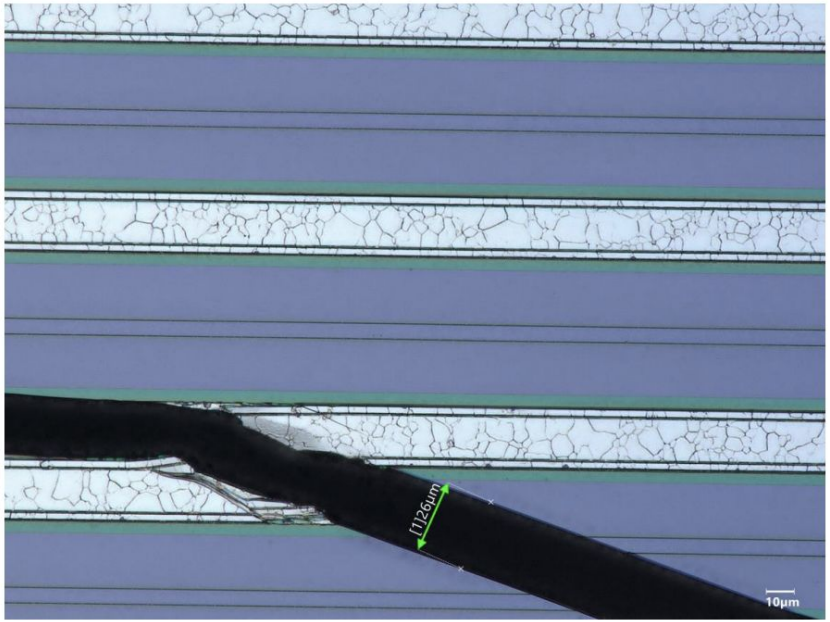
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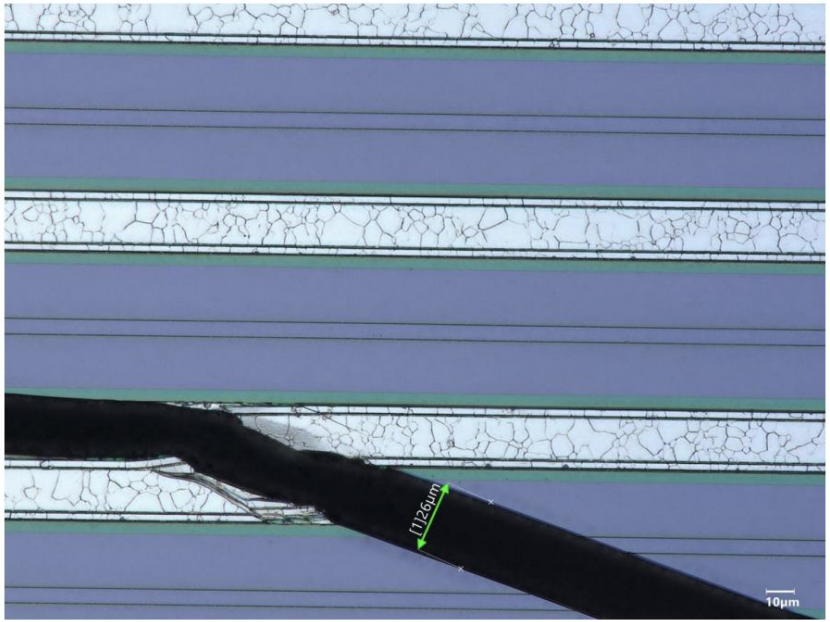
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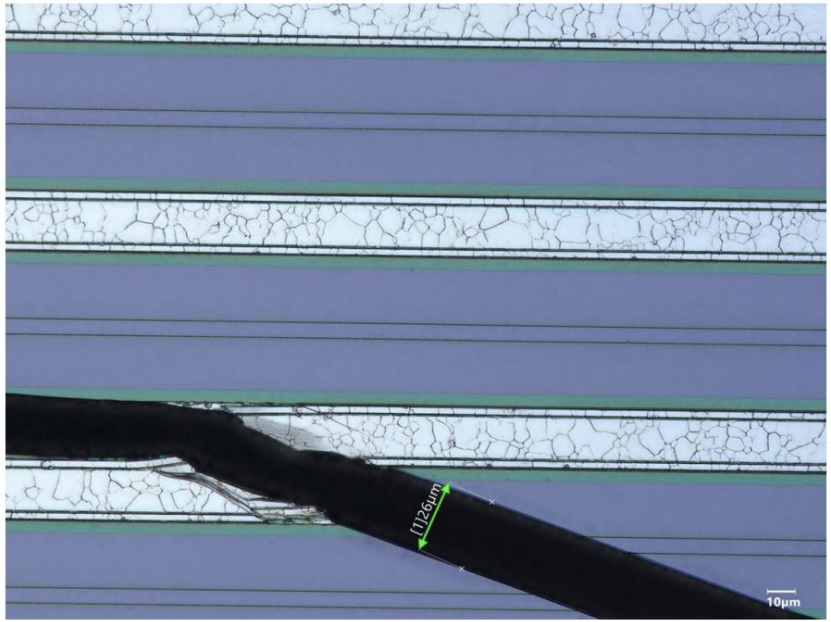
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- Upon investigation, found sensors **physically fracturing**
- Due to **Coefficient of Thermal Expansion mismatch**, close proximity of electronics.
- **Halted module production** while we investigated mitigation strategies

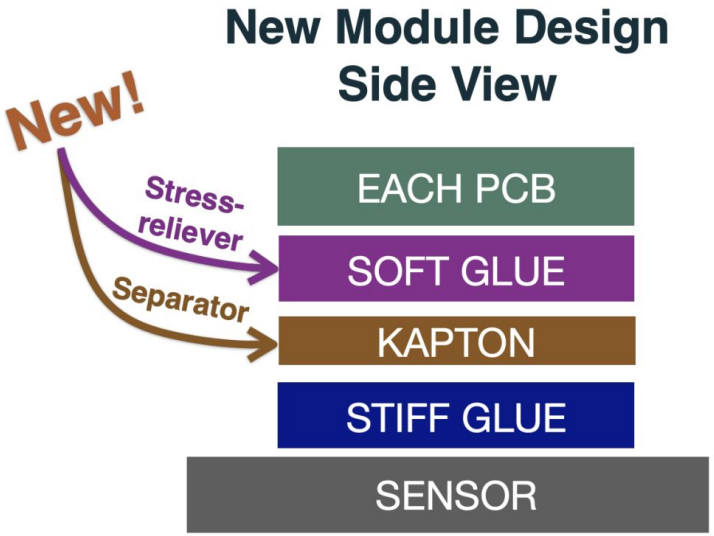


Interposers

- Multiple module **re-designs** tested.

Interposers

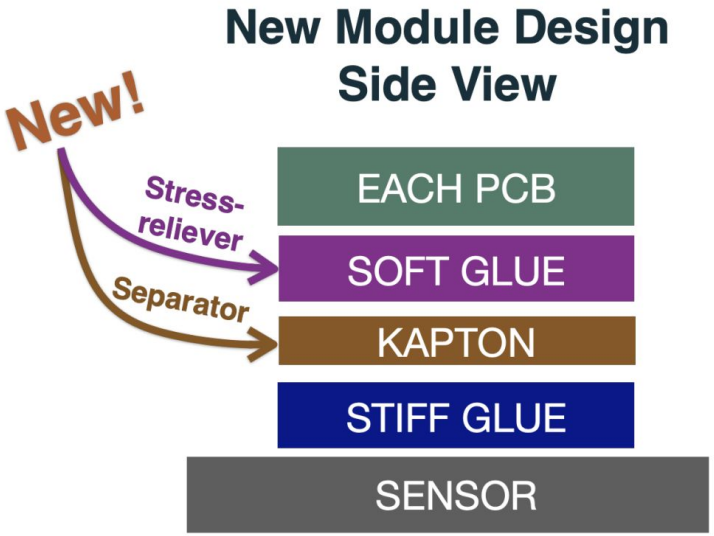
- Multiple module **re-designs** tested.
- **Interposer** design: Includes additional layer of **soft glue** to **absorb impact** from high stress



Stack-up diagram - A. Fortman, CPAD 2025

Interposers

- Multiple module **re-designs** tested.
- **Interposer** design: Includes additional layer of **soft glue** to **absorb impact** from high stress
- Rapidly ramped up interposer **module assembly program** to fill staves and test



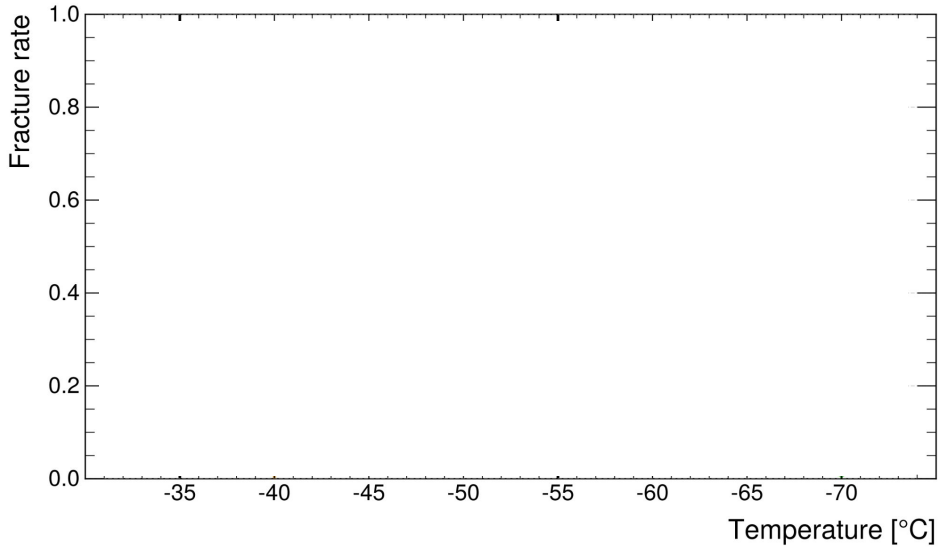
Stack-up diagram - A. Fortman, CPAD 2025

Extended-QC

- **Extended-QC** performed on **mitigation objects**

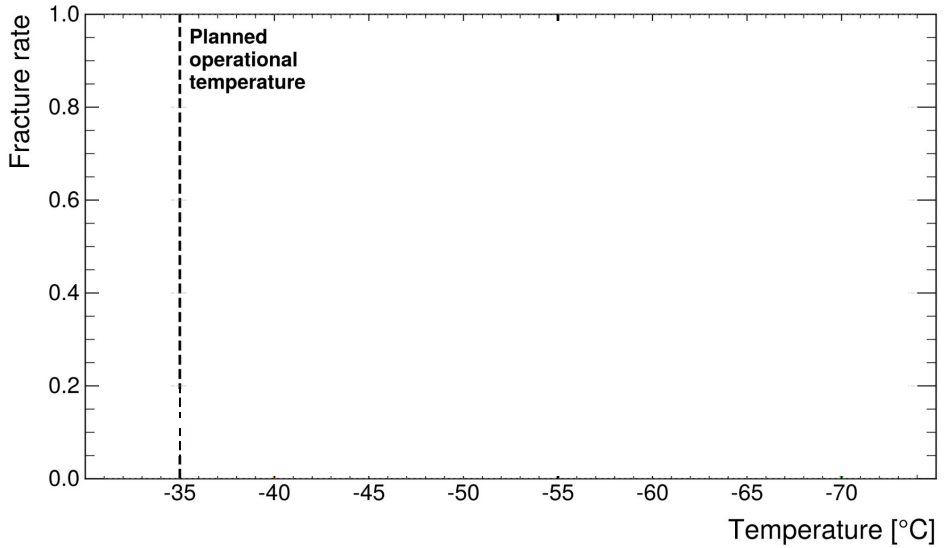
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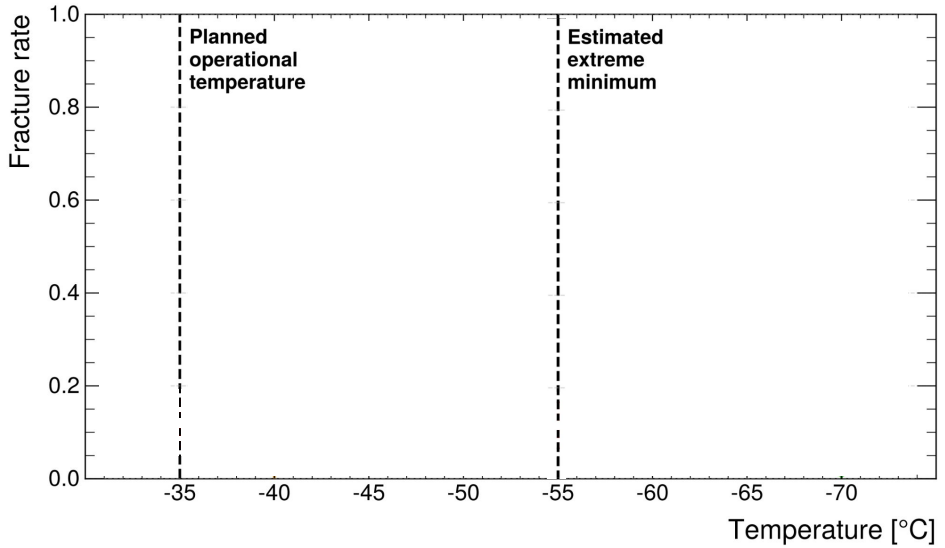
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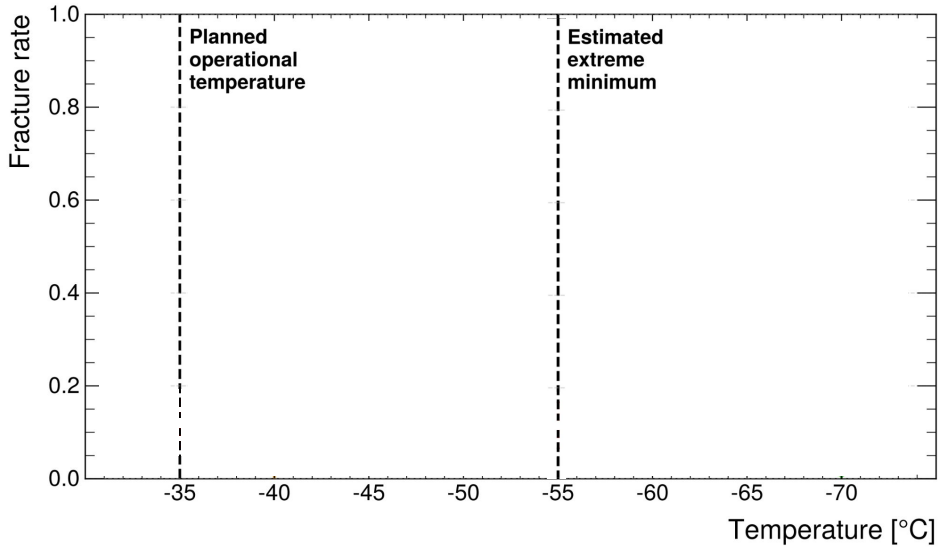
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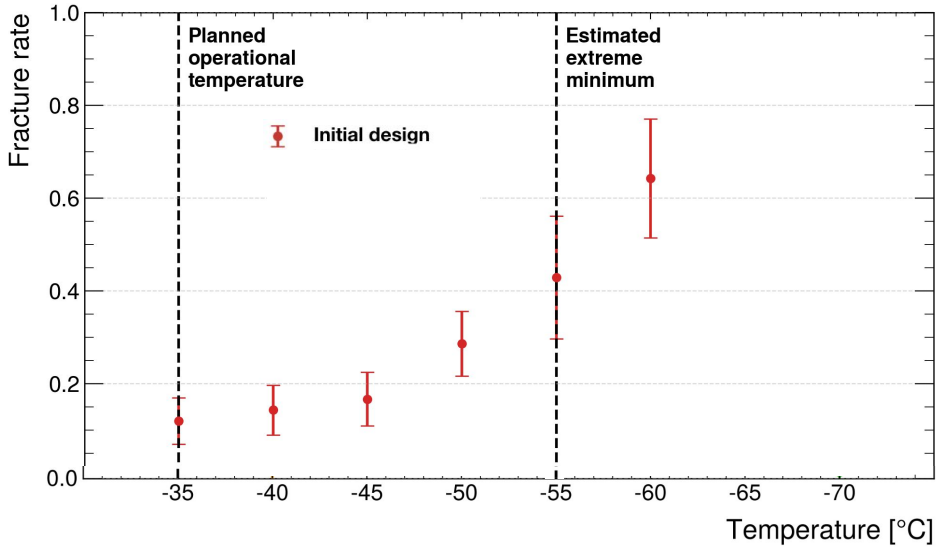
Extended-QC

- **Extended-QC** performed on **mitigation objects**
- Extreme testing performed down to **-70℃**



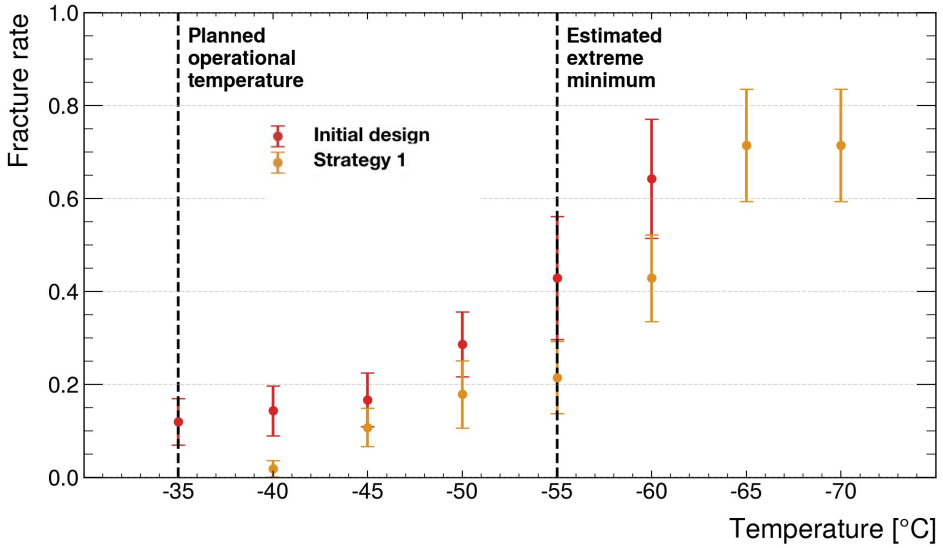
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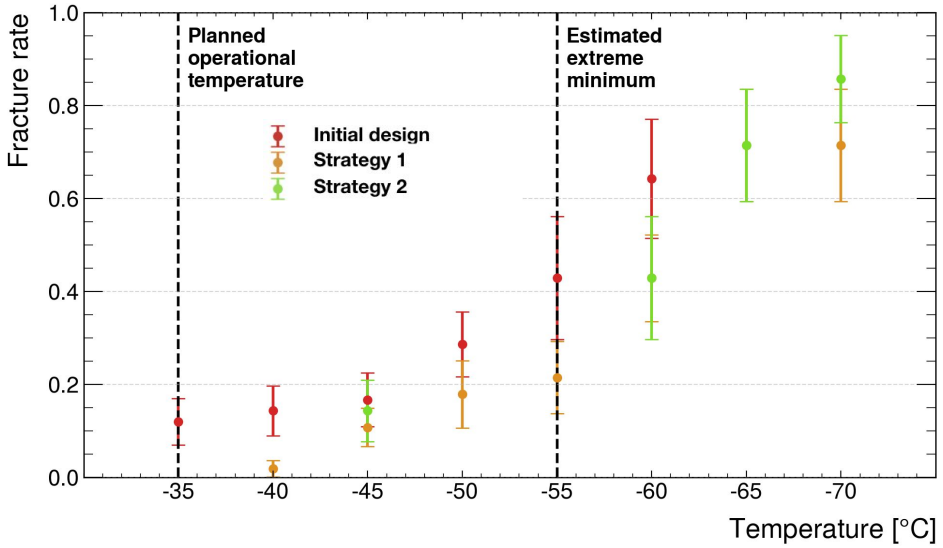
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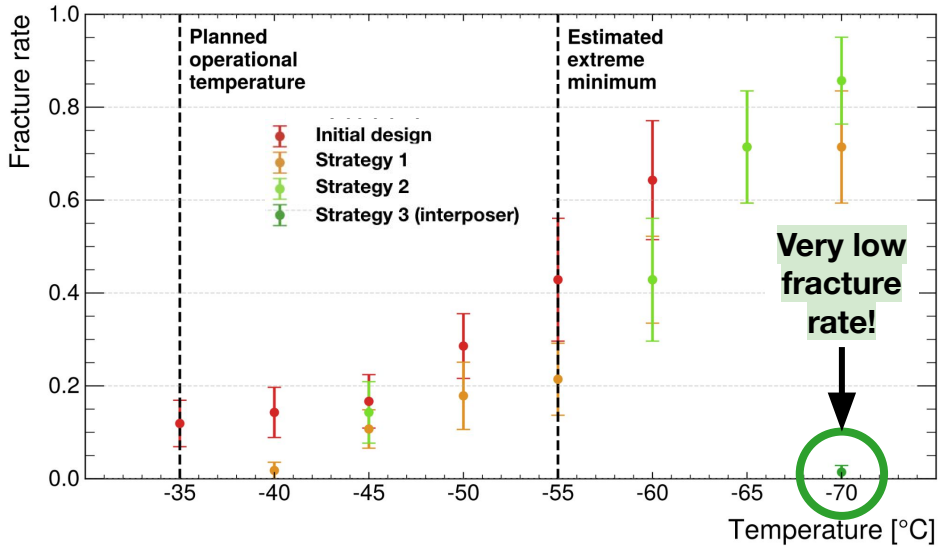
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2025 *JINST* 20 P09025



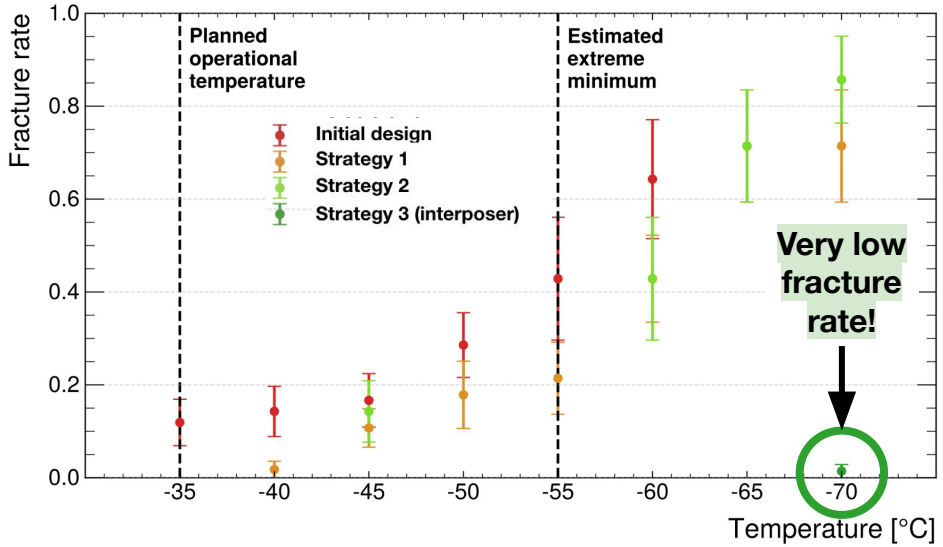
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- Moving forward with this solution, what **physics results** will be enabled at the HL-LHC?
  - Expect **HH discovery** (> **5σ**!)
  - **~ 30%** precision on **self-coupling**
  - **~5x** improvement on X→SH



2025 *JINST* 20 P09025



HL-LHC projections: ATLAS+CMS

Brookhaven National Laboratory

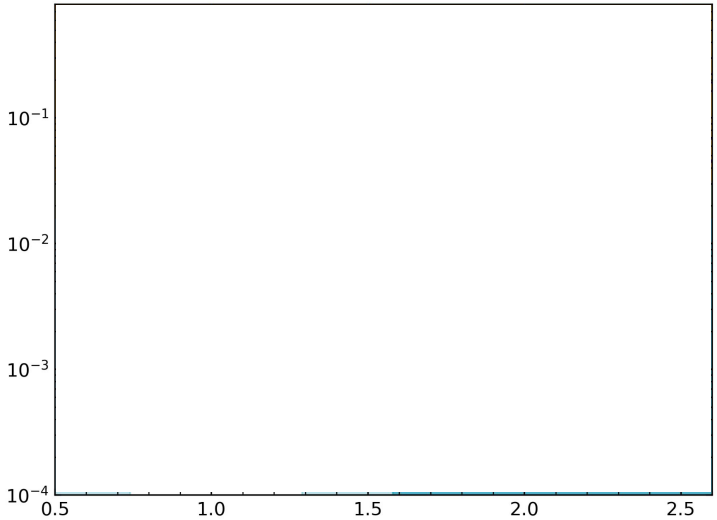
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Higgs → ZZ coupling precision

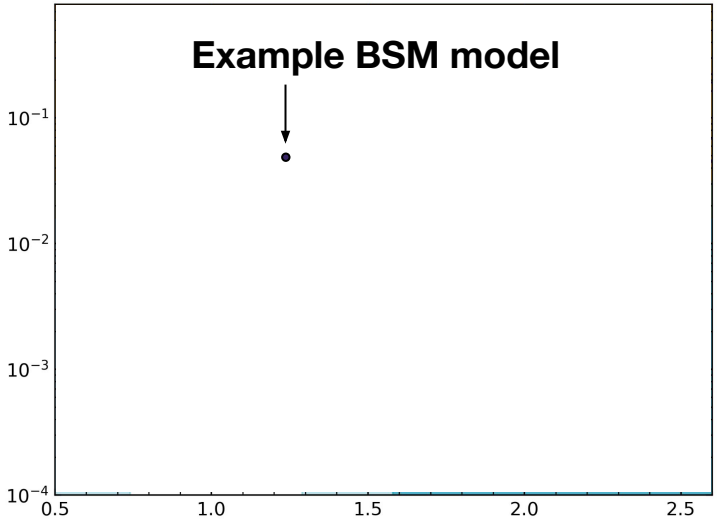


Higgs self-coupling measurement / standard model value

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- **Blue points:** Phase space where strong first-order phase transition is possible

Higgs → ZZ coupling precision



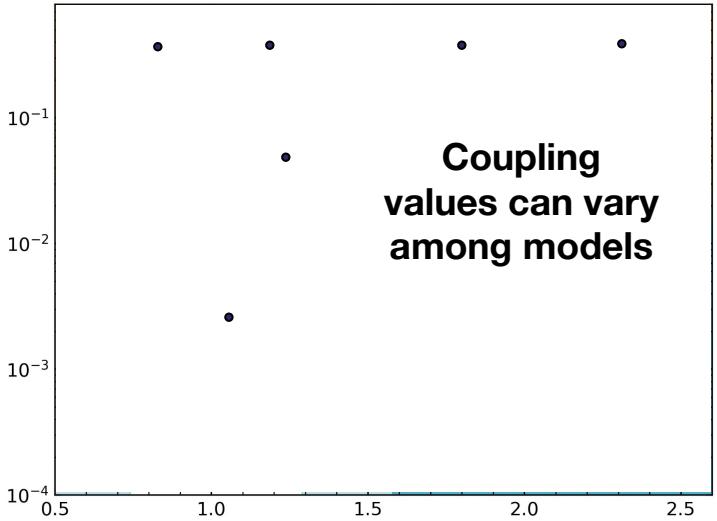
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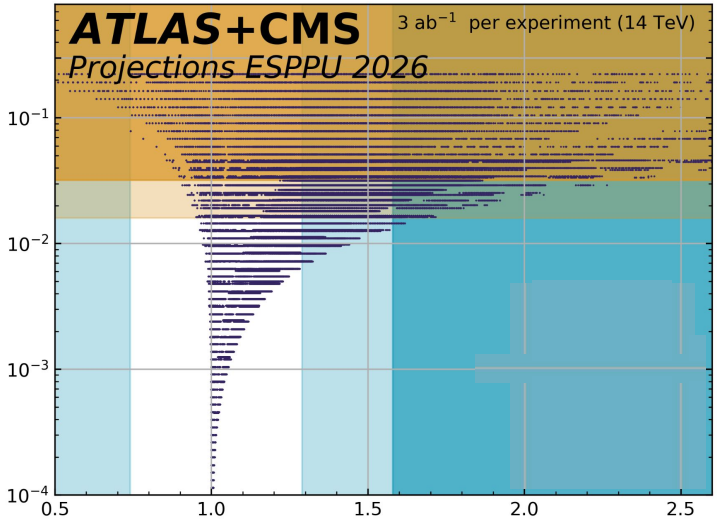


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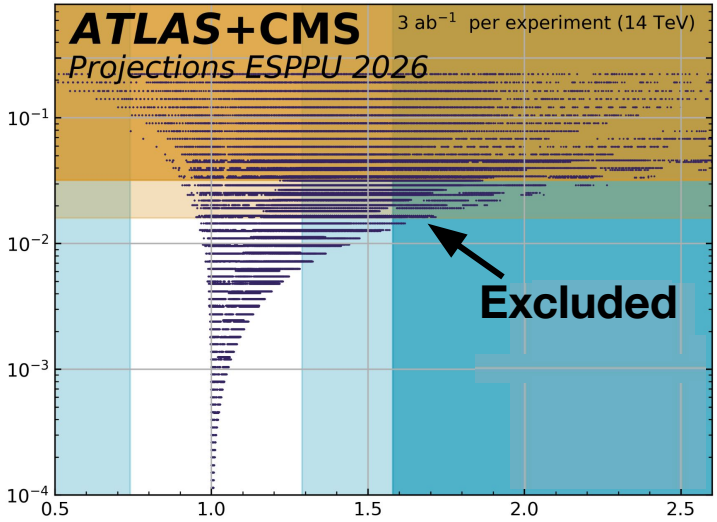
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arXiv:2504.00672

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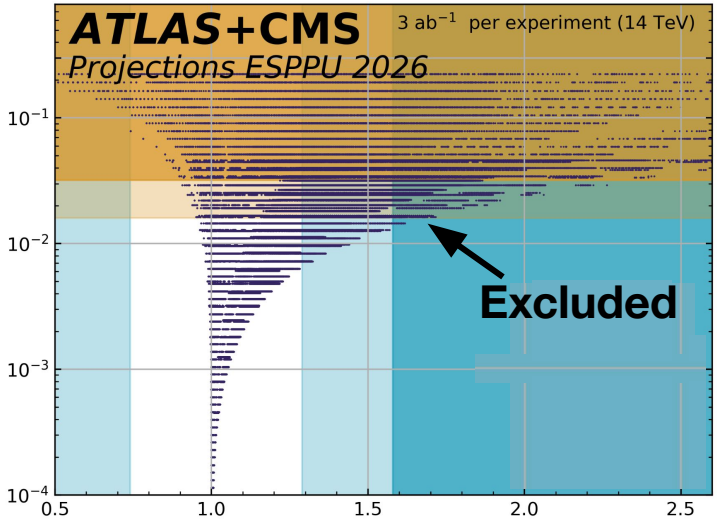
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HL-LHC projections: ATLAS+CMS

Brookhaven National Laboratory

- What can improved precision bring us?
- **Blue points:** Phase space where strong first-order phase transition is possible
- Precise Higgs coupling measurements, including the self-coupling and coupling to ZZ, **will exclude BSM scenarios**
- **Complementary to direct searches for new scalars!**

Higgs → ZZ coupling precision



Higgs self-coupling measurement / standard model value

arXiv:2504.00672

Next section



LHC

I

Higgs discovery

2009-2026

HL-LHC

II

di-Higgs discovery

2030-2041

Future

III

Future discoveries

2041-...

FCC

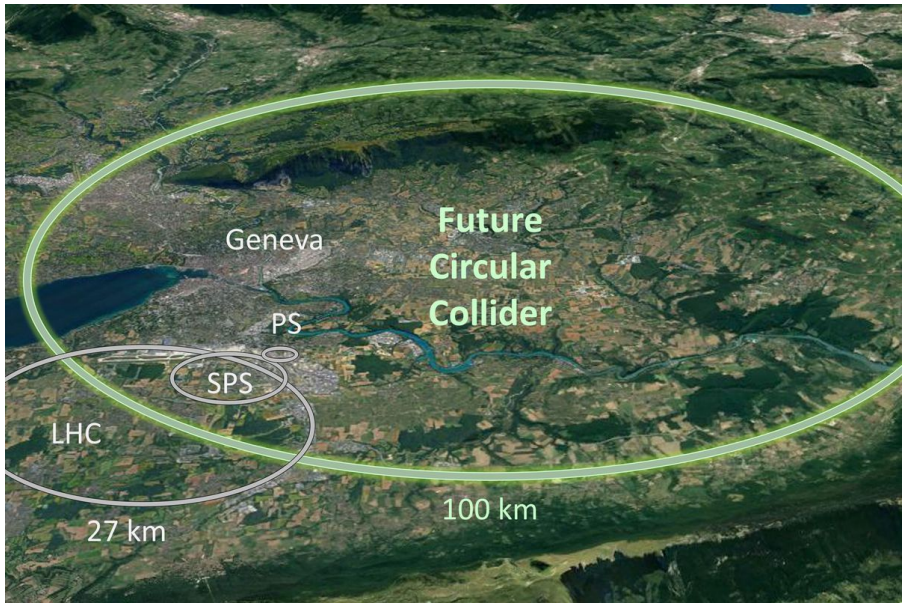
- What about after LHC?

FCC

- What about after LHC?
- Planning next collider with physics goals in mind

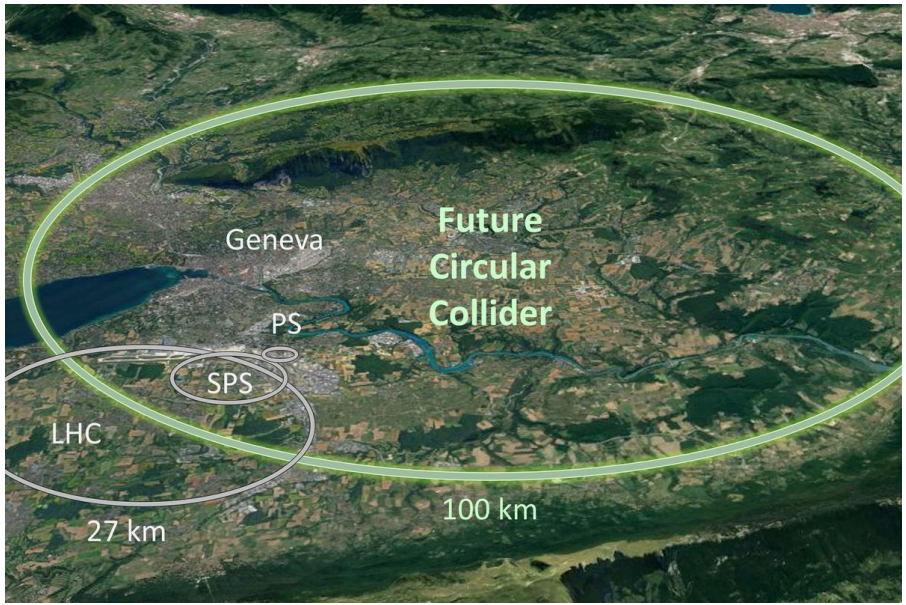
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FCC

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- Detectors need to meet specifications:
  - **FCC-ee:** Clean environment, need **efficient detectors**.
  - **FCC-hh:** Need unprecedented **radiation hardness**



SiC

- Is there a way to **improve tracker technology** to meet these challenges?

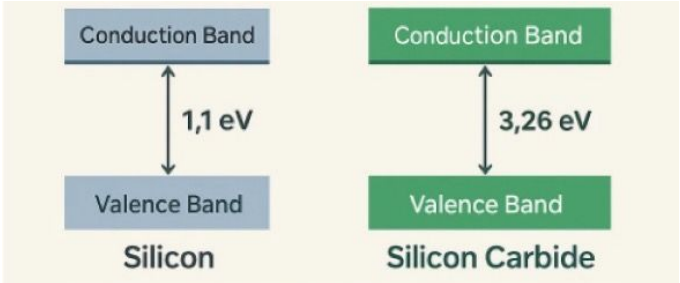
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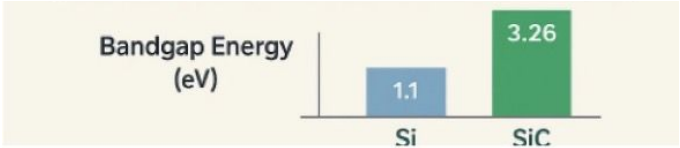
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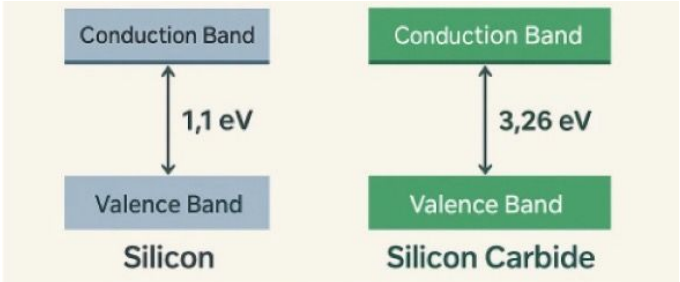
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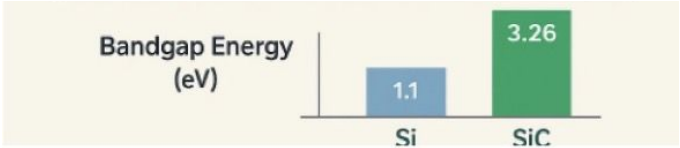




SiC

- Is there a way to **improve tracker technology** to meet these challenges?
- **Silicon Carbide** (SiC): Promising alternative to silicon detectors
- **Wide band-gap** material: Less noise!
- Can operate at **room temperature:**
  - **No need for cooling system!**
  - **Huge practical benefits**. Sensor fracturing delayed ITk strips production by **about 1.5 years**





Future results

- How can SiC help improve future **Higgs sector searches**?

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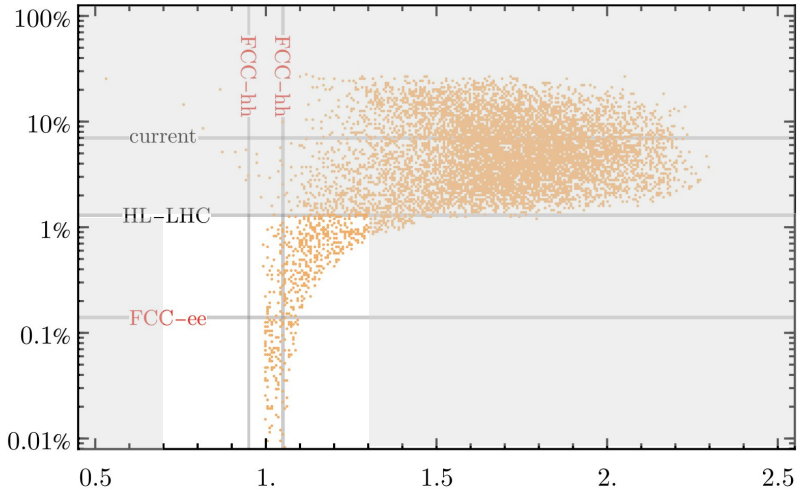
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Higgs → ZZ coupling precision



Higgs self-coupling measurement / standard model value

arXiv:2505.00272

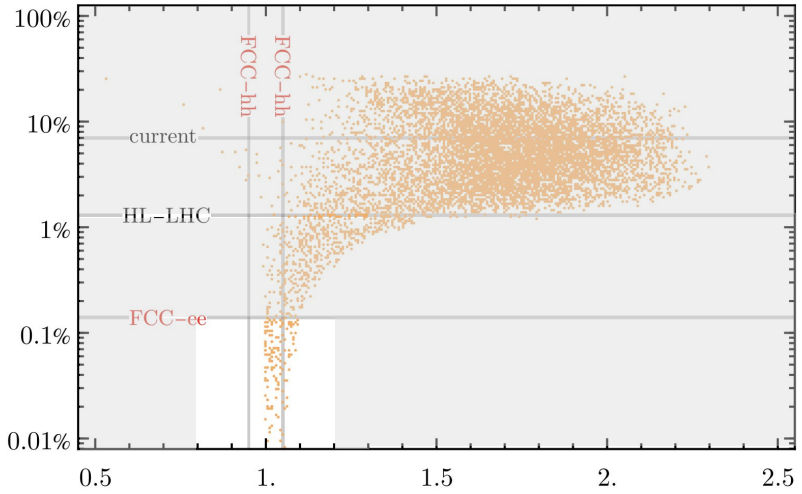
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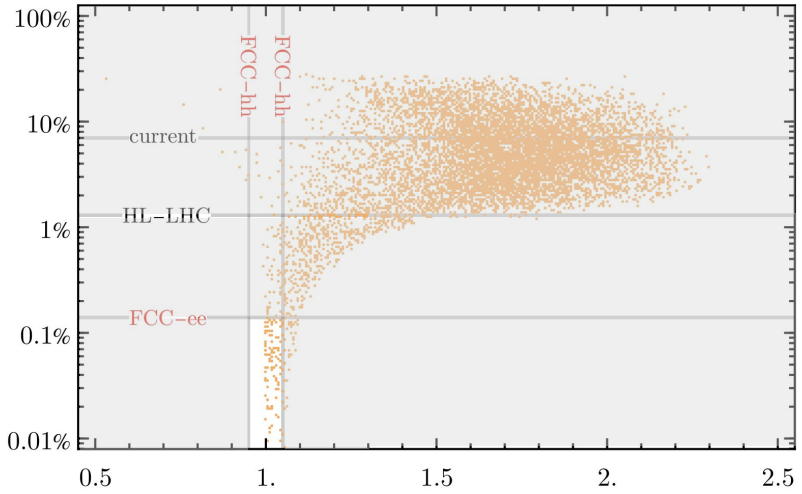
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Summary

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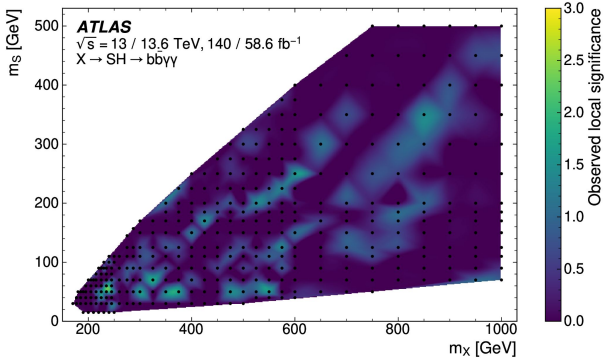
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  → **Tracking performance crucial** to achieve this

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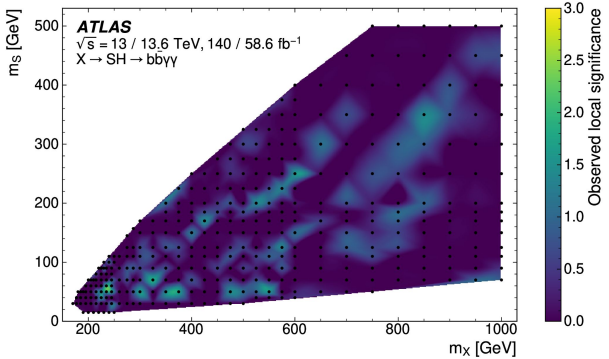
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LHC

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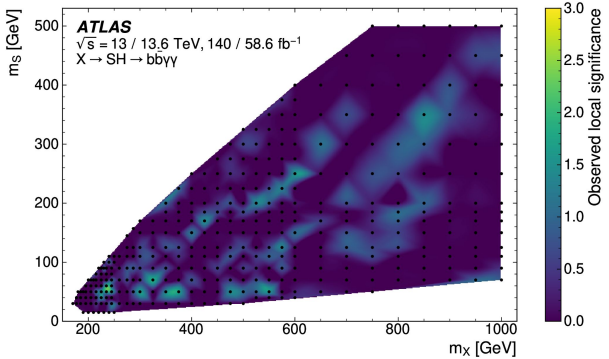
LHC



HL-LHC

Summary

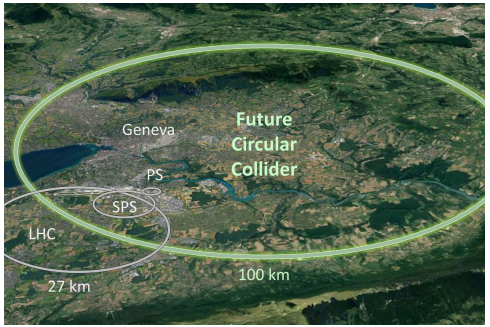
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LHC



HL-LHC



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Brookhaven National Laboratory

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Yes (so far!)

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Thank you for your attention!